Optimality of Public Persuasion for Single-Good Allocation*

Chen Chen, Xuyuanda Qi

New York University Shanghai
cc8029@nyu.edu, xq622@nyu.edu

First version: August 28, 2024
This version: September 8, 2026

Abstract

We study an information design problem in which a sender seeks to allocate an indivisible good to n receivers by strategically disclosing information. The good can be allocated to at most one receiver, and each receiver decides whether to accept it based on self-interest. Relevant applications include school advisors promoting students for job placements, incubators introducing startups to potential investors, and biotech firms seeking partnerships with larger pharmaceutical companies. In this setting, the sender can either send different signals to different receivers (i.e., *private persuasion*) or broadcast the same signal to all receivers (i.e., *public persuasion*). After receiving signals, receivers may communicate with one another in their self-interest to further reduce uncertainty about the good. We demonstrate that when the sender has a known preference among the receivers, public persuasion is optimal regardless of how the receivers communicate. The optimal public persuasion can be derived from a first-best relaxation problem that imposes only the receivers' participation constraints. We then focus on a special case in which the good's characteristics are captured by a one-dimensional variable, and all receivers' utility functions are linear in this variable. Using a dual approach, we derive optimality conditions for persuasion mechanisms. These conditions reveal that the persuasion problem decouples over linear segments of a piecewise-linear upper envelope function, which is derived from the Lagrangian relaxation of the first-best relaxation problem. We explicitly characterize this upper envelope function. This dual approach yields closed-form optimal mechanisms for the two-receiver case and an explicit characterization of the set of optimal mechanisms in the general case, thereby enhancing our understanding of their structural properties.

Subject classifications: Bayesian persuasion, public information, single-item allocation, multiple receivers, post-signal communication, Lagrangian dual

*An earlier version of this paper was circulated under the title "Optimality of Public Persuasion in Job Seeking."

1 Introduction

In this paper, we study a Bayesian persuasion problem in which a sender seeks to allocate an indivisible good among n receivers. The sender has a known preference over the receivers and can allocate the good to at most one receiver. Moreover, the sender holds private information about the good's characteristics, which determine the receivers' utilities. To maximize her expected payoff, the sender can commit to an information disclosure mechanism that strategically discloses these characteristics to the receivers. Notably, the sender can use either a *public* persuasion mechanism to share the same information with all receivers, or a *private* persuasion mechanism to send tailored information to different receivers based on their specific acceptance criteria. We illustrate our setting with three motivating examples.

Example 1: Job Seeking Consider a school advisor promoting a student in a job market with multiple potential employers (e.g., schools with open junior faculty positions or PhD programs at multiple universities). The advisor holds private information about the student's characteristics relevant to the employers' hiring requirements (e.g., research potential, teaching experience, and communication skills). The advisor can commit to an information disclosure mechanism (e.g., through targeted recommendation letters) that strategically discloses the candidate's characteristics to employers to maximize the candidate's expected payoff. Each employer decides whether to extend a job offer to the student, and the student can accept at most one offer. Finally, the student has a known preference among the employers. For example, candidates often prioritize employers based on alignment of interests, institutional reputation, and geographic location, which employers commonly recognize.[1]

Example 2: Startup Fundraising Consider an incubator introducing a startup to potential investors. The startup has achieved initial market traction and is seeking a significant round of venture capital (VC) funding (e.g., Series A). To attract investment, the incubator helps the startup approach potential VC investors and strategically share pertinent information about the startup's performance and product through either private or public channels.[2] Each VC firm decides whether

[1]For example, in academic job markets, top-ranked institutions typically do not spend much time worrying about whether a candidate will accept their offer because they are generally highly attractive to any candidate. Conversely, mid-ranked institutions will factor the chance that a student will accept an offer into consideration, as they know that a coveted candidate might decline their offer if a better offer comes.

[2]For example, a startup might hold private meetings with certain VC firms to disclose financial information and showcase prototypes. Alternatively, it could release a public signal like a press release highlighting key metrics, such as user growth or market traction, that every investor sees simultaneously.

to extend a term sheet (i.e., an investment offer). Typically, the startup selects only one investor as the lead for a funding round; thus, only one offer is ultimately accepted. Furthermore, it is widely recognized that VC funding is not just about money. Startups often prefer lead investors who can bring additional value beyond capital, such as mentorship, prestigious VC brand recognition, specialized industry expertise, and valuable industry networks (Hsu 2004, Sorensen 2007). These preferences are usually visible, either explicitly or through inference based on the startup's industry context and observable attributes.

Example 3: Biotech Company Seeking Partnership In the pharmaceutical industry, small biotech firms often develop new drug candidates and then seek partnership agreements, such as license-out arrangements, with larger pharmaceutical companies (Excedr 2025). Each pharmaceutical company decides whether to pursue such a partnership, and the biotech typically enters into an exclusive agreement with a single partner to jointly bring the drug to market. In this setting, the biotech firm holding a promising drug candidate acts as the information sender, and potential pharmaceutical partners serve as receivers. The biotech can strategically disclose information about the drug's efficacy, safety profile, or clinical trial results through either private or public channels.[3] Moreover, biotech firms tend to have clear preferences among potential partners based on factors such as available resources, strategic alignment, and commercialization expertise (Excedr 2025, Rothaermel and Boeker 2008).[4] These preferences are usually known to pharmaceutical companies or can be inferred based on observable characteristics of the drug candidate (e.g., its therapeutic target) and the biotech firm's profile.

Our model also allows receivers to communicate among themselves after receiving signals from the sender, which is common in practice. Specifically, receivers may communicate with each other (either simultaneously or sequentially, using either cheap talk or some degree of commitment) to reduce uncertainty about the good in their self-interest.[5] Then, based on the signal received from

[3]For example, a biotech may provide detailed clinical data to selected pharmaceutical companies during private meetings, or publicly disclose key trial outcomes by publishing in academic journals or presenting at conferences, thereby revealing information to all potential partners simultaneously.

[4]For instance, biotechs often prefer large pharmaceutical companies with substantial R&D budgets to support expensive late-stage trials, rich expertise in relevant therapeutic areas and regulatory processes, and established sales channels in target markets to maximize the drug's likelihood of success.

[5]For example, at academic job market conferences, colleagues from different institutions often exchange information about job market candidates during informal conversations. Analogously, the VC community is characterized by extensive formal and informal communication networks. VC partners frequently co-invest in deals, attend industry events, and leverage their networks for deal sourcing and due diligence. As a result, information and opinions about startups may be shared among VCs, sometimes selectively. Finally, in the biotech setting, pharmaceutical companies may communicate through various industry channels. For example, rumors about a drug's efficacy might circulate through shared contacts, such as scientists who consult for multiple firms.

the sender and the additional information from other receivers, each receiver decides whether to extend an offer to the sender. We note that the receivers in this context face both cooperative and competitive incentives. Communication reduces uncertainty about the good's characteristics, thereby benefiting each receiver. However, since the sender can accept only one offer, competition among the receivers arises. In particular, if a receiver knows that the good is of high quality, he may withhold this information from other receivers to avoid competition, especially if the sender prefers other receivers. Therefore, the potential for subsequent communication among receivers substantially complicates the information design problem, making it unclear what an optimal persuasion mechanism is.

As our first main result, we demonstrate that public persuasion is always optimal regardless of the detailed communication protocol used by the receivers, as long as the sender has a fixed preference ordering over the receivers (Section 3).[6] Since all receivers receive the same information under a public persuasion mechanism, subsequent communication cannot convey any payoff-related information and therefore becomes irrelevant. As a result, the sender eliminates any room for the receivers to communicate and infer further about the good for the sender's own benefit.

Furthermore, we show that the optimal public persuasion mechanism can be solved from a first-best relaxation problem that imposes only the receivers' participation constraints. Specifically, in the first-best problem, a central planner allocates goods to receivers based on the goods' characteristics. A receiver accepts the good when it is allocated to him. The first-best relaxation problem solves the optimal randomized allocation to maximize the sender's expected payoff, ensuring only a nonnegative expected utility for each receiver. We show that an optimal public persuasion mechanism can be derived from an optimal solution to the first-best relaxation problem and that its expected payoff matches the first-best upper bound.

Finally, we conduct a numerical study reflecting a realistic setting in which the sender's preferences over receivers are uncertain. In practice, although the sender prefers top-tier receivers to mid-tier receivers, and mid-tier receivers to lower-tier receivers, the sender's ranking of receivers within the same tier can be uncertain. Our numerical results indicate that, in this setting, the suboptimality gap of public persuasion mechanisms is negligible (Section 3.3.3).

Although an optimal public persuasion mechanism can be derived from the aforementioned first-best relaxation problem when the sender's preferences over receivers are fixed, the problem

[6]This is perhaps striking because the optimal persuasion mechanism differs among receivers when considering each receiver in isolation. Moreover, public persuasion is optimal even when the sender knows that receivers cannot communicate (but are aware of each other's existence), as we elaborate further in Remark 3.1.

becomes an infinite-dimensional linear program (LP) when the state space is infinite, making it challenging to solve. As our second main result, we study the structural properties and efficient computation of an optimal public persuasion mechanism for a special case in which the good's characteristics can be summarized by a one-dimensional variable, and all receivers' utility functions are linear in this variable; consequently, each receiver is concerned only about the mean quality of the good allocated to him (Section 4). This setup has been extensively studied in the literature; see, for example, Candogan (2022), Dworczak and Martini (2019), Kleiner et al. (2021), and Arieli et al. (2023), and we compare with these works in detail in Section 1.1.

Under the linear utility assumption, we provide optimality conditions for persuasion mechanisms based on the Lagrangian dual of the first-best relaxation problem, in which we dualize the participation constraints. Central to this dual approach is characterizing an upper envelope function arising from the geometric interpretation of the Lagrangian dual. This envelope function is increasing, convex, and piecewise linear. The optimality conditions state that once this upper envelope function is identified, the persuasion problem decouples across its linear segments (Theorem 4.3).

As a direct application of the optimality conditions, we derive optimal persuasion mechanisms in closed form when there are two receivers, where one receiver has a higher acceptance threshold but also brings a higher payoff (Section C). The main trade-off is that an offer from a more competitive receiver brings a higher payoff; however, targeting this receiver more aggressively is costly because it reduces the overall probability of receiving an offer. Depending on the relative desirability of the two receivers and their acceptance thresholds, the optimal persuasion mechanism carefully balances this trade-off.

For the general case with n receivers (Section 5), we explicitly characterize the upper envelope function (Proposition 5.2). This characterization is based on establishing a dual connection between our first-best relaxation problem and a convex optimization formulation proposed in Candogan (2022). Once the upper envelope function is characterized, the set of optimal persuasion mechanisms is fully determined. Finally, we also demonstrate various ways to construct optimal persuasion mechanisms in Section D. These include (i) a randomized persuasion mechanism with a monotone structure (i.e., a higher-quality good is more likely to be placed in a better position) and (ii) a deterministic persuasion mechanism with a double-interval structure, as illustrated in Candogan (2022).

The rest of the paper is organized as follows. Section 1.1 reviews related literature. Section 2 formulates the problem. In Section 3, we establish that public persuasion mechanisms are optimal in our setup, irrespective of how the receivers communicate. Moreover, the optimal public persuasion mechanism can be obtained from a first-best relaxation problem that requires only the receivers' participation constraints. Section 3.3.3 numerically examines a practically relevant setting in which the sender's preference ordering over receivers within the same tier is uncertain. We demonstrate that the suboptimality gap of public persuasion is negligible in this case. Section 4 analyzes the case in which the receivers possess linear utility functions. We provide optimality conditions based on duality in Section 4.2, which reveal that the persuasion problem decouples over linear segments of a piecewise-linear upper envelope function obtained from the Lagrangian relaxation of the first-best relaxation problem. Section 5 explicitly characterizes this upper envelope function. Section 6 concludes.

1.1 Related Literature

Our work contributes to the literature on Bayesian persuasion and information design, particularly in contexts involving multiple receivers. As Kamenica (2019) highlights, "if sender can send separate signals to each receiver, and if either (a) a receiver's optimal action depends on what other receivers do, or (b) sender's utility is not separable across receivers' actions, then the problem becomes significantly more difficult." Our setup incorporates both features. Below, we review relevant topics from the information design literature.

Persuasion with Spillovers Much of the existing literature does not incorporate post-signal communication among receivers as we do. Notable exceptions include Babichenko et al. (2022), Galperti and Perego (2023), and Candogan et al. (2023), which consider informational spillovers among receivers. In these studies, the spillover structure is prespecified, and public persuasion is typically suboptimal. By contrast, our model allows receivers to communicate in their self-interest via arbitrary communication processes. Babichenko et al. (2022) characterize the conditions under which one spillover structure universally yields higher expected utility for the sender than another. Galperti and Perego (2023) characterize all possible equilibrium outcomes that can arise from an information structure given spillover and seeding constraints. Candogan et al. (2023) show that the optimal information design problem is generally computationally challenging under information spillovers, except for some specific cases.

LP Formulation Bergemann and Morris (2016) and Bergemann and Morris (2019) relate the multi-receiver persuasion problem to the game-theoretic concept of Bayes correlated equilibrium (BCE). This relationship leads to a natural LP formulation for obtaining an optimal persuasion mechanism.

Specifically, the decision variables in the LP are joint probabilities of states and receivers' actions, and the constraints completely characterize the set of BCEs.[7] Our first-best relaxation problem (2) is also an LP. However, in our LP, the decision variables are marginal allocation probabilities under a mechanism. The LP imposes only participation constraints that any mechanism must satisfy, and thus, does not precisely characterize the set of equilibrium outcomes.

Optimality of Public Persuasion Bergemann and Morris (2019) also explore when public persuasion mechanisms are optimal (Section 4.1 therein). They show that public persuasion mechanisms are optimal when receivers' actions are strategic complements, as these mechanisms induce positively correlated actions. However, in our setup, receivers' actions are not strategic complements. For a binary state space without payoff externalities among receivers, Arieli and Babichenko (2019) provide necessary and sufficient conditions for the optimality of public persuasion (Theorem 3 therein). Consequently, we have identified distinct conditions under which public persuasion remains optimal in a strong, robust sense—specifically, when the sender allocates a single good and has known preferences over receivers.

Finally, Lingenbrink and Iyer (2018) study a scenario in which a seller offers a product over two periods. The seller knows the current inventory level and the number of customers, whereas customers lack this information. They demonstrate that public information disclosure is optimal for the seller when customers are identical. By contrast, Drakopoulos et al. (2021) study a similar setting, but with customers having heterogeneous valuations for the product. They demonstrate that personalized (or private) signaling significantly increases early purchases and profits; indeed, personalized information can be as effective as personalized pricing.

Linear Utilities In Section 4, we consider a special case in which the good's characteristics w is one-dimensional, and each receiver's utility function is linear in this variable. Consequently, each receiver is concerned only about the mean quality of the good allocated to him. This assumption simplifies the information design problem, as it becomes equivalent to designing a distribution of posterior means of the good's quality. Dworczak and Martini (2019), Kleiner et al. (2021), and Arieli et al. (2023) study a general optimization problem in which the sender's utility depends on the posterior mean of the underlying state in an arbitrary way. They design the distribution of posterior means, which must be a mean-preserving contraction of the prior distribution, to maximize the sender's expected utility. The persuasion problem studied in Sections 4 and 5 corresponds to

[7] That is, a joint distribution sustains a BCE if and only if it is feasible to the LP.

a special case of their problem, in which the sender's utility function is increasing and piecewise constant in the posterior mean, as in Candogan (2022).

Candogan (2022), Kleiner et al. (2021), and Arieli et al. (2023) study the persuasion problem through an extreme points approach.[8] Their results are an optimal persuasion mechanism that has a double-interval or a more general bi-pooling structure. Our work, on the contrary, uses a dual approach to establish optimality conditions for any persuasion mechanism, and characterizes the set of *all* optimal persuasion mechanisms.

Dworczak and Martini (2019) also use a dual approach to derive optimality conditions for their general optimization problem (see their Theorem 1 and Proposition 1). Specifically, they show that a distribution of posterior means sustains an optimal persuasion mechanism if and only if it, together with a price function $p(x)$, satisfies conditions (2)–(4) therein, and they interpret this price function $p(x)$ as the Walrasian equilibrium price in a persuasion economy. We study very different formulations of the persuasion problem (our problem (5) versus their problem (1)) and dualize different constraints. Specifically, we dualize the participation constraints in (5), whereas Dworczak and Martini (2019) dualize the second-order stochastic dominance constraints in their (1). Consequently, their optimality conditions do not directly imply our optimality conditions in Theorem 4.3. Nevertheless, we observe that the upper envelope function obtained from our Lagrangian coincides precisely with the equilibrium price function $p(x)$ in Dworczak and Martini (2019). As a result, we fully characterize this equilibrium price, which is generally challenging to specify, for a special case of Dworczak and Martini (2019), in which the sender's utility function is increasing and piecewise constant in the posterior mean. Moreover, for this special case, our approach is simpler and yields more direct optimality conditions. We provide further discussion in Remark 4.2.

Information Design in Other Applications In addition to the single-good allocation studied in this paper, numerous works have investigated efficient information disclosure in various operational applications. For example, Papanastasiou et al. (2018) study how online platforms can strategically disclose consumer reviews to incentivize exploration of alternative options. Lingenbrink and Iyer (2018) and Drakopoulos et al. (2021) examine how sellers can signal product availability to persuade customers to purchase earlier and increase profits. Lingenbrink and Iyer (2019) and Anunrojwong

[8] Specifically, Kleiner et al. (2021) and Arieli et al. (2023) independently characterize the extreme points of the set of mean-preserving contractions of a given prior distribution. These extreme points correspond to bi-pooling distributions (termed by Arieli et al. 2023). The double-interval structure in Candogan (2022) is a special case of bi-pooling distribution.

et al. (2023) explore how to strategically reveal queue congestion to maximize revenue or social welfare. Yang et al. (2019) and Candogan and Wu (2023) investigate how platforms can strategically reveal information to incentivize spatial resource repositioning. Alizamir et al. (2020) and De Véricourt et al. (2021) study how public agencies can strategically inform pandemic severity to induce preventive behavior or compliance with confinement measures. We direct interested readers to Candogan (2020) for a comprehensive review.

1.2 Notation and Terminology

We let $\mathbb{N}$ denote the set of nonnegative integers and $\mathbb{N}_+$ the set of strictly positive integers. For any two integers $a, b \in \mathbb{N}$ with $a \leq b$, we let $[a : b] = \{a, a+1, \ldots, b-1, b\}$ denote a sequence of integers starting from a and ending with b, and we denote $[n] = [1 : n]$ for any $n \in \mathbb{N}_+$. For any real number $x \in \mathbb{R}$, we let $(x)^+ \triangleq \max\{x, 0\}$ denote the maximum of x and 0.

2 Problem Formulation

We consider a sender (referred to as "she") who promotes a good among n potential receivers (each referred to as "he") via strategic information disclosure. The sender can allocate the good to at most one receiver and has a known preference among the receivers. Specifically, we let $v_i > 0$ denote the sender's utility from allocating the good to receiver i, and we rank receivers in decreasing preference; that is, $v_i > v_j$ if $i < j$, as assumed in Assumption 2.1. If the good is not allocated, the sender's utility is normalized to zero.

Assumption 2.1. The sender's utility from allocating the good satisfies $0 < v_n < \cdots < v_2 < v_1$.

Let $w \in \Omega$ represent the characteristics of the good, where Ω is a general state space.[9] Although the sender privately observes the realization of w, receivers possess only a prior distribution $G(w)$ regarding the good's characteristics. For each receiver i, let $u_i(w)$ denote the utility of receiving a good with characteristics w; the utility of not receiving the good is zero.

Information Disclosure Mechanism We study a Bayesian persuasion setup in which the sender, who has commitment power, designs an information disclosure mechanism to promote her good to the n receivers. Let S_i denote the set of signals the sender uses to interact with receiver i and

[9] For example, we may have $\Omega \subseteq \mathbb{R}^m$, where m represents the number of attributes relevant to the receivers' acceptance standards, such as research potential, teaching experience, and communication skills in the student promotion example.

$\boldsymbol{S} = \bigotimes_{i=1}^{n} S_i$ the set of all signals. Upon observing the characteristics w, the sender sends a signal $s_i \in S_i$ to each receiver i according to a joint distribution $f(\boldsymbol{s}|w)$, where $\boldsymbol{s} = (s_1, \ldots, s_n) \in \boldsymbol{S}$ denotes the vector of signals. We define the information disclosure mechanism $f(\cdot|w)$ as a *public* mechanism if the following are met:

1. The signals share a common signal space S, that is, $S_i = S_j = S$ for all $i, j \in [n]$.
2. The signals $(s_i)_{i\in[n]}$ are perfectly correlated, that is, $f(\boldsymbol{s}|w) = 0$ for any signal $\boldsymbol{s} = (s_i)_{i\in[n]}$, where $s_i \neq s_j$ for some $i, j \in [n]$.

With a public mechanism, receivers always receive the same signal, eliminating the need for further communication. Conversely, if $f(\cdot|w)$ can send different signals to different receivers, we refer to it as a *private* information mechanism. In this case, the receivers may receive different signals, leading to varied information about the good's characteristics w.

Communication among Receivers We assume that receivers may communicate with each other after receiving signal $\boldsymbol{s}$. We do not formally model how receivers will communicate. Notably, receivers may or may not be able to communicate, and if they do, it could be either simultaneously or sequentially, using either cheap talk or with some degree of commitment. Any of these communication methods can be reasonable in specific scenarios. However, as we demonstrate in Section 3, the optimal persuasion mechanism will be independent of the detailed communication method. This is because, regardless of how receivers communicate, a public information disclosure mechanism will always be optimal for the sender, leaving nothing for the receivers to communicate.

However, some notations are helpful in describing the problem. Given a specific communication protocol, let C_i denote the set of information that receiver i can receive from other receivers and $\mathbf{C} = \bigotimes_{i=1}^{n} C_i$ represent the communication space. Denote the communication outcome as $\mathbf{c} = (c_1, \ldots, c_n) \in \mathbf{C}$, where c_i is the information receiver i receives through communication. Given a signal $\boldsymbol{s}$, suppose the cumulative distribution function of $\mathbf{c}$ is $C(\mathbf{c}|\boldsymbol{s})$, and the probability density function of $\mathbf{c}$ is $c(\mathbf{c}|\boldsymbol{s}) = \frac{\mathrm{d}C(\mathbf{c}|\boldsymbol{s})}{\mathrm{d}\mathbf{c}}$, possibly derived from the receivers' equilibrium strategies.

Sender's Problem The game proceeds as follows:

1. The sender commits to an information disclosure mechanism $f(\cdot|w)$ and a signal space $\mathbf{S} = \bigotimes_{i=1}^{n} S_i$.
2. The good's characteristics w are drawn from the cumulative probability distribution $G(w)$. A signal $\mathbf{s} = (s_i)_{i\in[n]}$ is then generated according to the disclosure mechanism $f(\cdot|w)$ and sent

to the receivers.

3. Receivers communicate with each other after receiving the signal $\boldsymbol{s}$ using $C(\cdot|\boldsymbol{s})$, which may represent an equilibrium communication strategy in a specific scenario. After communication, each receiver i decides whether to accept the good (or extend an offer) based on the signal s_i and the communication outcome c_i.

4. The sender accepts the offer that maximizes her payoff, which corresponds to the receiver with the smallest index among those sending offers, according to Assumption 2.1.

Given a signal $s \in S_i$ and communication outcome $c \in C_i$, we define $\mathbf{S}^i(s) = \{\boldsymbol{s} \in \mathbf{S} : s_i = s\}$ and $\mathbf{C}^i(c) = \{\mathbf{c} \in \mathbf{C} : c_i = c\}$ as the sets of possible signals and communications, respectively. Upon observing s and c, receiver i understands that the signal must be in set $\mathbf{S}^i(s)$ and that the communication outcome must be in set $\mathbf{C}^i(c)$. He then updates his beliefs about the good's characteristics w, the signal $\boldsymbol{s}$, and the communication outcome $\mathbf{c}$ using Bayes's rule whenever possible. Specifically, let $f_i(s,c)$ denote the probability that receiver i receives a signal s and communication outcome c:

$$f_i(s,c) = \int_{w\in\Omega}\int_{\boldsymbol{s}\in\mathbf{S}^i(s)}\int_{\mathbf{c}\in\mathbf{C}^i(c)} c(\mathbf{c}|\boldsymbol{s})\, f(\boldsymbol{s}|w)\, d\mathbf{c}\, d\boldsymbol{s}\, dG(w).$$

If $f_i(s,c) > 0$, the receiver i's posterior belief on the tuple $(w, \boldsymbol{s}, \mathbf{c})$ is defined as

$$f_i(w,\boldsymbol{s},\mathbf{c}|s,c) = \begin{cases} \frac{c(\mathbf{c}|\boldsymbol{s}) f(\boldsymbol{s}|w)\, \mathrm{d}G(w)}{f_i(s,c)}, & \text{if } \boldsymbol{s} \in \mathbf{S}^i(s) \text{ and } \mathbf{c} \in \mathbf{C}^i(c), \\ 0, & \text{otherwise.} \end{cases}$$

Denote receiver i's equilibrium strategy by $\delta_i(s,c)$, which represents his probability of extending an offer after receiving a signal $s \in S_i$ and communication outcome $c \in C_i$. The optimality of receiver i's strategy implies that $\delta_i(s,c)$ satisfies the following equation:

$$\delta_i(s,c) = \begin{cases} 0, & \text{if } \mathbb{E}\left[u_i(w)\cdot \mathbb{1}[a_j^* = 0,\, \forall\, j < i] \mid s, c\right] < 0, \\ \delta \in [0,1], & \text{if } \mathbb{E}\left[u_i(w)\cdot \mathbb{1}[a_j^* = 0,\, \forall\, j < i] \mid s, c\right] = 0, \\ 1, & \text{if } \mathbb{E}\left[u_i(w)\cdot \mathbb{1}[a_j^* = 0,\, \forall\, j < i] \mid s, c\right] > 0, \end{cases}$$

where the binary variable $a_j^* \in \{0,1\}$ represents receiver j's action of extending an offer in an equilibrium and satisfies $\mathbb{P}[a_j^* = 1|s_j, c_j] = \delta_j(s_j, c_j)$, and the expectation $\mathbb{E}[\cdot|s,c]$ is taken over the posterior distribution $f_i(w,\boldsymbol{s},\mathbf{c}|s,c)$. Note that the sender accepts receiver i's offer if and only if none of the receivers $j < i$ extends an offer, as represented by $\mathbb{1}[a_j^* = 0, \forall\, j < i]$.

Finally, let the random set $I(\boldsymbol{s}, \mathbf{c})$ denote the receivers who extend an offer, and let $i(\boldsymbol{s}, \mathbf{c}) \triangleq \min I(\boldsymbol{s}, \mathbf{c})$ denote the index of the offer to accept, given the signal realization $\boldsymbol{s} \in \mathbf{S}$, communication outcome $\mathbf{c} \in \mathbf{C}$, and the receivers' equilibrium strategies. If $I(\boldsymbol{s}, \mathbf{c}) = \varnothing$, that is, the good receives no offer, we let $i(\boldsymbol{s}, \mathbf{c}) = \varnothing$ and set $v_\varnothing = 0$ as the corresponding utility of the sender. The sender selects an information disclosure mechanism $f(\cdot|w)$ that maximizes her expected payoff by solving

$$V^* \triangleq \max_{f(\cdot|w)} \int_{w\in\Omega} \int_{\boldsymbol{s}\in\boldsymbol{S}} \int_{\mathbf{c}\in\mathbf{C}} \mathbb{E}_{i(\boldsymbol{s},\mathbf{c})}\left[v_{i(\boldsymbol{s},\mathbf{c})}\right] \cdot c(\mathbf{c}|\boldsymbol{s}) \cdot f(\boldsymbol{s}|w) \cdot d\mathbf{c}\; d\boldsymbol{s}\; dG(w). \tag{1}$$

In (1), the expectation $\mathbb{E}_{i(\boldsymbol{s},\mathbf{c})}[\cdot]$ is taken over the possible randomness in the receivers' equilibrium offer-extending strategies when the signal and communication realizations are $\boldsymbol{s}$ and $\mathbf{c}$, respectively, and V^* denotes the expected payoff of an optimal information disclosure mechanism.

3 Optimality of Public Persuasion

In this section, we demonstrate that a public persuasion mechanism solves the sender's optimal information disclosure problem (1), regardless of how the receivers communicate. We begin by introducing a relaxation of the sender's problem (1) in Section 3.1, which provides an upper bound on her optimal expected payoff V^*.

3.1 First-Best Problem with Participation Constraints

In this section, we introduce the first-best relaxation (2) for the sender's information design problem, where we impose only the receivers' participation constraints.

$$\begin{aligned} \bar{V} = \max_{q(i|w)\geq 0} \quad & \sum_{i=1}^{n} v_i \int_{w\in\Omega} q(i|w)\, dG(w) \\ \text{s.t.} \quad & \int_{w\in\Omega} u_i(w)\, q(i|w)\, dG(w) \geq 0,\; \forall\, i \in [n], \\ & \sum_{i\in[n]} q(i|w) \leq 1,\; \forall\, w \in \Omega. \end{aligned} \tag{2}$$

In (2), a central planner allocates the good with characteristics w to receiver i with probability $q(i|w)$, and requires the receiver to accept the good when the latter is allocated to him. The allocation $q(i|w)$ must ensure a nonnegative expected utility for each receiver, as indicated by the first constraint in (2). This reflects the fact that each receiver should at least break even in expectation from accepting the good. In addition, a good is allocated to at most one receiver, as indicated by the second constraint in (2). The central planner chooses $q(i|w)$, which satisfies these two constraints, to maximize the sender's expected payoff, and the optimal value is denoted by $\bar{V}$.

Lemma 3.1 demonstrates that (2) provides an upper bound on the sender's optimal expected payoff V^*, regardless of how the receivers communicate.

Lemma 3.1. *We have $\bar{V} \geq V^*$, regardless of how receivers communicate.*

We prove Lemma 3.1 in Section A.1. Intuitively, given any disclosure mechanism $f(\cdot|w)$, let $q(i|w)$ denote the ex-ante probability that the good with characteristics w is allocated to receiver i under the receivers' equilibrium strategies induced by $f(\cdot|w)$. These $\{q(i|w)\}$ are feasible to (2) and have an objective value no larger than $\bar{V}$.

3.2 Optimality of Public Persuasion

In this section, we construct a public persuasion mechanism $f^*(\cdot|w)$ from the optimal solution of (2) and show that its expected payoff attains the first-best upper bound $\bar{V}$. Therefore, the mechanism $f^*(\cdot|w)$ is optimal to (1), and this optimality does not depend on the communication protocol among the receivers.

Let $\{q^*(i|w)\}$ denote an optimal solution to (2). We consider a public persuasion mechanism $f^*(\cdot|w)$ with signal space $S_i = S \triangleq [n] \cup \{\varnothing\}$ for all receivers $i \in [n]$. When the good's characteristics are w, the mechanism broadcasts the signal $s = i$ to all receivers with probability $q^*(i|w)$ for any $i \in [n]$ and the signal $s = \varnothing$ to all receivers with probability $1 - \sum_{i \in [n]} q^*(i|w)$. We can interpret the signal $s = i$ as a recommendation for only receiver i to extend an offer, and the signal $s = \varnothing$ as a recommendation for none of the receivers to extend an offer. Theorem 3.2 shows that this persuasion mechanism achieves the first-best upper bound $\bar{V}$.

Theorem 3.2 (Optimality of Public Persuasion). *Under the public persuasion mechanism $f^*(\cdot|w)$, it is an equilibrium for each receiver $i \in [n]$ to extend an offer only upon receiving the signal $s = i$. Moreover, the expected payoff of the mechanism $f^*(\cdot|w)$, denoted by V^{P}, satisfies $V^{\mathrm{P}} = \bar{V}$.*

We prove Theorem 3.2 in Section A.2. To understand the equilibrium in Theorem 3.2, suppose that the sender recommends the good to receiver i. Receiver i is willing to extend an offer because (i) his offer will be accepted with certainty, given that no other receiver will extend an offer, and (ii) he can break even from his offer in expectation, as indicated by the first constraint in (2). Any receiver $j > i$ cannot benefit from extending an offer because the sender will accept the more attractive offer from receiver i. Any receiver $j < i$ is unwilling to extend an offer because (i) his offer, if extended, will be accepted with certainty given that no better offer will be extended, and (ii) $\{q^*(i|w)\}$ being an optimal solution of (2) implies that receiver j cannot break even from his offer in expectation—otherwise, the central planner in (2) can strictly improve the sender's payoff by allocating the good to receiver j instead of receiver i without violating any constraints in (2).

Since the mechanism $f^*(\cdot|w)$ achieves the first-best upper bound $\bar{V}$, Lemma 3.1 implies that the first-best upper bound is tight (i.e., $\bar{V} = V^*$) and that $f^*(\cdot|w)$ is an optimal persuasion mechanism. Since the sender sends the same information to all receivers with mechanism $f^*(\cdot|w)$, communication becomes irrelevant. Therefore, the sender eliminates any communication among the receivers for her own benefit, regardless of the way the receivers can communicate. This holds true even when the receivers cannot communicate but are aware of each other's existence. Additionally, the vanilla private persuasion mechanism is suboptimal in this setting, as elaborated in Remark 3.1.

Remark 3.1 (Suboptimality of Vanilla Private Persuasion without Communication)**.** When the receivers cannot communicate, one might be inclined to treat them in isolation, sending each receiver separate signals using their respective optimal persuasion strategies. We refer to this mechanism as the *vanilla private persuasion mechanism* and demonstrate its suboptimality. First, note that even without communication, a public persuasion mechanism remains optimal by Theorem 3.2. To illustrate the suboptimality of the vanilla private persuasion mechanism, consider a concrete example with two receivers who cannot communicate. The underlying state $w \sim \text{Unif}[0, 1]$ is one-dimensional and follows a uniform distribution on $[0, 1]$. The sender's payoffs from the two receivers are $v_1 = 2$ and $v_2 = 1$. Each receiver's utility function is linear in w and takes the form $u_i(w) = w - \alpha_i$, with $\alpha_1 = 0.9$ and $\alpha_2 = 0.7$. Under the vanilla private persuasion mechanism, the sender recommends receiver 1 to extend an offer (signal $s = 1$) whenever $w \geq 0.8$, and recommends receiver 2 to extend an offer (signal $s = 2$) whenever $w \geq 0.4$.

Despite the lack of communication, receiver 2, aware of the presence of a more preferred receiver 1, will refrain from extending an offer upon receiving signal $s = 2$. This is because receiver 1's offer adversely selects the goods recommended to receiver 2, resulting in negative expected utility for receiver 2. Notably, if receiver 2 extends an offer, only goods with quality $w \in [0.4, 0.8)$ would eventually be allocated to him. The expected quality of these goods satisfies

$$\mathbb{E}[w \mid 0.4 \leq w < 0.8] < \mathbb{E}[w \mid w \geq 0.4] = 0.7,$$

which is below receiver 2's acceptance threshold α_2. Given receiver 2's equilibrium strategy of not extending an offer, only goods with quality $w \in [0.8, 1]$ end up being allocated. This renders the vanilla private persuasion mechanism suboptimal, as demonstrated in Example C.1 in the Appendix, where we characterize the optimal (public) persuasion mechanism. Specifically, the vanilla private persuasion mechanism yields the sender an expected payoff of 0.4, whereas the optimal public persuasion (as characterized in Example C.1) yields an expected payoff of 0.625, which is 56% higher.

Finally, we remark that our results remain valid even when we relax the strict preference assumption (Assumption 2.1) and allow weak preferences among receivers (i.e., $v_i \geq v_j$ for any $i < j$), as detailed in Remark 3.2.

Remark 3.2 (Weak Preference)**.** Public persuasion remains optimal and continues to achieve first-best performance even when the sender is indifferent among certain receivers (i.e., $v_i \geq v_j$ for any $i < j$). Specifically, the first-best problem (2) can still be implemented through a public persuasion mechanism in an identical way, provided that the sender breaks ties according to the optimal solution of (2), and the receivers anticipate this tie-breaking rule.

3.3 Extension: Uncertain Sender Offer Values

In Section 3.2, we demonstrate that public persuasion is broadly optimal when the sender derives a deterministic utility v_i from allocating the good to receiver i (Assumption 2.1), regardless of how receivers can communicate. In this section and the next, we carefully examine the assumptions underlying our base model to delineate the boundaries within which public persuasion remains optimal.

We first relax the assumption of fixed offer values. Specifically, in this section, we consider a general model in which the offer values $\{v_i\}$ are uncertain and potentially correlated with the good's characteristics w. We describe the general model in Section 3.3.1. In Section 3.3.2, we demonstrate that public persuasion remains optimal as long as the sender's ordinal ranking of receivers remains fixed. Finally, in Section 3.3.3, we conduct a numerical experiment that reflects a realistic setting in which the sender's ordinal ranking of receivers may vary. Our numerical results show that the suboptimality gap of public persuasion mechanisms is negligible in this example.

3.3.1 The General Model

In this section, we describe a more general model in which the offer values $\{v_i\}$ are uncertain and potentially correlated with the good's characteristics w.

To this end, define a general state space $\Phi = \Omega \times \Theta$. As before, Ω represents the space of the good's characteristics. Each receiver $i \in [n]$ derives utility $u_i(w)$ from receiving a good with characteristics $w \in \Omega$, and zero from not receiving the good. Θ represents the sender's preference space. Specifically, a sender with preference $\theta \in \Theta$ obtains utility $v_i(\theta)$ from allocating the good to receiver $i \in [n]$, and zero utility if the good remains unallocated.

The sender privately observes both the realization of the good's characteristics w and the preference type θ, whereas the receivers know only the prior joint distribution over the underlying state $(w, \theta) \in \Phi$. Let $H(\theta)$ denote the marginal cumulative distribution function of the sender's preference type θ and $G(w|\theta)$ denote the cumulative distribution function of the good's characteristics w conditional on θ. The sender can commit to an information disclosure mechanism to reveal information about the underlying state $(w, \theta) \in \Phi$ to the receivers.

We can adapt the first-best relaxation problem (2) to the general model by replacing v_i with $v_i(\theta)$ and $q(i|w)$ with $q(i|w, \theta)$, and taking expectations over both the good's characteristics and the sender's preferences, as formalized in (3).

$$
\begin{aligned}
\bar{V} = \max_{q(i|w,\theta) \geq 0} \quad & \sum_{i=1}^{n} \int_{w \in \Omega} \int_{\theta \in \Theta} v_i(\theta)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \\
\text{s.t.} \quad & \int_{w \in \Omega} \int_{\theta \in \Theta} u_i(w)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \geq 0, \quad \forall\, i \in [n], \\
& \sum_{i \in [n]} q(i|w,\theta) \leq 1, \quad \forall\, w \in \Omega,\, \theta \in \Theta.
\end{aligned} \tag{3}
$$

Problem (3) has an interpretation analogous to problem (2). In (3), a central planner allocates a good with characteristics w to receiver i with probability $q(i|w, \theta)$ when the sender's preference type is θ. The planner aims to maximize the sender's expected payoff while ensuring a nonnegative expected utility for each receiver. Analogous to Lemma 3.1, (3) provides an upper bound on the sender's optimal expected payoff V^* in the general model, regardless of how receivers communicate.

Lemma 3.3 (Extension of Lemma 3.1). *Problem (3) provides an upper bound on the sender's optimal expected payoff V^* in the general model, regardless of how receivers communicate.*

The proof mimics the proof of Lemma 3.1 presented in Section A.1. Therefore, we omit the details. Intuitively, for any disclosure mechanism $f(\cdot|w, \theta)$, we define $q(i|w, \theta)$ as the ex-ante probability that the good is allocated to receiver i, given that the good's characteristics are w and the sender's preference type is θ, under the equilibrium strategies induced by $f(\cdot|w, \theta)$. Analogous to the proof of Lemma 3.1, these probabilities $\{q(i|w, \theta)\}$ are feasible to (3) and yield an objective value no larger than $\bar{V}$.

We conclude this section with a remark on the tightness of the first-best relaxation (3).

Remark 3.3 (Implementability of the First-Best Relaxation (3))**.** Let $\{q^*(i|w,\theta)\}$ be an optimal solution to (3). This solution can be implemented[10]—and hence the first-best relaxation bound $\bar{V}$ is attained—under either of the following two scenarios.

1. Suppose that $v_i(\theta) > 0$ for all $i \in [n]$ and $\theta \in \Theta$, so that the sender always prefers that the good be allocated. Moreover, suppose that for each receiver $i \in [n]$, the following condition holds:

$$\int_{w\in\Omega}\int_{\theta\in\Theta} u_i(w)\left[1-\sum_{j=1}^{n} q^*(j|w,\theta)+\sum_{j\neq i}\mathbb{1}\Big[v_i(\theta)>v_j(\theta)\Big]\cdot q^*(j|w,\theta)\right] dG(w|\theta)\, dH(\theta) \le 0\,,$$

which ensures that no receiver can profitably deviate by extending an offer to goods not allocated to him under the proposed allocation rule. Then, $\{q^*(i|w,\theta)\}$ can be implemented by a private persuasion mechanism when the receivers cannot communicate.

2. Alternatively, $\{q^*(i|w,\theta)\}$ can be implemented by a public persuasion mechanism if the sender has commitment power over which offer to accept.

We provide further details on Remark 3.3 in Section A.3.1. To interpret the condition in Part 1, consider receiver i after receiving a private recommendation not to extend an offer: if he nevertheless extends an unsolicited offer, his offer would be accepted for goods left unallocated under q^*, captured by $1-\sum_{j=1}^{n} q^*(j|w,\theta)$, and for goods assigned to another receiver j only in preference states in which the sender prefers receiver i's offer to receiver j's, captured by $\mathbf{1}\{v_i(\theta) > v_j(\theta)\}q^*(j|w,\theta)$. The inequality therefore requires that this entire off-recommendation pool yield receiver i negative expected utility. Under private persuasion, this condition deters receivers from extending offers to goods not intended for them, because doing so risks attracting inferior-quality goods. Under public persuasion, however, this deterrent may no longer be effective because the public signal reveals the allocation destination of each good. Receivers may therefore distinguish inferior goods based on their allocation destinations, information that is unavailable under private persuasion. In the remainder of Section 3.3, we focus on public persuasion and assume that the sender has no commitment power in accepting offers. That is, as assumed in Section 2, the sender always accepts the best offer she receives.

[10]The solution is implementable in the sense that there exists a mechanism under which, in equilibrium, the good is allocated to receiver i with probability $q^*(i|w,\theta)$ conditional on (w,θ).

3.3.2 Optimality of Public Persuasion under Fixed Ordinal Ranking

In this section, we demonstrate that public persuasion remains optimal when the sender's ordinal ranking over receivers is fixed. We formally state this assumption in Assumption 3.1.

Assumption 3.1. We have $v_i(\theta) \geq v_j(\theta)$ for any receivers $i < j$ and realization $\theta \in \Theta$.

We now demonstrate that an optimal solution to (3) can be implemented via a public persuasion mechanism, whose expected payoff attains the upper bound $\bar{V}$. Consequently, public persuasion remains optimal. Let $\{q^*(i|w,\theta)\}$ denote an optimal solution to (3). Analogous to Section 3.2, consider a public persuasion mechanism $f^*(\cdot|w,\theta)$ with signal space $S_i = S \triangleq [n] \cup \{\varnothing\}$ for all receivers $i \in [n]$. When the good's characteristics are w and the sender's preference type is θ, the mechanism broadcasts the signal $s = i$ to all receivers with probability $q^*(i|w,\theta)$ for each $i \in [n]$, and broadcasts the signal $s = \varnothing$ to all receivers with the remaining probability $1 - \sum_{i\in[n]} q^*(i|w,\theta)$. We can interpret the signal $s = i$ as a recommendation for only receiver i to extend an offer and signal $s = \varnothing$ as a recommendation for none of the receivers to extend an offer. Theorem 3.4 shows that this persuasion mechanism achieves the first-best upper bound $\bar{V}$.

Theorem 3.4 (Extension of Theorem 3.2). *Suppose Assumption 3.1 holds. Under the public persuasion mechanism $f^*(\cdot|w,\theta)$, it is an equilibrium for each receiver $i \in [n]$ to extend an offer if and only if he receives the signal $s = i$. Moreover, the expected payoff of the mechanism $f^*(\cdot|w,\theta)$ attains the upper-bound value $\bar{V}$.*

We prove Theorem 3.4 in Section A.3.2 using an argument analogous to the proof of Theorem 3.2. Assumption 3.1 provides a sufficient condition for public persuasion to be optimal. More generally, public persuasion remains optimal as long as no receiver can obtain positive expected utility by cherry-picking goods intended for other receivers. However, without Assumption 3.1, public persuasion may no longer be optimal. We illustrate this through a simple example (Example A.1) in Section A.3.3.

3.3.3 Numerical Experiments: Public Persuasion Suboptimality Gap

In the absence of Assumption 3.1, public persuasion may no longer be optimal. In practice, although some receivers are regarded as substantially more desirable than others, the precise ranking among receivers within the same tier may vary. In this section, we numerically evaluate the suboptimality gap of public persuasion mechanisms in a realistic setting in which the sender's preference order may vary among receivers within the same tier, and we show that this suboptimality gap is negligible under mild assumptions.

Parameter Setup In this example, we assume that the good's characteristic w is one-dimensional and follows a discrete uniform distribution on $\left\{\frac{1}{100}, \frac{2}{100}, \ldots, \frac{99}{100}, 1\right\}$. That is, $\mathbb{P}_w(k/100) = 0.01$ for $k \in [100]$, where $\mathbb{P}_w(\cdot)$ denotes the probability mass function of w. There are $n = 6$ receivers, each with a linear utility function of the form $u_i(w) = w - \alpha_i$. Therefore, each receiver's decision depends solely on whether the posterior mean of w exceeds his acceptance threshold α_i.

We model receivers 1 and 2 as top-tier, receivers 3 and 4 as mid-tier, and receivers 5 and 6 as lower-tier. The acceptance thresholds are set to $\alpha_1 = \alpha_2 = 0.9$, $\alpha_3 = \alpha_4 = 0.8$, and $\alpha_5 = \alpha_6 = 0.7$. Accordingly, higher-tier receivers are more selective, while receivers within the same tier are equally selective.

Sender's Preference For simplicity, we assume that the sender's preferences, characterized by the offer values $v_i(\theta)$, are independent of the good's quality w. Furthermore, it is common knowledge that top-tier receivers are preferred over mid-tier receivers, which in turn are preferred over lower-tier receivers. However, there is uncertainty in the sender's preferences among receivers within the same tier.

To capture this structure, we set the mean offer values to $\bar{v}_1 = \bar{v}_2 = 3$, $\bar{v}_3 = \bar{v}_4 = 2$, and $\bar{v}_5 = \bar{v}_6 = 1$. We assume that for each receiver $i \in [n]$, the offer value $v_i(\theta)$ is uniformly distributed on $[\bar{v}_i - \epsilon, \bar{v}_i + \epsilon]$, where $\epsilon \in (0, 0.5)$ is a dispersion parameter. To facilitate numerical computation, we approximate the sender's preferences by discretizing the distributions as follows:

1. Draw T independent realizations $\{\theta_t\}_{t=1}^{T}$ by independently sampling $v_i(\theta_t) \sim \text{Unif}[\bar{v}_i - \epsilon, \bar{v}_i + \epsilon]$ for each receiver $i \in [n]$ and each $t \in [T]$.
2. Assume that each θ_t occurs with equal probability $1/T$, that is, $\mathbb{P}_\theta(\theta_t) = 1/T$ for all $t \in [T]$, where $\mathbb{P}_\theta(\cdot)$ denotes the probability mass function of the preference type θ.

Candidate Public Persuasion Mechanism Since Assumption 3.1 is not satisfied here, problem (3) is only a relaxation and may not be implementable by a public persuasion mechanism. By contrast, the LP (4) below, which includes one additional constraint (namely, the second-to-last constraint), can always be implemented via a public persuasion mechanism.

$$V^{\mathrm{P}} = \max_{q(i|w,\theta) \geq 0} \quad \sum_{i=1}^{n} \int_{w \in \Omega} \int_{\theta \in \Theta} v_i(\theta)\, q(i|w, \theta)\, dG(w|\theta)\, dH(\theta)$$

$$
\begin{aligned}
\text{s.t.} \quad & \int_{w\in\Omega}\int_{\theta\in\Theta} u_i(w)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \geq 0, \quad \forall\, i \in [n], \\
& \int_{w\in\Omega}\int_{\theta\in\Theta} \mathbb{1}[v_i(\theta) > v_j(\theta)]\, u_i(w)\, q(j|w,\theta)\, dG(w|\theta)\, dH(\theta) \leq 0, \quad \forall\, i,j \in [n],\, i \neq j, \\
& \sum_{i\in[n]} q(i|w,\theta) \leq 1, \quad \forall\, w \in \Omega,\, \theta \in \Theta.
\end{aligned} \tag{4}
$$

Problem (4) follows the same notation as in problem (3). In (4), the second-to-last constraint requires that no receiver i can obtain positive expected utility from making offers to goods intended for any other receiver j. Therefore, the optimal solution to (4) can be implemented via a public persuasion mechanism in the same manner as described in Section 3.3.2: the sender publicly announces the receiver to whom the good should be allocated, and a receiver extends an offer if and only if he receives the signal $s = i$. We denote this public persuasion mechanism by π^{P}. The sender's expected payoff under this mechanism equals the optimal value of (4), denoted by V^{P}. As a consequence, V^{P} is a lower bound on the performance of the optimal public persuasion. We note that, in general, π^{P} is not necessarily the optimal public persuasion mechanism, and we illustrate this point further in Section A.3.5.

Numerical Results In the numerical experiments, we set the dispersion parameter $\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$ and the number of draws $T \in \{20, 50, 100, 200\}$. For each combination of ϵ and T, we generate $N = 500$ independent samples for Monte Carlo simulation. For each sample, we compute: (a) the first-best relaxation upper bound $\bar{V}$ by solving the LP (3), and (b) the performance V^{P} of the public persuasion mechanism π^{P}, obtained by solving the LP (4).

Table 1 reports the median (left panel) and the maximum (right panel) of the performance-loss ratio $1 - V^{\mathrm{P}}/\bar{V}$ over $N = 500$ independent samples for each combination of (ϵ, T). We also visualize the empirical distribution of the ratio $V^{\mathrm{P}}/\bar{V}$ using scatter plots in Figure 2 of Section A.3.4. From Table 1, the performance loss of the public persuasion mechanism π^{P} increases as the dispersion parameter ϵ increases (i.e., the sender's preferences over receivers within the same tier become more variable) or as the number of preference scenarios T decreases (i.e., the sender's preferences become more idiosyncratic and deviate further from a uniform distribution). However, in all cases the performance loss is negligible: the loss relative to the first-best relaxation upper bound $\bar{V}$ is on the order of 10^{-5} in most instances and on the order of 10^{-4} in the worst case.

T	ϵ 0.1	0.2	0.3	0.4
20	1.064	2.037	2.281	2.540
50	0.581	1.132	1.415	2.034
100	0.366	0.684	0.789	1.305
200	0.206	0.404	0.485	0.758

(a) Median of $10^5 \times \left(1 - V^{\mathrm{P}}/\bar{V}\right)$

T	ϵ 0.1	0.2	0.3	0.4
20	3.489	8.989	11.966	14.298
50	1.732	3.379	3.517	6.762
100	0.819	1.867	1.976	2.284
200	0.277	0.625	0.611	1.302

(b) Maximum of $10^4 \times \left(1 - V^{\mathrm{P}}/\bar{V}\right)$

Table 1: Summary statistics of the performance-loss ratio $1 - V^{\mathrm{P}}/\bar{V}$ over $N = 500$ independent samples for each combination of (ϵ, T). Panel (a) reports the median (scaled by 10^5), and panel (b) reports the maximum (worst case, scaled by 10^4).

3.4 Discussion of Other Assumptions

In this section, we examine other assumptions in our base model. First, our model assumes that receivers take binary actions (i.e., accepting or rejecting the good). In practice, a receiver may have more than two possible actions to choose from.[11] In Remark 3.4, we demonstrate that public persuasion remains optimal even with multiple possible actions.

Remark 3.4 (Multiple Actions)**.** Suppose that each receiver can select from multiple actions regarding the good and that the sender has a known preference for these actions. Public persuasion remains optimal in this scenario. Specifically, analogous to Section 3.3, we can again adapt the first-best relaxation problem (2) and show that its optimal solution can be implemented via a public persuasion mechanism. We provide further details in Section A.4.

Finally, we note in Remark 3.5 that the assumption of a single-good allocation is critical. If multiple goods are available instead, public persuasion may no longer be optimal.

Remark 3.5 (Multiple Goods)**.** Public persuasion may no longer be optimal if the sender has multiple goods to allocate. To illustrate, consider an example with one sender and two receivers. The sender has two goods to allocate, whose characteristics are perfectly correlated (i.e., $w_1 = w_2$ with probability one), and each receiver can accept at most one good. In this case, competition between the receivers vanishes. Consequently, if the receivers cannot communicate, the sender's persuasion problem decomposes into independent problems for each receiver. Therefore, it is optimal for the sender to send private signals to each receiver using their respective optimal persuasion strategies, rather than employing public persuasion.

[11] For example, in the student promotion context, an employer can extend a regular offer, provide an offer with additional benefits, or decline to make an offer.

4 Structural Properties of the One-Dimensional Linear-Utility Case

In the remainder of the paper, we return to the base model. From Theorem 3.2, it suffices for the sender to consider public persuasion mechanisms when solving the optimal persuasion problem (1). Moreover, the optimal public persuasion mechanism can be derived from (2) and achieves the first-best performance (i.e., the optimal value of (2)). However, when the state space for w is infinite, (2) becomes an infinite-dimensional LP, which can be challenging to solve. In this section, we focus on the special case in which the state variable w is one-dimensional and all receivers' utility functions are linear in w. We derive optimality conditions for a persuasion mechanism based on the Lagrangian dual of (2), in which we dualize the participation constraints. As the main result of this section, we show that the persuasion problem decouples over the linear segments of a piecewise linear upper envelope function induced by the Lagrangian dual. We introduce the upper envelope function in this section and provide a complete characterization of it in Section 5.

4.1 The Setup

In this section, we formally describe the one-dimensional linear utility setup that will be used in this and the following sections. First, we assume that the good's characteristics can be summarized by a one-dimensional state variable w within a finite interval. Without loss of generality, let $w \in \Omega = [0, 1]$. Additionally, w follows a continuous distribution with a strictly increasing cumulative distribution function $G(\omega)$ and a density function $g(w) > 0$ for all $w \in (0, 1)$. We summarize these in Assumption 4.1.

Assumption 4.1. The good's characteristic w belongs to the one-dimensional interval $\Omega = [0, 1]$ and follows a continuous distribution. Let $G(w)$ and $g(w)$ denote the cumulative distribution function and density function of w, respectively. The function $G(w)$ is strictly increasing, so its inverse, denoted by $G^{-1}(\cdot)$, exists.

Second, we assume that for each receiver $i \in [n]$, his utility function $u_i(w)$ for receiving a good with characteristic w is linear and increasing in w; that is, $u_i(w) = w - \alpha_i$ for some positive constant $\alpha_i > 0$. Under this assumption, each receiver i cares only about the mean value of the characteristic w among the goods potentially allocated to him. In particular, receiver i accepts the good only if this mean value exceeds his threshold α_i. We state this assumption in Assumption 4.2.

Assumption 4.2. For each receiver $i \in [n]$, his utility function $u_i(w)$ for a good with characteristic w is $u_i(w) = w - \alpha_i$, where α_i is a positive constant.

Note that since the receivers are ranked in decreasing preference by Assumption 2.1, there is no loss of generality in assuming that the threshold values α_i also decrease as the receiver index i increases.[12] This is because, if receiver i is more preferred than j ($v_i > v_j$) but also easier to get into ($\alpha_i \leq \alpha_j$), then receiver j will never be targeted and can be dropped from consideration. We state this assumption in Assumption 4.3.

Assumption 4.3. The receivers' threshold values α_i satisfy $0 < \alpha_n < \cdots < \alpha_2 < \alpha_1 < 1$.

Finally, given the linear-utility Assumption 4.2, the first-best problem (2) can be written as (5):

$$\begin{aligned} \bar{V} = \max_{q(i|w)\geq 0} \quad & \sum_{i=1}^{n} v_i \cdot \int_0^1 q(i|w)\, g(w)\, dw \\ \text{s.t.} \quad & \int_0^1 w \cdot q(i|w)\, g(w)\, dw \geq \alpha_i \int_0^1 q(i|w)\, g(w)\, dw,\ \forall\, i \in [n], \\ & \sum_{i\in[n]} q(i|w) \leq 1,\ \forall\, w \in [0,1]. \end{aligned} \tag{5}$$

4.2 Lagrangian Dual and Optimality Conditions

In this section, we introduce the Lagrangian dual problem of (5), in which we dualize the receivers' participation constraints. We then interpret the Lagrangian from a geometric perspective and derive the optimality conditions for a persuasion mechanism as the main result of Section 4. Specifically, we show that the persuasion problem decouples across the linear segments of a piecewise linear envelope function induced by the Lagrangian of (5).

Denote by $\mu_i \geq 0$ the Lagrange multiplier for the participation constraint of receiver $i \in [n]$. The Lagrangian relaxation, denoted by $V^{\mathrm{LR}}(\boldsymbol{\mu})$ with $\boldsymbol{\mu} = (\mu_i)_{i\in[n]} \in \mathbb{R}^n_+$, is given as follows:

$$\begin{aligned} V^{\mathrm{LR}}(\boldsymbol{\mu}) &= \max_{\substack{q(i|w)\geq 0,\\ \sum_{i\in[n]} q(i|w)\leq 1}} \int_0^1 \sum_{i=1}^{n} \Big\{ v_i + \mu_i \big(w - \alpha_i\big) \Big\}\, q(i|w)\, g(w)\, dw \\ &= \int_0^1 \left(\max_{\substack{q(i|w)\geq 0,\\ \sum_{i\in[n]} q(i|w)\leq 1}} \sum_{i=1}^{n} \Big\{ v_i + \mu_i \big(w - \alpha_i\big) \Big\} \cdot q(i|w) \right) g(w)\, dw\,. \end{aligned} \tag{6}$$

After dualizing the participation constraints, the Lagrangian decouples over state w. Specifically, define

$$\ell_i\big(w; \mu_i\big) \triangleq v_i + \mu_i \big(w - \alpha_i\big)$$

[12] In the student promotion context, this means that a more preferred employer is also more difficult to get into.

as the line associated with receiver $i \in [n]$. This line passes through point (α_i, v_i) and has slope $\mu_i \geq 0$. In addition, let

$$h\big(w; \boldsymbol{\mu}\big) \triangleq \max_{i \in [n]} \ell_i\big(w; \mu_i\big) = \max_{i \in [n]} \Big\{ v_i + \mu_i \big(w - \alpha_i\big) \Big\}$$

denote the upper envelope of these n lines, and let $\bar{h}(w; \boldsymbol{\mu}) \triangleq \max\big\{h(w; \boldsymbol{\mu}), 0\big\}$ denote the upper envelope of these n lines and the x-axis. Both the functions $h(w; \boldsymbol{\mu})$ and $\bar{h}(w; \boldsymbol{\mu})$ are convex, increasing (since $\mu_i \geq 0$), and piecewise linear in w. Finally, let $\mathbf{Q}^{\mathrm{LR}}(\boldsymbol{\mu})$ denote the set of optimal solutions $\{q(i|w)\}$ to $V^{\mathrm{LR}}(\boldsymbol{\mu})$. According to (6), the set $\mathbf{Q}^{\mathrm{LR}}(\boldsymbol{\mu})$ can be characterized as follows:

$$\begin{aligned} \mathbf{Q}^{\mathrm{LR}}(\boldsymbol{\mu}) = \Bigg\{ q(i|w) : q(i|w) \geq 0, \ \sum_{i \in [n]} q(i|w) \leq 1, \qquad & \forall\, w \in [0,1]\,, \\ \sum_{i \in [n]} q(i|w) = 1, \qquad & \forall\, w \in [0,1] \ \text{ s.t. } h\big(w; \boldsymbol{\mu}\big) > 0\,, \\ q(i|w) > 0 \ \Rightarrow \ \ell_i\,(w; \mu_i) = \bar{h}\,(w; \boldsymbol{\mu})\,, \qquad & \forall\, i \in [n] \Bigg\}. \end{aligned} \tag{7}$$

That is, an optimal solution of $V^{\mathrm{LR}}(\boldsymbol{\mu})$ allocates a good with quality w to receiver i with positive probability only if receiver i's line, $\ell_i(w; \mu_i)$, lies above the x-axis and is not dominated at w by the lines of other receivers, $\ell_j(w; \mu_j)$ for $j \neq i$. Finally, from (6), we have

$$V^{\mathrm{LR}}(\boldsymbol{\mu}) = \int_0^1 \bar{h}\big(w; \boldsymbol{\mu}\big)\, g(w)\, dw.$$

Since every feasible solution to (5) is also feasible to (6) and attains an objective value in (6) that is no smaller than its value in (5), it follows that $\bar{V} \leq V^{\mathrm{LR}}(\boldsymbol{\mu})$ for any $\boldsymbol{\mu} \in \mathbb{R}^n_+$. We formally state this weak duality property in Lemma 4.1.

Lemma 4.1 (Weak Duality). *We have $\bar{V} \leq V^{\mathrm{LR}}(\boldsymbol{\mu})$ for any dual variable $\boldsymbol{\mu} \in \mathbb{R}^n_+$.*

4.2.1 The Optimality Conditions

Since the Lagrangian $V^{\mathrm{LR}}(\boldsymbol{\mu})$ is a convex function of $\boldsymbol{\mu}$ by (6), we can solve a convex optimization problem

$$V^{\mathrm{LR}} \triangleq \min_{\boldsymbol{\mu} \in \mathbb{R}^n_+} V^{\mathrm{LR}}(\boldsymbol{\mu}) \geq \bar{V} \tag{8}$$

to obtain the tightest Lagrangian relaxation bound V^{LR}. Let $\boldsymbol{\mu}^* = (\mu^*_i)_{i \in [n]} \in \mathrm{argmin}_{\boldsymbol{\mu} \in \mathbb{R}^n_+} V^{\mathrm{LR}}(\boldsymbol{\mu})$ denote an optimal Lagrangian dual variable. In Remark 4.1, we show that $\boldsymbol{\mu}^*$, and hence the upper envelope function $h(w; \boldsymbol{\mu}^*)$, can be efficiently computed.

Remark 4.1 (Computing $\boldsymbol{\mu}^*$). According to Danskin's theorem (see Theorem 9.27 in Shapiro et al. 2021), combined with the fact that a convex combination of any two optimal solutions to (6) is also optimal to (6), the sub-differential of $V^{\mathrm{LR}}(\boldsymbol{\mu})$ at any $\boldsymbol{\mu} \in \mathbb{R}^n_+$ (i.e., set of subgradients), denoted by $\partial V^{\mathrm{LR}}(\boldsymbol{\mu})$, can be expressed as

$$\partial V^{\mathrm{LR}}(\boldsymbol{\mu}) = \left\{ (g_i)_{i\in[n]} \text{ with } g_i \triangleq \int_0^1 (w - \alpha_i)\, q(i|w)\, g(w)\, dw : \{q(i|w)\} \in \mathbf{Q}^{\mathrm{LR}}(\boldsymbol{\mu}) \right\}.$$

Since both $V^{\mathrm{LR}}(\boldsymbol{\mu})$ and its subgradients can be efficiently computed, we can apply subgradient-based algorithms (e.g., the subgradient method or the cutting-plane method) to solve the convex program (8) and obtain an optimal Lagrangian dual variable $\boldsymbol{\mu}^*$ efficiently.

Furthermore, Lemma 4.2 shows that strong duality holds, which follows from standard strong duality results for linear optimization in vector spaces.

Lemma 4.2 (Strong Duality). *Problem (5) and its Lagrangian relaxation (6) satisfy the following:*

1. *Strong duality holds, and there exists an optimal dual variable $\boldsymbol{\mu}^* \in \mathbb{R}^n_+$ such that $\bar{V} = V^{\mathrm{LR}} = V^{\mathrm{LR}}(\boldsymbol{\mu}^*)$.*
2. *$\boldsymbol{\mu} \in \mathbb{R}^n_+$ is an optimal dual variable and $\{q(i|w)\}$ is an optimal solution to (5) if and only if*
   - *(a) $\{q(i|w)\} \in \mathbf{Q}^{\mathrm{LR}}(\boldsymbol{\mu})$, and*
   - *(b) $\{q(i|w)\}$ satisfies all participation constraints in (5), and the participation constraint for receiver i is binding for all $i \in [n]$ with $\mu_i > 0$.*

We prove Lemma 4.2 in Section B.2. Part 2 of Lemma 4.2, together with (7), establishes optimality conditions for a persuasion mechanism. Specifically, once the upper envelope function $h(w; \boldsymbol{\mu}^*)$ associated with an optimal dual variable $\boldsymbol{\mu}^*$ is specified, the persuasion problem (5) decouples across its linear segments, as we demonstrate in Theorem 4.3. We also provide a geometric illustration of the optimality conditions in Figure 1.

Theorem 4.3 (Optimality Conditions). *Let $\boldsymbol{\mu}^*$ be an optimal dual variable for (5) and let $h(w; \boldsymbol{\mu}^*)$ be the corresponding upper envelope function. The persuasion problem decouples across the linear segments of $h(w; \boldsymbol{\mu}^*)$. Specifically, assume that $h(w; \boldsymbol{\mu}^*)$ comprises m linear segments above the x-axis. For each segment $i \in [m]$, let $[z_i, z_{i-1}] \subseteq [0, 1]$ be its range and $T_i \subseteq [n]$ the set of receivers whose points (α_j, v_j) lie strictly within segment i.*[13] *A solution $\{q(j|w)\}$ is optimal to* (5) *if and only if the following conditions hold:*

[13] That is, point (α_j, v_j) lies on segment i of $h(w; \boldsymbol{\mu}^*)$ and satisfies $z_i < \alpha_j < z_{i-1}$.

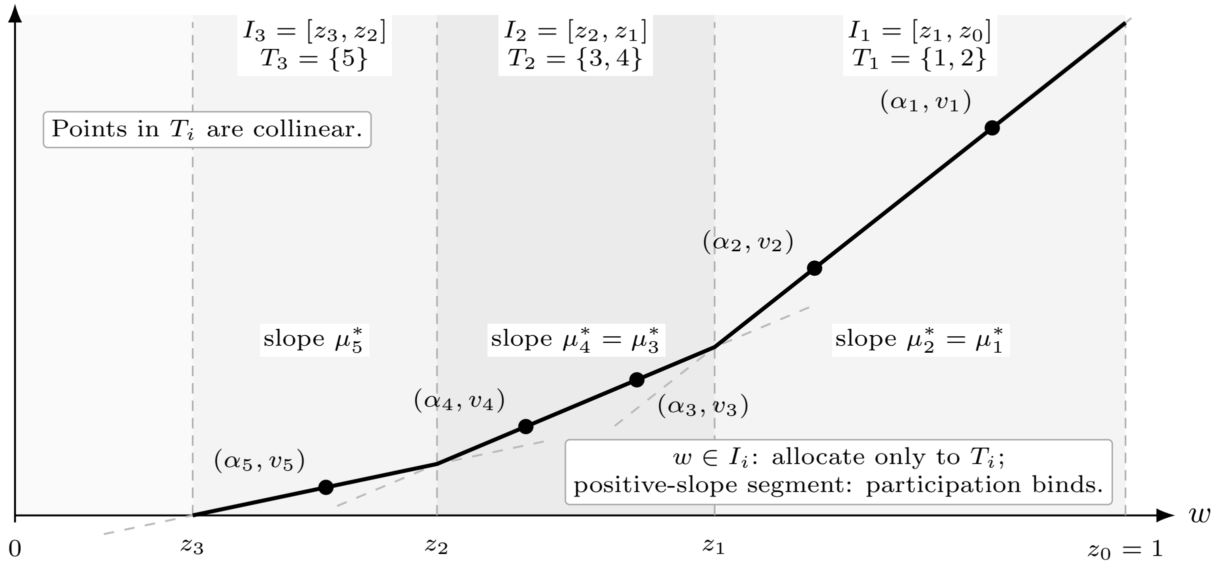


Figure 1: Geometric illustration of Theorem 4.3. Each receiver j is associated with a line $\ell_j(w; \mu_j^*)$ that passes through (α_j, v_j) and has slope $\mu_j^* \geq 0$. The solid curve represents the positive part of the upper envelope $h(w; \boldsymbol{\mu}^*) = \max_{j \in [n]} \ell_j(w; \mu_j^*)$, which partitions the state space into intervals $I_i = [z_i, z_{i-1}]$. On each interval I_i, the good can be allocated only to receivers in T_i, whose points lie on the corresponding segment of the envelope. In the illustration, there are $n = 5$ receivers, and the envelope consists of $m = 3$ segments above the x-axis. All receiver points (α_j, v_j) lie on the envelope. A persuasion mechanism is optimal if and only if it allocates states $w \in I_1$ exclusively to receivers 1 and 2, states $w \in I_2$ exclusively to receivers 3 and 4, and states $w \in I_3$ exclusively to receiver 5, while ensuring that the participation constraints bind for all five receivers, because every segment has a strictly positive slope.

1. *For each segment $i \in [m]$, $q(j|w)$ allocates $w \in [z_i, z_{i-1}]$ exclusively to receivers in set T_i:*

$$\sum_{j \in T_i} q(j|w) = 1, \, \forall\, w \in (z_i, z_{i-1}), \, i \in [m].$$

2. *If segment i has a positive slope, the participation constraints bind for all receivers in T_i:*

$$\int_{z_i}^{z_{i-1}} w \cdot q(j|w)\, g(w)\, dw = \alpha_j \int_{z_i}^{z_{i-1}} q(j|w)\, g(w)\, dw, \, \forall\, j \in T_i.$$

   Otherwise, if segment i has zero slope (which must be the leftmost segment because $h(w; \boldsymbol{\mu}^*)$ is increasing and convex), then T_i contains exactly one receiver, and $q(j|w)$ allocates $w \in [z_i, z_{i-1}]$ exclusively to this receiver.

3. *Any receiver $j \in [n] \setminus \bigcup_{i \in [m]} T_i$ whose point (α_j, v_j) either lies strictly below $h(w; \boldsymbol{\mu}^*)$ or lies at a kink point of $h(w; \boldsymbol{\mu}^*)$ receives the good with zero probability.*

Theorem 4.3 follows directly from Lemma 4.2 Part 2 and the fact that the envelope function

$h(w; \boldsymbol{\mu}^*)$ is convex, increasing, and piecewise linear. Detailed proof is provided in Section B.3. Finally, recall that the envelope function $h(w; \boldsymbol{\mu}^*)$ can be efficiently computed using a subgradient-based approach, as described in Remark 4.1. In Section 5, we further provide a constructive characterization of $h(w; \boldsymbol{\mu}^*)$, which yields additional structural insights. Once $h(w; \boldsymbol{\mu}^*)$ is obtained, the set of optimal persuasion mechanisms is fully determined by Theorem 4.3.

We conclude this section with two remarks: one on the connection with Dworczak and Martini (2019), and the other one on the closed-form characterization of optimal persuasion mechanisms in the two-receiver case.

Remark 4.2 (Connection to Dworczak and Martini 2019)**.** The linear-utility case studied in our paper is equivalent to the setting of Dworczak and Martini (2019), in which the sender's utility is increasing and piecewise constant in the posterior mean. In this special case, we observe that our envelope function

$$\bar{h}(w; \boldsymbol{\mu}^*) = \max\left\{h(w; \boldsymbol{\mu}^*), 0\right\}$$

coincides precisely with the equilibrium price function $p(x)$ in Dworczak and Martini (2019) (see their Theorem 1). Thus, we fully characterize their equilibrium price in this special case, which is generally challenging to specify. We emphasize, however, that because we study different formulations of the persuasion problem (our problem (5) versus their problem (1)) and dualize different constraints (we dualize the participation constraints in (5), whereas Dworczak and Martini (2019) dualize the second-order stochastic dominance constraints in their (1)), their optimality conditions (see their Theorem 1 and Proposition 1) do not directly imply our optimality conditions in Theorem 4.3. In addition, our optimality conditions are expressed in terms of the allocation probabilities $q(j|w)$ that define a persuasion mechanism, rather than in terms of the induced distribution of posterior means as in Dworczak and Martini (2019); in this sense, our characterization is more direct. Finally, our dual approach is also simpler[14] and may be of independent methodological interest.

Remark 4.3 (Optimal Persuasion with Two Receivers)**.** As a direct application of the optimality conditions in Theorem 4.3 (or equivalently, Part 2 of Lemma 4.2), we derive optimal persuasion mechanisms in closed form for the two-receiver case in Section C, which provide valuable insights for the general setting. In this case, receiver 1 has a higher acceptance threshold but also brings a higher payoff. Intuitively, this creates a trade-off: An offer from receiver 1 yields a higher payoff,

[14] In contrast, the analysis in Dworczak and Martini (2019) requires dualizing a more complex second-order stochastic dominance constraint and constructing a sequence of problems approximating their optimization problem (1), along with establishing appropriate convergence properties.

but targeting receiver 1 more aggressively reduces the overall probability of securing an offer. The optimal persuasion mechanisms take the following form (Theorem C.2). If receiver 1's offer value v_1 is sufficiently large, prioritizing receiver 1 is optimal. Alternatively, if v_1 is sufficiently small, it is optimal to allocate the good exclusively to receiver 2. Finally, if v_1 lies between the thresholds defining these two regimes, an optimal mechanism involves a nontrivial balance between the two receivers and exhibits the structure described in Proposition C.1.

5 Characterizing the Upper Envelope Function

In this section, we characterize the upper envelope function $h(w;\boldsymbol{\mu}^*)$ associated with an optimal dual variable $\boldsymbol{\mu}^*$. Our analysis builds on a dual connection between the first-best relaxation (5) and a convex optimization formulation proposed in Candogan (2022). We first introduce the convex optimization problem (9) and demonstrate its equivalence to (5) in Section 5.1. We then establish their connection from a dual perspective and use this connection to construct the envelope function $h(w;\boldsymbol{\mu}^*)$ in Section 5.2. Finally, Section 5.3 discusses ways to construct optimal persuasion mechanisms based on $h(w;\boldsymbol{\mu}^*)$.

5.1 The Convex Optimization Formulation

In this section, we introduce a convex optimization problem (9) with n decision variables and constraints and establish its equivalence to (5). Problem (9) is analogous to the problem (OPT) in Candogan (2022), albeit with n fewer decision variables and constraints.[15]

$$
\begin{aligned}
V^{\mathrm{CR}} = \max_{q_i \geq 0} \quad & \sum_{i=1}^{n} v_i q_i \\
\text{s.t.} \quad & \sum_{i\leq k} \alpha_i q_i \leq \sum_{i\leq k} q_i \cdot \mathbb{E}\left[w \,\middle|\, G(w) \geq 1 - \sum_{i\leq k} q_i \right] = \int_{1-\sum_{i\leq k} q_i}^{1} G^{-1}(x)\, dx, \ \forall\, k \in [n], \qquad (9) \\
& \sum_{i\in[n]} q_i \leq 1.
\end{aligned}
$$

In (9), the decision variables q_i represent the ex-ante probabilities that the good is allocated to receiver $i \in [n]$; specifically, q_i corresponds to $\int_0^1 q(i|w)g(w)dw$ in (5). The first constraint captures the receivers' participation constraints. Only a limited portion of qualified goods meet

[15]Note that in the problem (OPT) in Candogan (2022), given the optimal values $\{p_k^*\}$, it is optimal to set $z_k^* = b_k p_k^*$ there.

the receivers' acceptance standards. This constraint requires that goods within the top $\sum_{i \le k} q_i$ quantile be sufficient to meet the acceptance thresholds (α_i) of the top k receivers, given that each receiver $i \in [k]$ recruits a proportion q_i of goods. This condition is necessary to maintain the participation of the first k receivers. The equality in this constraint follows from the fact that for any random variable w with the cumulative distribution function $G(\cdot)$, the transformed variable $G(w)$ has a uniform distribution on $[0, 1]$. Finally, we note that (9) is a convex optimization problem. To see this, define $\Phi(x) \triangleq \int_{1-x}^{1} G^{-1}(s)\, ds$. This function is concave because its derivative, $\Phi'(x) = G^{-1}(1 - x)$, is decreasing in x. Consequently, the right-hand side of the first constraint is concave in $\{q_i\}$ because it is the composition of $\Phi(\cdot)$ with an affine mapping.

Given a feasible solution $\{q(i|w)\}$ to (5), the set $\{q_i\}$ with $q_i = \int_0^1 q(i|w)g(w)dw$ is feasible to (9) and attains the same objective value. Therefore, (9) is a relaxation of (5). Conversely, analogous to the two-receiver case (Section C), the optimal aggregate allocation probabilities $\{q_i^*\}$, along with the binding participation constraints for receivers with a positive dual variable $\mu_i^* > 0$, characterize an optimal mechanism. Specifically, given an optimal solution $\{q_i^*\}$ to (9), we can construct a persuasion mechanism that obtains the optimal value V^{CR}. Therefore, the relaxation (9) is tight. We state the above in Proposition 5.1 and provide its proof in Section B.4.

Proposition 5.1 (Primal Equivalence). *The optimal values of (5) and (9) are equal; that is,* $\bar{V} = V^{\mathrm{CR}}$. *Furthermore, let* $\{q^*(i|w)\}$ *be an optimal solution to (5). Then,* $\{q_i^*\}$*, where* $q_i^* = \int_0^1 q^*(i|w)g(w)dw$*, is an optimal solution to (9). Conversely, if* $\{q_i^*\}$ *is an optimal solution to (9), then there exists an optimal solution* $\{q^*(i|w)\}$ *to (5) such that* $q_i^* = \int_0^1 q^*(i|w)g(w)dw$.

5.2 Characterization of the Upper Envelope Function $h(w; \boldsymbol{\mu}^*)$

In this section, we characterize the upper envelope function $h(w; \boldsymbol{\mu}^*)$ associated with an optimal dual variable $\boldsymbol{\mu}^*$. Let $\{q_i^*\}$ be an optimal solution to (9). We assume $q_i^* > 0$ for all $i \in [n]$; this does not lose generality because receivers with $q_i^* = 0$ can be disregarded from consideration. We formalize this assumption in Assumption 5.1 and maintain it throughout Section 5.2.

Assumption 5.1. There exists an optimal solution $\{q_i^*\}$ to (9) such that $q_i^* > 0$ for all $i \in [n]$.

We first introduce several parameters needed to characterize the envelope function $h(w; \boldsymbol{\mu}^*)$. Let $\boldsymbol{\lambda}^* = \left(\lambda_k^*\right)_{k \in [n]} \in \mathbb{R}_+^n$ denote an optimal dual variable associated with the participation constraints in (9), and let γ^* denote an optimal dual variable for the constraint $\sum_{i \in [n]} q_i \le 1$ in (9). Define the set

$$T \triangleq \left\{ k \in [n] : \lambda_k^* > 0 \right\}$$

as the indices corresponding to positive entries in the optimal dual variable $\boldsymbol{\lambda}^*$. According to complementary slackness, the participation constraint in (9) is binding with the top k receivers for any $k \in T$. We assume $\lambda_n^* > 0$. The case where $\lambda_n^* = 0$ is discussed in Remark 5.1. In that scenario, we have $\lambda_{n-1}^* > 0$ (and hence, $n-1 \in T$), and the characterization of the upper envelope function $h(w; \boldsymbol{\mu}^*)$ remains essentially unchanged; see Remark 5.1 for additional details.[16]

Suppose that the set $T = \{t_1 < t_2 < \cdots < t_m = n\}$ consists of m receivers. These receivers partition the n receivers into m groups $\{T_i\}_{i \in [m]}$, where $T_1 = [t_1]$ and $T_i = [t_{i-1}+1 : t_i]$ for all $i \in [2:m]$. Define the endpoints as

$$z_i \triangleq G^{-1}\left(1 - \sum_{j=1}^{t_i} q_j^*\right), \quad \text{for each } i \in [m], \quad \text{and set } z_0 = 1.$$

Every receiver group T_i is associated with a state interval $I_i = [z_i, z_{i-1}]$. In Proposition 5.2, we establish the connection between the optimal dual variables of problems (5) and (9), and we characterize the upper envelope function $h(w; \boldsymbol{\mu}^*)$.

Proposition 5.2 (Characterization of the Envelope Function $h(w; \boldsymbol{\mu}^*)$)**.** *Suppose Assumption 5.1 holds. Let $\boldsymbol{\lambda}^* = (\lambda_k^*)_{k \in [n]}$ be an optimal dual variable associated with the participation constraints in (9). Define $\boldsymbol{\mu} = (\mu_i)_{i \in [n]}$ by setting $\mu_i = \sum_{k \geq i} \lambda_k^*$ for all $i \in [n]$. Then, $\boldsymbol{\mu}$ is an optimal dual variable for (5); that is, $V^{\mathrm{LR}}(\boldsymbol{\mu}) = \bar{V}$. Moreover, suppose $\lambda_n^* > 0$. The receivers' lines $\ell_j(w; \mu_j)$ and the upper envelope function $h(w; \boldsymbol{\mu}) = \max_{j \in [n]} \ell_j(w; \mu_j)$ are characterized as follows:*

1. *For each group T_i and receiver $j \in T_i$, the threshold satisfies $\alpha_j \in (z_i, z_{i-1})$. Moreover, within each group T_i, the lines $\ell_j(w; \mu_j)$ coincide for all $j \in T_i$ and pass through the points (α_j, v_j) for all $j \in T_i$.*
2. *For any two receivers $j \in T_i$ and $k \in T_{i+1}$ from adjacent groups (where $i \leq m-1$), their lines $\ell_j(w; \mu_j)$ and $\ell_k(w; \mu_k)$ intersect at $w = z_i$. Additionally, for every receiver $j \in T_m$ (the last group), line $\ell_j(w; \mu_j)$ intersects the x-axis at $w = z_m > 0$ if $\sum_{j \in [n]} q_j^* < 1$, and intersects the y-axis at point $\gamma^* \in [0, v_n]$ if $\sum_{j \in [n]} q_j^* = 1$ (which implies that $z_m = 0$).*
3. *The envelope function $h(w; \boldsymbol{\mu})$ satisfies $h(w; \boldsymbol{\mu}) = \ell_j(w; \mu_j)$ for all group T_i, receiver $j \in T_i$, and state $w \in [z_i, z_{i-1}]$. Additionally, $h(w; \boldsymbol{\mu}) = \ell_j(w; \mu_j)$ for all $j \in T_m$ and $w \in [0, z_m]$.*

We prove Proposition 5.2 in Section B.5 by comparing the optimality conditions of (5) and (9). Proposition 5.2 shows that the optimal dual variables for the participation constraints in (9)

[16] When $\lambda_n^* = 0$, the participation constraint for receiver n does not bind. The persuasion problem simplifies to the one involving only the first $n-1$ receivers, with any unallocated goods assigned to receiver n.

correspond to the differences between the optimal dual variables for the participation constraints in (5). Additionally, the envelope function $h(w; \boldsymbol{\mu}^*)$, as constructed in Proposition 5.2, is linear on each interval $[z_i, z_{i-1}]$ and is positive if and only if $w \geq z_m$. Finally, we address the case where $\lambda_n^* = 0$ in Remark 5.1.

Remark 5.1 (The case of $\lambda_n^* = 0$)**.** Suppose that Assumption 5.1 holds and that $\lambda_n^* = 0$. Then, the following properties hold: (i) $\lambda_{n-1}^* > 0$, and hence $n-1 \in T$; (ii) $\sum_{j \in [n]} q_j^* = 1$; and (iii) $\gamma^* = v_n$. Suppose that the set $T = \{t_1 < t_2 < \cdots < t_m = n-1\}$ consists of m receivers. These receivers partition the n receivers into $m+1$ groups $\{T_i\}$, where $T_1 = [t_1]$, $T_i = [t_{i-1}+1 : t_i]$ for each $i \in [2:m]$, and $T_{m+1} = \{n\}$. Define the endpoint $z_i \triangleq G^{-1}\left(1 - \sum_{j=1}^{t_i} q_j^*\right)$ for each $i \in [m]$, and set $z_0 = 1$ and $z_{m+1} = 0$. The dual variable $\boldsymbol{\mu}$ constructed in Proposition 5.2 remains optimal for (5). Moreover, the characterization of $h(w; \boldsymbol{\mu}^*)$ remains identical to that in Proposition 5.2, with an additional feature that $h(w; \boldsymbol{\mu}^*) = \gamma^* = v_n > 0$ is constant (horizontal) on the interval $[0, z_m]$. Further details are provided in Section B.5.5.

As a final remark, to characterize the upper envelope function $h(w; \boldsymbol{\mu}^*)$, it suffices to identify its breakpoints $\{z_i\}$ and the slopes $\{\mu_j^*\}$ of each of its linear segments, as established in Proposition 5.2 and Remark 5.1. In Algorithm 2 in Section E, we provide a step-by-step procedure for constructing the upper envelope function $h(w; \boldsymbol{\mu}^*)$ based on these results.

5.3 Characterization of Optimal Persuasion Mechanisms

Once the upper envelope function $h(w; \boldsymbol{\mu}^*)$ is characterized (see Proposition 5.2 and Remark 5.1), the set of optimal persuasion mechanisms is fully determined by Theorem 4.3. Specifically, suppose we have identified an optimal solution $\{q_i^*\}$ and an optimal dual variable $\boldsymbol{\lambda}^*$ to (9) and obtained the corresponding partition $\{T_i\}_{i \in [m]}$ of receivers. The persuasion problem then decouples across groups.

Within each group T_i, an optimal mechanism allocates the goods with characteristics $w \in I_i = [z_i, z_{i-1}]$ exclusively to receivers in this group such that all participation constraints are binding. If a group contains only one receiver, we simply allocate all goods with characteristics in the interval I_i to the receiver. However, if a group contains multiple receivers, the allocation must be handled more carefully. As in the two-receiver case (Section C.2), there are multiple ways to construct an optimal mechanism. In particular, we can iteratively build an optimal solution as follows. Suppose that we have already allocated a quantity q_ℓ^* of goods with mean quality α_ℓ from the interval I_i to each of the first k receivers ℓ in group T_i. We can then allocate a quantity q_j^* of goods with mean quality α_j from the remaining goods in interval I_i to the next receiver j, where j denotes the $(k+1)$-th receiver in T_i. This procedure is repeated until we reach the final receiver in the group, receiver t_i. Finally, we allocate the remaining quantity $q^*_{t_i}$, with mean quality α_{t_i}, to receiver t_i.

In Section D, we specify a particular allocation approach at each iteration step to obtain an optimal solution $\{q^*(j|w)\}$ to (5) that exhibits a monotone structure: A good with higher quality w is more likely to be allocated to a more preferred place. We also demonstrate how a deterministic persuasion mechanism with a double-interval structure, as described in Candogan (2022), can be easily derived using the results of our dual analysis.

6 Conclusions

We have studied a Bayesian persuasion problem in which a sender allocates an indivisible good to n receivers by strategically disclosing information. We demonstrate that as long as the sender has a known preference over the receivers and can allocate the good to at most one receiver, public persuasion is optimal regardless of how the receivers can communicate. Moreover, the optimal public persuasion mechanism can be derived from the first-best relaxation problem that imposes only participation constraints. Extending this strong result to more general settings or characterizing the suboptimality gap when public persuasion is not optimal is a promising direction for future research.

We also investigate a specific setting in which the state variable is one-dimensional and the receivers' utility functions are linear; therefore, a receiver cares only about the good's mean quality. We derive optimality conditions for persuasion mechanisms using a dual approach, which dualizes the participation constraints of the first-best relaxation problem. The optimality conditions state that the persuasion problem decouples over the linear segments of a piecewise linear upper envelope function arising from the geometric interpretation of the Lagrangian dual. Based on the dual approach, we obtain closed-form optimal mechanisms for the two-receiver case and an explicit characterization of all optimal persuasion mechanisms for the general case. This dual-based analysis enriches our understanding of the structures of optimal persuasion mechanisms.

Acknowledgements

We thank the area editor Rene Caldentey, an anonymous associate editor, and two anonymous reviewers for providing a number of valuable comments that greatly improved the paper.

References

Alizamir, S., de Véricourt, F. and Wang, S. (2020), 'Warning against recurring risks: An information design approach', *Management Science* **66**(10), 4612–4629.

Anunrojwong, J., Iyer, K. and Manshadi, V. (2023), 'Information design for congested social services: Optimal need-based persuasion', *Management Science* **69**(7), 3778–3796.

Arieli, I. and Babichenko, Y. (2019), 'Private bayesian persuasion', *Journal of Economic Theory* **182**, 185–217.

Arieli, I., Babichenko, Y., Smorodinsky, R. and Yamashita, T. (2023), 'Optimal persuasion via bi-pooling', *Theoretical Economics* **18**(1), 15–36.

Babichenko, Y., Talgam-Cohen, I., Xu, H. and Zabarnyi, K. (2022), 'Multi-channel bayesian persuasion', *arXiv preprint arXiv:2111.09789* .

Bergemann, D. and Morris, S. (2016), 'Bayes correlated equilibrium and the comparison of information structures in games', *Theoretical Economics* **11**(2), 487–522.

Bergemann, D. and Morris, S. (2019), 'Information design: A unified perspective', *Journal of Economic Literature* **57**(1), 44–95.

Bertsekas, D. P. (2016), *Nonlinear programming*, 3 edn, Athena Scientific.

Candogan, O. (2020), *Information Design in Operations*, INFORMS, chapter 8, pp. 176–201.

Candogan, O. (2022), 'Persuasion in networks: Public signals and cores', *Operations Research* **70**(4), 2264–2298.

Candogan, O., Guo, Y. and Xu, H. (2023), 'On information design with spillovers', *Available at SSRN 3537289* .

Candogan, O. and Wu, M. (2023), 'Information design for spatial resource allocation', *Available at SSRN 4505414* .

De Véricourt, F., Gurkan, H. and Wang, S. (2021), 'Informing the public about a pandemic', *Management Science* **67**(10), 6350–6357.

Drakopoulos, K., Jain, S. and Randhawa, R. (2021), 'Persuading customers to buy early: The value of personalized information provisioning', *Management Science* **67**(2), 828–853.

Dworczak, P. and Martini, G. (2019), 'The simple economics of optimal persuasion', *Journal of Political Economy* **127**(5), 1993–2048.

Excedr (2025), 'Biotech partnerships: How partnering with big pharma can support r&d'. Available at `https://www.excedr.com/blog/how-biotech-partnerships-support-research`. Accessed: 2025-04-29.

Galperti, S. and Perego, J. (2023), 'Games with information constraints: seeds and spillovers', *Available at SSRN 3340090* .

Hsu, D. H. (2004), 'What do entrepreneurs pay for venture capital affiliation?', *The journal of finance* **59**(4), 1805–1844.

Kamenica, E. (2019), 'Bayesian persuasion and information design', *Annual Review of Economics* **11**, 249–272.

Kleiner, A., Moldovanu, B. and Strack, P. (2021), 'Extreme points and majorization: Economic applications', *Econometrica* **89**(4), 1557–1593.

Lingenbrink, D. and Iyer, K. (2018), Signaling in online retail: Efficacy of public signals, *in* 'Proceedings of the 13th Workshop on Economics of Networks, Systems and Computation', NetEcon '18, Association for Computing Machinery, pp. 1–1.

Lingenbrink, D. and Iyer, K. (2019), 'Optimal signaling mechanisms in unobservable queues', *Operations research* **67**(5), 1397–1416.

Luenberger, D. G. (1997), *Optimization by vector space methods*, John Wiley & Sons.

Papanastasiou, Y., Bimpikis, K. and Savva, N. (2018), 'Crowdsourcing exploration', *Management Science* **64**(4), 1727–1746.

Rothaermel, F. T. and Boeker, W. (2008), 'Old technology meets new technology: Complementarities, similarities, and alliance formation', *Strategic management journal* **29**(1), 47–77.

Shapiro, A., Dentcheva, D. and Ruszczynski, A. (2021), *Lectures on Stochastic Programming: Modeling and Theory, Third Edition*, Society for Industrial and Applied Mathematics.

Sorensen, M. (2007), 'How smart is smart money? a two-sided matching model of venture capital', *The Journal of Finance* **62**(6), 2725–2762.

Yang, P., Iyer, K. and Frazier, P. (2019), 'Information design in spatial resource competition', *arXiv preprint arXiv:1909.12723* .

A Proofs and Additional Details for Section 3

A.1 Proof of Lemma 3.1

Fix any information disclosure mechanism $f(\cdot|w)$. For any $i \in [n]$, let

$$
\begin{aligned}
q(i|w) &= \mathbb{P}\left[a_i^* = 1 \text{ and } a_j^* = 0,\, \forall\, j < i \mid w\right] \\
&= \int_{\boldsymbol{s}} \int_{\mathbf{c}} \delta_i(s_i, c_i)\, \Pi_{j<i} \left(1 - \delta_j(s_j, c_j)\right) c(\mathbf{c}|\boldsymbol{s})\, f(\boldsymbol{s}|w)\, d\mathbf{c}\, d\boldsymbol{s}
\end{aligned}
$$

denote the probability that receiver i extends an offer and the good is allocated to him under the receivers' equilibrium strategies when the good's characteristics are w. The random binary variable $a_i^* \in \{0, 1\}$ represents receiver i's action of extending an offer in the equilibrium of the game induced by the mechanism $f(\cdot|w)$. Note that the good will be allocated to receiver i if and only if none of the receivers $j < i$ extends an offer.

We first prove that the participation constraint in (2) holds; that is,

$$
\int_{w \in \Omega} u_i(w)\, q(i|w)\, dG(w) = \mathbb{E}\big[u_i(w) \cdot \mathbb{1}[a_i^* = 1 \text{ and } a_j^* = 0,\, \forall\, j < i]\big] \geq 0.
$$

To see this, note that

$$
\begin{aligned}
&\mathbb{E}\big[u_i(w) \cdot \mathbb{1}[a_i^* = 1 \text{ and } a_j^* = 0,\, \forall\, j < i] \mid c_i, s_i\big] \\
=&\mathbb{E}\big[\mathbb{1}[a_i^* = 1] \mid c_i, s_i\big] \cdot \mathbb{E}\big[u_i(w) \cdot \mathbb{1}[a_j^* = 0,\, \forall\, j < i] \mid c_i, s_i\big] \geq 0,
\end{aligned}
$$

where the equality follows from the fact that the action a_i^* is independent of a_j^* and w conditional on the signal-communication-information pair (c_i, s_i), and the inequality follows from the optimality of the receiver's equilibrium strategy—that is, receiver i extends an offer only if doing so provides nonnegative expected utility to him. Taking expectation over (c_i, s_i) on both sides of the above inequality yields the desired result.

For the second constraint, note that for any $w \in \Omega$, we have

$$
\sum_{i \in [n]} q(i|w) = \mathbb{P}\left[\, \exists\, i \in [n] \text{ such that } a_i^* = 1 \mid w\right] \leq 1.
$$

Finally, the expected payoff of the mechanism $f(\cdot|w)$ can be expressed as

$$
\sum_{i=1}^{n} v_i \int_{w \in \Omega} q(i|w)\, dG(w),
$$

which is the objective function of (2). Since $\{q(i|w)\}$ is feasible to (2) given any mechanism $f(\cdot|w)$, we have $V^* \leq \bar{V}$.

A.2 Proof of Theorem 3.2

Let $\{q^*(i|w)\}$ denote an optimal solution to (2). We first show that for any two receivers j and k with $j < k$, we have

$$
\int_{w \in \Omega} u_j(w)\, q^*(k|w)\, dG(w) < 0. \tag{10}
$$

We prove this by contradiction. Assume that there exists j and k with $j < k$ such that

$$\int_{w\in\Omega} u_j(w)\, q^*(k|w)\, dG(w) \geq 0.$$

Consider the new allocation rule $\tilde{q}(i|w)$ defined as:

$$\tilde{q}(i|w) = \begin{cases} q^*(j|w) + q^*(k|w) & \text{if } i = j, \\ 0 & \text{if } i = k, \\ q^*(i|w) & \text{if } i \notin \{j, k\}. \end{cases}$$

$\{\tilde{q}(i|w)\}$ is feasible to (2), and because $v_j > v_k$, $\{\tilde{q}(i|w)\}$ achieves a strictly larger objective value than $\{q^*(i|w)\}$. This contradicts the fact that $\{q^*(i|w)\}$ is optimal to (2). Thus, our assumption fails.

Since a public persuasion mechanism leaves no payoff-related information for the receivers to communicate, receivers make decisions based only on the public signal and ignore potential communication among themselves. We now show that it is an equilibrium for each receiver $i \in [n]$ to extend an offer only upon receiving the signal $s = i$. To do so, suppose all receivers other than receiver i follow this strategy; we verify that it is optimal for receiver i to do the same.

First, suppose receiver i receives the signal $s = i$. The expected payoff for extending an offer is nonnegative because

$$\int_{w\in\Omega} u_i(w)\, dG\big(w|s = i\big) = \frac{1}{\int_w q^*(i|w)\, dG(w)} \int_{w\in\Omega} u_i(w)\, q^*(i|w)\, dG(w) \geq 0,$$

where $dG\big(w|s = i\big) = \frac{q^*(i|w)\, dG(w)}{\int_w q^*(i|w)\, dG(w)}$ denotes the posterior belief of w given $s = i$, and the inequality follows from the participation constraint in (2). Therefore, it is optimal for the receiver i to extend an offer.

Second, suppose receiver i receives the signal $s = k$ with $k > i$. The expected payoff for extending an offer is negative because

$$\int_{w\in\Omega} u_i(w)\, dG\big(w|s = k\big) = \frac{1}{\int_w q^*(k|w)\, dG(w)} \int_{w\in\Omega} u_i(w)\, q^*(k|w)\, dG(w) < 0,$$

where the inequality follows from (10). Therefore, receiver i will not extend an offer.

Finally, suppose receiver i receives the signal $s = j$ with $j < i$. Since the sender will never accept receiver i's offer (because receiver j will extend an offer), receiver i is indifferent between extending an offer or not.

Note that the expected payoff for the sender is $\bar{V}$ under this equilibrium. Therefore, the public mechanism $f^*(\cdot|w)$ is optimal to (1).

A.3 Additional Analysis for Section 3.3

A.3.1 Details of Remark 3.3

Proof of Scenario 1 Consider the private persuasion mechanism $f^*(\cdot|w, \theta)$ with signal space $S_i = \{\text{Yes}, \text{No}\}$ for each receiver $i \in [n]$. When the good's characteristics are w and the sender's preference type is θ, the mechanism operates as follows. With probability $q^*(i|w, \theta)$ for each $i \in [n]$, the mechanism privately sends the signal $s_i = \text{Yes}$ to receiver i and the signal $s_j = \text{No}$ to all other receivers $j \neq i$. With the remaining probability $1 - \sum_{i=1}^{n} q^*(i|w,\theta)$, the mechanism privately sends the signal $s_k = \text{No}$ to all receivers $k \in [n]$. We can interpret the signal $s_i = \text{Yes}$ as a recommendation for receiver i to extend an offer, and the signal $s_i = \text{No}$ as a recommendation for him not to extend an offer.

In the following, we show that under the private persuasion mechanism $f^*(\cdot|w,\theta)$, it is an equilibrium for each receiver $i \in [n]$ to extend an offer if and only if he receives the signal $s_i = \text{Yes}$. To do so, suppose that all receivers other than receiver i follow this strategy; we verify that it is optimal for receiver i to do the same. As preparation, for any $i \in [n]$, let

$$q_i \triangleq \int_{w\in\Omega} \int_{\theta\in\Theta} q^*(i|w,\theta)\, dG(w|\theta)\, dH(\theta)$$

denote the ex-ante probability that receiver i receives the signal $s_i = \text{Yes}$; in this event, all other receivers $j \neq i$ receive the signal $s_j = \text{No}$. Moreover, let

$$q_\varnothing \triangleq 1 - \sum_{i\in[n]} q_i = \int_{w\in\Omega} \int_{\theta\in\Theta} \left(1 - \sum_{j=1}^{n} q^*(j|w,\theta)\right) dG(w|\theta)\, dH(\theta)$$

denote the ex-ante probability that all receivers i receive the signal $s_i = \text{No}$.

First, suppose that receiver i receives the signal $s_i = \text{Yes}$, which happens with probability q_i. The expected payoff for extending an offer is

$$\mathbb{E}\left[u_i(w) \mid s_i = \text{Yes}\right] = \frac{1}{q_i} \int_{w\in\Omega} \int_{\theta\in\Theta} u_i(w)\, q^*(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \geq 0\,,$$

where the inequality follows from the participation constraint in (3). Therefore, it is optimal for receiver i to extend an offer.

Second, suppose that receiver i receives the signal $s_i = \text{No}$, which happens with probability $q_\varnothing + \sum_{j\neq i} q_j$. If receiver i were to extend an offer, the expected utility from doing so would be

$$\frac{1}{q_\varnothing + \sum_{j\neq i} q_j} \int_{w\in\Omega} \int_{\theta\in\Theta} u_i(w) \Bigg(1 - \sum_{j=1}^{n} q^*(j|w,\theta) + \sum_{j\neq i} \mathbb{1}\Big\{v_i(\theta) > v_j(\theta)\Big\}\, q^*(j|w,\theta)\Bigg)\, dG(w|\theta)\, dH(\theta) \leq 0\,.$$

To see this, note that conditional on $s_i = \text{No}$, all other receivers also observe No with probability $\frac{q_\varnothing}{q_\varnothing + \sum_{j\neq i} q_j}$ and hence do not extend offers; in this case, receiver i's offer is accepted. Alternatively, with probability $\frac{q_j}{q_\varnothing + \sum_{j\neq i} q_j}$, some receiver $j \neq i$ receives Yes and therefore extends an offer; in this case, receiver i's offer is accepted if $v_i(\theta) > v_j(\theta)$. The inequality above follows from the assumption imposed in this scenario. Since the expected payoff from extending an offer is non-positive, receiver i will not extend an offer.

Finally, we note that the expected payoff for the sender is $\bar{V}$ under this equilibrium. Hence, the first-best relaxation bound $\bar{V}$ is attained.

Proof of Scenario 2 Consider the following public persuasion mechanism $f^*(\cdot|w,\theta)$ with signal space $S_i = S \triangleq [n] \cup \{\varnothing\}$ for all receivers $i \in [n]$. When the good's characteristics are w and the sender's preference type is θ, the mechanism broadcasts the signal $s = i$ to all receivers with probability $q^*(i|w,\theta)$ for each $i \in [n]$, and the signal $s = \varnothing$ to all receivers with probability $1 - \sum_{i\in[n]} q^*(i|w,\theta)$. Moreover, the sender commits to the following offer-acceptance rule: conditional on the signal $s = i$ being sent, the sender accepts only receiver i's offer, even if a better offer is received from another receiver.

In the following, we show that under the public persuasion mechanism $f^*(\cdot|w,\theta)$ and given the sender's commitment power over which offer to accept, it is an equilibrium for each receiver $i \in [n]$ to extend an offer only upon receiving the signal $s = i$. To do so, suppose all receivers other than receiver i follow this strategy; we verify that it is optimal for receiver i to do the same.

First, suppose that receiver i receives the signal $s = i$. The expected payoff for extending an offer is nonnegative by the participation constraint in (3). Therefore, it is optimal for receiver i to extend an offer.

Second, suppose that receiver i receives the signal $s = j$ with some $j \neq i$. Since the sender will never accept receiver i's offer in this case, receiver i is indifferent between extending an offer or not.

Finally, we note that the expected payoff for the sender is $\bar{V}$ under this equilibrium. Hence, the first-best relaxation bound $\bar{V}$ is attained.

A.3.2 Proof of Theorem 3.4

The proof is analogous to the proof of Theorem 3.2 in Section A.2. In the proof, we impose the following tie-breaking rule for the sender: whenever there is a tie, the sender accepts the offer from the lowest-index receiver among those whose offers attain the highest value.

Let $\{q^*(i|w,\theta)\}$ denote an optimal solution to (3). We assume that, for any two receivers j and k with $j < k$, if receiver k is allocated the good with positive probability under the first-best allocation,[17] then the following holds:

$$\int_{w\in\Omega}\int_{\theta\in\Theta} u_j(w)\, q^*(k|w,\theta)\, dG(w|\theta)\, dH(\theta) < 0\,. \tag{11}$$

Otherwise, if (11) fails for some pair (j,k), we can define an alternative allocation rule $\{\tilde{q}(i|w,\theta)\}$ with

$$\tilde{q}(i|w,\theta) = \begin{cases} q^*(j|w,\theta) + q^*(k|w,\theta) & \text{if } i = j\,, \\ 0 & \text{if } i = k\,, \\ q^*(i|w,\theta) & \text{if } i \notin \{j,k\}\,. \end{cases}$$

This allocation rule $\{\tilde{q}(i|w,\theta)\}$ is feasible to (3) and, by Assumption 3.1, achieves a weakly higher objective value than $\{q^*(i|w,\theta)\}$. Hence, $\{\tilde{q}(i|w,\theta)\}$ is also optimal to (3) and it satisfies (11).[18]

Since a public persuasion mechanism leaves no payoff-related information for the receivers to communicate, receivers make decisions based only on the public signal. We now show that it is an equilibrium for each receiver $j \in [n]$ to extend an offer only upon receiving the signal $s = j$. To do so, suppose all receivers other than receiver j follow this strategy; we verify that it is optimal for receiver j to do the same.

First, suppose receiver j receives the signal $s = j$. The expected payoff for extending an offer is nonnegative by the participation constraint in (3). Therefore, it is optimal for receiver j to extend an offer.

[17] That is, $q_k^* \triangleq \int_{w\in\Omega}\int_{\theta\in\Theta} q^*(k|w,\theta)\, dG(w|\theta)\, dH(\theta) > 0$.

[18] We note that, with a more complex tie-breaking rule, any optimal solution $\{q^*(i|w,\theta)\}$ to (3) (i.e., without the above modification) can sustain a Nash equilibrium under the public persuasion mechanism $f^*(\cdot|w,\theta)$ and achieves the first-best performance $\bar{V}$.

Second, suppose receiver j receives the signal $s = k$ with $j < k$. Since (11) holds, the expected payoff for extending an offer (in this case, the sender will accept receiver j's offer) is negative. Therefore, receiver j will not extend an offer.

Third, suppose receiver j receives the signal $s = i$ with $i < j$. In this case, the sender will never accept receiver j's offer, because receiver i will extend an offer, and the sender prefers receiver i's offer by Assumption 3.1 or her tie-breaking rule. Hence, receiver j is indifferent between extending an offer or not.

Finally, we note that the expected payoff for the sender is $\bar{V}$ under this equilibrium.

A.3.3 A Negative Example

In this section, we show that under the general model, public persuasion may no longer be optimal when the offer values $\{v_i\}$ correlate arbitrarily with the good's characteristics w. We illustrate this point through the following simple example.

Example A.1. Consider an example with two receivers, each having linear utilities as in Section 4. The good's characteristic w is one-dimensional and uniformly distributed on $[0, 1]$. The receivers' acceptance thresholds are $\alpha_1 = 0.9$ and $\alpha_2 = 0.7$, respectively. The sender's offer values correlate with the good's quality w as follows. When w exceeds α_1, the sender slightly prefers receiver 1 over receiver 2, with offer values $v_1 = 1 + \epsilon$ and $v_2 = 1$, where $\epsilon > 0$ is small (e.g., $\epsilon = 0.1$). However, when w is below α_1, the sender strongly prefers receiver 2, with offer values $v_1 = 1$ and $v_2 = 100$.

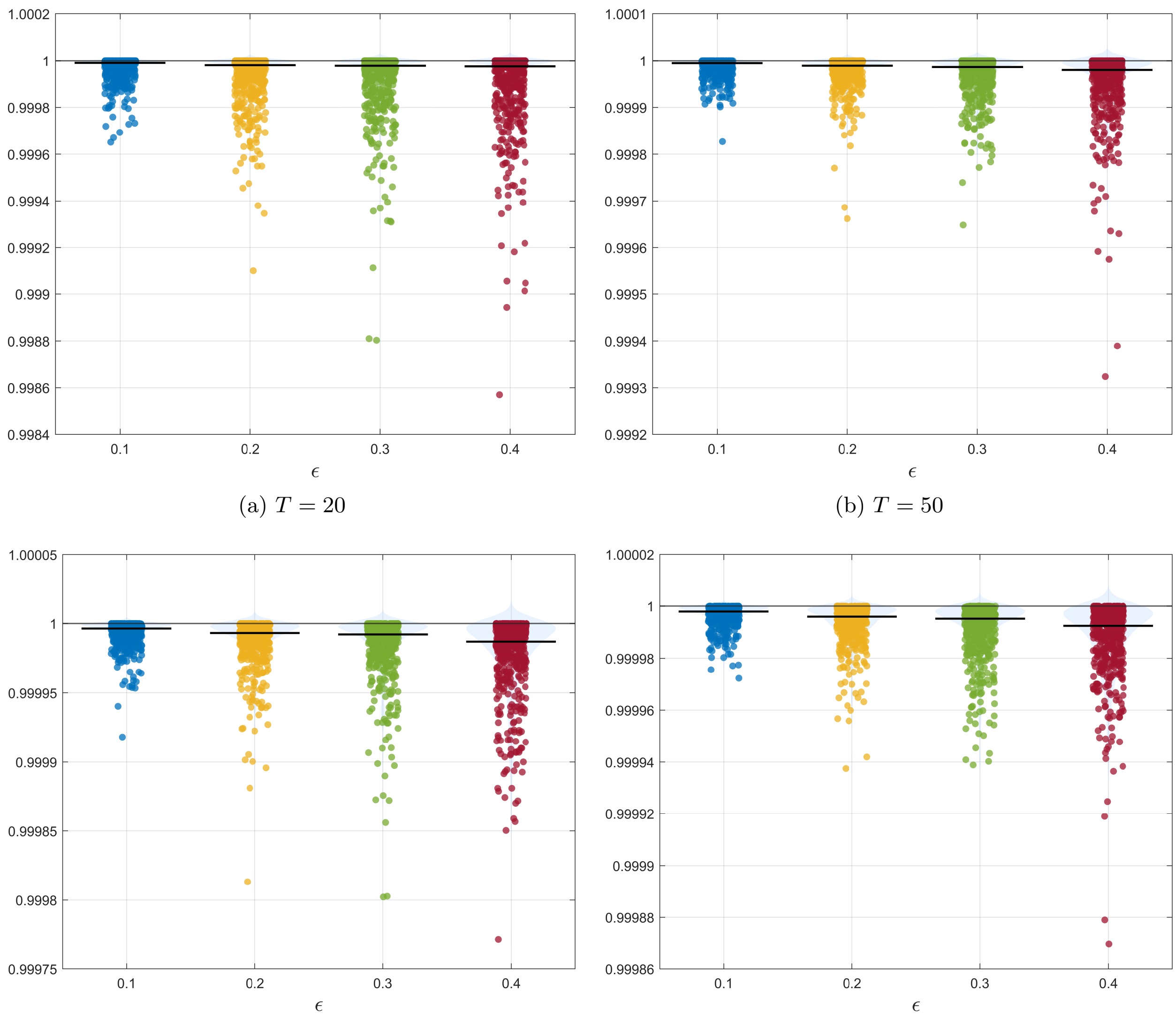
Private Persuasion Attains First-Best Value Since v_2 is significantly large when the good's quality w is below α_1, the optimal solution to the first-best relaxation (3) exclusively targets receiver 2, i.e., allocating all goods with $w \in [0.4, 1]$ to receiver 2 and none to receiver 1. This solution can be implemented through private persuasion when the receivers cannot communicate. Specifically, the sender provides no information to receiver 1 and informs receiver 2 whether the good's quality exceeds 0.4. Consequently, receiver 1 refrains from making offers, while receiver 2 extends an offer only when $w \geq 0.4$.[19]

Failure of Public Persuasion We next show that public persuasion cannot achieve the first-best outcome. Achieving the first-best outcome requires that the public signal induces receiver 2 to extend an offer with probability one whenever $w \geq 0.4$. However, since the signal is public, receiver 1 observes the same information as receiver 2, including the goods recommended to receiver 2. Therefore, receiver 1 can target these goods with competing offers. Consequently, all goods with quality $w \in [0.9, 1]$ are allocated to receiver 1, because the sender prefers receiver 1 whenever $w \geq \alpha_1$. This selection effectively removes the highest-quality goods from receiver 2's pool, resulting in negative utility for receiver 2 and making the first-best outcome unsustainable under public persuasion.

A.3.4 Empirical Distribution of the Performance Ratio in Section 3.3.3

In this section, we visualize the empirical distribution of the performance ratio $V^{\mathrm{P}}/\bar{V}$ from Section 3.3.3 for each combination of (ϵ, T) using scatter plots, as illustrated in Figure 2.

[19] If receiver 1 were to extend an offer, given receiver 2's strategy, only goods with quality $w \in [0, 0.4) \cup [0.9, 1]$ would ultimately be allocated to receiver 1, whose expected quality falls below receiver 1's acceptance threshold α_1.



(a) $T = 20$ (b) $T = 50$

(c) $T = 100$ (d) $T = 200$

Figure 2: Scatter plot of the performance ratio $V^{\mathrm{P}}/\bar{V}$ based on $N = 500$ independent samples for each (ϵ, T) combination. Each point represents one sample, and the thick black horizontal line indicates the sample median.

A.3.5 Discussion on Optimal Public Persuasion

When the sender's preference order is fixed (i.e., Assumption 3.1 holds), Theorem 3.4 shows that a public outcome-revealing persuasion mechanism (as defined in Definition A.1)—in which the sender publicly announces the receiver to whom the good should be allocated, and each receiver i finds it incentive compatible to extend an offer if and only if he receives the signal $s = i$—achieves the first-best relaxation bound $\bar{V}$ and is therefore optimal. We note that the optimal public outcome-revealing persuasion mechanism can be computed by solving the LP (4).

Definition A.1 (Public Outcome-Revealing Persuasion Mechanism)**.** A public persuasion mechanism is called a *public outcome-revealing persuasion mechanism* if: (i) the signal space is $S \triangleq [n] \cup \{\varnothing\}$, (ii) under the mechanism, the sender publicly announces the receiver to whom the good should be allocated, and (iii) it is incentive compatible for each receiver i to extend an offer if and only if he receives the signal $s = i$.

A natural question under the general model of Section 3.3.1 is whether one can, without loss of optimality, restrict attention to outcome-revealing mechanisms when computing an optimal public persuasion mechanism. In this section, we provide a negative answer: a public outcome-revealing persuasion mechanism can be strictly suboptimal relative to the optimal public persuasion mechanism.

We illustrate this point through a simple example (Example A.2), in which no public outcome-revealing persuasion mechanism can implement the first-best outcome, whereas a suitably designed alternative public persuasion mechanism can achieve it. Finally, we illustrate how an optimal public persuasion mechanism can be computed in the general setting.

Example A.2. Consider an example in which the good's quality w is one-dimensional. There are $n = 3$ receivers, each with a linear utility function of the form $u_i(w) = w - \alpha_i$, with $\alpha_1 = \alpha_2 = 0.9$ and $\alpha_3 = 0.99$.

The good can have four possible types. Each type fully characterizes both the quality of the good and the sender's payoff from the receivers' offers, as summarized in Table 2 (where each row corresponds to one good type). For example, a type-one good has quality $w = 1$ and occurs with probability 0.2. When the good is of type one, the sender's payoff from the receivers' offers are $v_1 = v_2 = 1$ and $v_3 = 1.01$, indicating that the sender slightly prefers receiver 3's offer. The remaining rows of Table 2 can be interpreted analogously.

w	$\mathbb{P}_w(w)$	v_1	v_2	v_3
1.0	0.2	1	1	1.01
0.8	0.1	0	1	1.01
0.8	0.1	1	0	0
0.0	0.6	0	0	0

Table 2: Distribution of the good's quality w and the sender's offer values.

The First-Best Allocation The first-best allocation (i.e., the optimal solution to (3)) in Example A.2 is as follows. Type-one goods are allocated to receiver 1 with probability 0.5 and to receiver 2 with probability 0.5. All type-two goods are allocated to receiver 2, and all type-three goods are allocated to receiver 1. Finally, type-four goods are not allocated to any receiver, as their quality is too low to meet any receiver's acceptance threshold.

Under this allocation, receivers 1 and 2 break even, while receiver 3 is not allocated any goods. This is because receiver 3 has a very stringent quality requirement, which cannot be satisfied without allocating almost exclusively type-one goods. By contrast, allocating type-one goods to receivers 1 and 2 enables the sender to also allocate type-two and type-three goods to these receivers, which in aggregate generates a higher total payoff. Therefore, it is optimal to exclude receiver 3 in the first-best allocation.

Public Outcome-Revealing Persuasion Mechanism Cannot Attain the First-Best Outcome We remark that no public outcome-revealing persuasion mechanism can achieve the first-best outcome. To see this, note that achieving the first-best outcome requires that half of the type-one goods be allocated to receiver 1. However, conditional on the good being type-one, the sender slightly prefers the offer from receiver 3. Therefore, under any public outcome-revealing persuasion mechanism, receiver 3 can cherry-pick type-one goods from receiver 1 by submitting a competing offer whenever the sender publicly announces the signal $s = 1$. As a consequence, type-one goods cannot be allocated to receiver 1, and thus no public outcome-revealing persuasion mechanism can achieve the first-best outcome.

An Alternative Public Persuasion Mechanism that Attains the First-Best Outcome Consider the following public persuasion mechanism, in which the sender recommends to each receiver the action to take, rather than directly revealing the final allocation outcome as in an outcome-revealing persuasion mechanism. Whenever the good is of type one, type two, or type three, the sender publicly announces that both receivers 1 and 2 should extend an offer. Whenever the good is of type four, the sender publicly announces that no receiver should extend an offer. In what follows, we show that this public persuasion mechanism is incentive compatible—that is, each receiver finds it optimal to follow the sender's recommendation—and that it induces the first-best outcome in equilibrium.

First, when the sender announces that no offer should be made, all receivers infer that the good is of type four and thus have no incentive to extend an offer.

Next, consider the case in which the sender announces that both receivers 1 and 2 should extend an offer. We show that it constitutes an equilibrium if (i) all receivers follow the recommendation (that is, receivers 1 and 2 extend an offer while receiver 3 refrains), and (ii) when the good is of type one, the sender breaks ties by allocating the good to receiver 1 with probability 0.5 and to receiver 2 with probability 0.5. Specifically:

- Receiver 3 will not extend an offer. If receiver 3 were to do so, goods of both type one and type two would be allocated to him, which would make the offer strictly loss-making given his stringent quality requirement.

- Given that receiver 3 does not extend an offer and given the tie-breaking rule, both receivers 1 and 2 break even when extending offers and thus are willing to do so.

Intuitively, mixing type-one, type-two, and type-three goods across receivers 1 and 2 deters receiver 3 from cherry-picking the type-one goods allocated to receiver 1, which strictly benefits the sender. The resulting equilibrium allocation then endogenously separates goods between receivers 1 and 2. Example A.2 demonstrates that public outcome-revealing persuasion mechanisms can be strictly suboptimal among public persuasion mechanisms.

Computation of Optimal Public Persuasion Mechanism As shown in Example A.2, when computing an optimal public persuasion mechanism, one cannot restrict attention to outcome-revealing mechanisms. On the other hand, by the revelation principle, it is without loss of optimality to restrict attention to the class of *action-revealing* mechanisms, as defined in Definition A.2.

Definition A.2 (Public Action-Revealing Persuasion Mechanism)**.** A public persuasion mechanism is called a *public action-revealing persuasion mechanism* if the following conditions hold:

1. The signal space is $S \triangleq 2^{[n]}$, the power set of $[n]$, which contains all subsets of $[n]$ and has 2^n elements.
2. Under the mechanism, the sender publicly recommends a set of receivers $A \subseteq [n]$ to extend offers.
3. For each receiver i, it is incentive compatible to extend an offer if and only if he is recommended to do so, that is, if and only if $i \in A$.

Consequently, the problem of computing an optimal public persuasion mechanism can be formulated as an LP whose numbers of decision variables and constraints are both exponential in the number of receivers n.

A.4 Details of Remark 3.4

In our base model, we assume that receivers take binary actions (i.e., either accepting or rejecting the good). In practice, a receiver may have multiple options. For example, in the student promotion example, an employer might extend a regular offer, provide an offer with additional benefits, or an offer for an alternative position, or decline to make an offer. In this section, we generalize our setting to allow receivers to choose among multiple possible actions regarding the good, and we demonstrate that our results from Section 3 continue to hold.

A.4.1 A Model with Multiple Actions

In this section, we extend our base model from Section 2 by allowing each receiver to select from multiple actions. Specifically, for each receiver $i \in [n]$, let $\bar{A}_i \triangleq A_i \cup \{\varnothing\}$ denote his action set, where A_i represents multiple acceptance options available to receiver i and $\varnothing$ denotes the action of rejection. For each acceptance action $a_i \in A_i$, let $u_i(w, a_i)$ denote receiver i's utility when he offers acceptance option a_i and a good with characteristics w is allocated to him. The utility of not receiving the good is zero; hence, $u_i(w, \varnothing) = 0$ for all $i \in [n]$ and $w \in \Omega$.

The sender allocates an indivisible good and obtains a deterministic utility $v(a_i)$ if it is accepted through option a_i. We assume these offer values $\{v(a_i)\}_{a_i \in A_i, i \in [n]}$ are distinct and positive in Assumption A.1. The sender's utility is zero if the good is not allocated.

Assumption A.1. The sender receives deterministic utility $v(a_i)$ when allocating the good to receiver i through acceptance option a_i. For simplicity, assume that the offer values $\{v(a_i)\}_{a_i \in A_i, i \in [n]}$ are distinct and positive.

Finally, since the offer values $\{v(a_i)\}$ are distinct, we can sort the acceptance options $\bigcup_{i \in [n]} A_i$ in descending order according to their offer values, and partition them into groups $\{B_k\}$ such that:

- Each group B_k consists exclusively of acceptance actions available to one receiver (i.e., $B_k \subseteq A_i$ for some $i \in [n]$);
- Any two adjacent groups B_{k-1} and B_k correspond to different receivers;

- For any two acceptance actions $a_i \in B_{k-1}$ and $a_j \in B_k$ from adjacent groups, we have $v(a_i) > v(a_j)$; that is, the sender strictly prefers actions in group B_{k-1} to those in group B_k.

For each group B_k, we define $i(B_k) \in [n]$ as the receiver to whom these actions belong (i.e., $B_k \subseteq A_{i(B_k)}$).

A.4.2 The Relaxation Problem

In Section A.4.3, we demonstrate that public persuasion remains optimal. To this end, we first reformulate the first-best relaxation problem (2) for the multiple-action setting by incorporating an additional incentive compatibility constraint for each receiver, as presented in (12).

$$
\begin{aligned}
\bar{V} = \max_{q(a_i|w)\geq 0} \quad & \sum_{i\in[n]}\sum_{a_i\in A_i}\int_{w\in\Omega} v(a_i)\, q(a_i|w)\, dG(w) \\
\text{s.t.} \quad & \int_{w\in\Omega} u_i(w,a_i)\, q(a_i|w)\, dG(w) \geq 0,\ \forall\, a_i \in A_i,\ i\in[n], \\
& \int_{w\in\Omega} u_i(w,a_i)\, q(a_i|w)\, dG(w) \geq \int_{w\in\Omega} u_i(w,\tilde{a}_i)\, q(a_i|w)\, dG(w),\ \forall\, a_i,\tilde{a}_i \in B_k,\ k\in\mathbb{N}_+, \\
& \sum_{i\in[n]}\sum_{a_i\in A_i} q(a_i|w) \leq 1,\ \forall\, w\in\Omega.
\end{aligned}
\tag{12}
$$

Analogous to (2), in (12), a planner allocates a good with characteristics w to receiver i under acceptance option a_i with probability $q(a_i|w)$. The planner maximizes the sender's expected payoff while ensuring nonnegative expected utility for each action and satisfying a new incentive compatibility constraints for each receiver, as captured by the second constraint in (12). To interpret this constraint, suppose that $a_i \in B_k$. When the sender recommends action a_i to receiver i, receiver i understands if accepting with option a_i ensures allocation of the good to him, any other acceptance option $\tilde{a}_i \in B_k$ would also guarantee him the allocation. As a result, receiver i will select the action within B_k that provides him with the highest expected utility based on his posterior belief about the good's characteristics. Therefore, the second constraint in (12) represents the incentive compatibility condition that receiver i follows the sender's recommendation, which is without loss of optimality due to the revelation principle.

Analogous to Lemma 3.1, problem (12) is a relaxation to the sender's information design problem, thereby providing an upper bound on the sender's optimal expected payoff V^* in the multiple-action setting, as we formally state in Lemma A.1.

Lemma A.1 (Extension of Lemma 3.1)**.** *Problem (12) provides an upper bound on the sender's optimal expected payoff V^* in the multiple-action setting, regardless of how receivers communicate.*

The proof closely follows that of Lemma 3.1 in Section A.1; hence, we omit the details here. Intuitively, for any disclosure mechanism $f(\cdot|w)$, define $q(a_i|w)$ as the ex-ante probability that the good is allocated to receiver i under acceptance option a_i, given characteristic w and equilibrium strategies induced by $f(\cdot|w)$. We can show that these probabilities $\{q(a_i|w)\}$ are feasible to (12) and yield an objective value not exceeding $\bar{V}$.

A.4.3 Public Persuasion Remains Optimal

In this section, we demonstrate that an optimal solution to (12) can be implemented through a public persuasion mechanism, whose expected payoff attains the upper bound value $\bar{V}$. Therefore, public persuasion remains optimal.

Specifically, let $\{q^*(a|w)\}_{a\in\bigcup_{i\in[n]} A_i}$ denote an optimal solution to (12). Consider a public persuasion mechanism $f^*(\cdot|w)$ with signal space $S_i = S \triangleq \bigcup_{i\in[n]} A_i \cup \{\varnothing\}$ for all receivers $i \in [n]$. When the good's characteristics are w, the mechanism broadcasts the signal $s = a$ to all receivers with probability $q^*(a|w)$ for any $a \in \bigcup_{i\in[n]} A_i$ and the signal $s = \varnothing$ to all receivers with probability $1 - \sum_{i\in[n]} \sum_{a\in A_i} q^*(a|w)$. We can interpret signal $s = a$ as a recommendation to accept the good under acceptance option a and signal $s = \varnothing$ as a recommendation for none of the receivers to extend any acceptance offer. Let $v(\varnothing) = 0$ denote the sender's utility when the good is unallocated. For any recommendation $a \in \bigcup_{i\in[n]} A_i \cup \{\varnothing\}$, let:

$$a_j(a) \triangleq \arg\max\{v(\tilde{a}) : \tilde{a} \in A_j \cup \{\varnothing\}, v(\tilde{a}) \le v(a)\}$$

represent the strongest competing offer receiver j can extend without exceeding the sender's utility for option a.[20] Note that $a_j(\varnothing) = \varnothing$ for all $j \in [n]$. Additionally, when $a = a_i \in A_i$ and each receiver j extends offer $a_j(a_i)$, only receiver i's offer a_i is accepted. Nevertheless, the competing offers from other receivers help sustain an equilibrium under the public persuasion mechanism $f^*(\cdot|w)$, whose performance achieves the first-best upper bound $\bar{V}$, as we demonstrate in Theorem A.2.

Theorem A.2 (Extension of Theorem 3.2). *Under the public persuasion mechanism $f^*(\cdot|w)$, it is an equilibrium for each receiver $j \in [n]$ to extend an offer $a_j(a)$ when he receives the signal $s = a$. Moreover, the expected payoff of the mechanism $f^*(\cdot|w)$ attains the upper bound value $\bar{V}$.*

Proof. The proof is analogous to that of Theorem 3.2 presented in Section A.2. Let $\{q^*(a|w)\}$ denote an optimal solution to (12). Analogous to the proof in Section A.2, for any two acceptance options $a \in B_k$ and $\tilde{a} \in B_\ell$ from different groups $k < \ell$, we have:

$$\int_{w\in\Omega} u_{i(B_k)}(w, a)\, q^*(\tilde{a}|w)\, dG(w) < 0\,. \tag{13}$$

Otherwise, we could reallocate the probability mass $\int_{w\in\Omega} q^*(\tilde{a}|w)\, dG(w)$ to actions within group B_k to obtain a new allocation that is feasible to (12). Since the sender prefers any option in B_k over option $\tilde{a}$, the new allocation brings a strictly higher objective value than $\{q^*(a|w)\}$, contradicting the optimality of $\{q^*(a|w)\}$. Using the same argument, we can also show that for any receiver i and any action $a \in A_i$, we have:

$$\int_{w\in\Omega} u_i(w, a)\, q^*(\varnothing|w)\, dG(w) < 0\,, \tag{14}$$

where $q^*(\varnothing|w) \triangleq 1 - \sum_{i\in[n]} \sum_{a\in A_i} q^*(a|w)$ denotes the probability of receiving signal $s = \varnothing$.

Since a public persuasion mechanism leaves no payoff-related information for the receivers to communicate, receivers make decisions based only on the public signal. We now show that it is an equilibrium for each receiver $i \in [n]$ to extend an offer $a_i(a)$ upon receiving the signal $s = a$. To do so, suppose all receivers except receiver i follow this strategy; we verify that it is optimal for receiver i to adopt the same strategy.

First, suppose receiver i receives the signal $s = \varnothing$. By (14), extending any acceptance offer $a_i \in A_i$ yields a negative expected payoff. Therefore, it is optimal for receiver i to select $a_i(\varnothing) = \varnothing$, i.e., reject the good.

Second, suppose receiver i receives the signal $s = a$ with $a = a_j \in A_j$ for some $j \neq i$; that is, the sender recommends another receiver j to extend an offer a_j. If receiver i extends a stronger offer a_i

[20] The functions $a_j(a)$ are well-defined since the offer values are distinct and positive, and $v(\varnothing) = 0$.

with $v(a_i) > v(a_j)$, the good will be allocated to him with certainty given other receivers' strategies. However, doing so only brings him a negative expected payoff by (13). Therefore, receiver i is better off not receiving the good and is indifferent about extending the weaker offer $a_i(a)$, as this offer will not be accepted given receiver j's strategy.

Finally, suppose receiver i receives the signal $s = a$ with $a = a_i \in A_i$; that is, the sender recommends that receiver i extend the acceptance offer a_i. We now show that following the sender's recommendation yields the highest expected payoff for receiver i. To see this, assume that $a_i \in B_k$ for some subset B_k. The argument proceeds as follows:

- First, the participation constraints in (12) ensure that action a_i yields a nonnegative expected payoff to receiver i. Moreover, the incentive compatibility constraints in (12) guarantee that action a_i provides receiver i with a weakly higher expected payoff than any other action in the set B_k.
- Second, if receiver i extends a stronger offer $\tilde{a} \in B_\ell$ for some $\ell < k$, he will receive the good with certainty. However, according to (13), this action yields a negative expected payoff. Therefore, receiver i is better off not choosing any action $\tilde{a} \in B_\ell$ for $\ell < k$.
- Finally, receiver i has no incentive to take any weaker action $\tilde{a} \in B_\ell$ with $\ell > k$: given that other receivers extending slightly weaker competing offers $a_j(a)$, doing so would cause receiver i to lose the good (because at least one of those competing offers will be more attractive to the sender than any offer in B_ℓ).

Consequently, the optimal strategy for receiver i is to follow the recommended action $a_i(a) = a_i$.

Therefore, it constitutes an equilibrium under the public persuasion $f^*(\cdot|w)$ when each receiver $j \in [n]$ extends an offer $a_j(a)$ upon receiving the signal $s = a$. Moreover, the sender's expected payoff under this equilibrium is $\bar{V}$, as $\{q^*(a|w)\}$ precisely correspond to the equilibrium allocation probabilities. $\square$

B Proofs and Additional Details for Sections 4 and 5

B.1 Preliminary Properties of Optimal Persuasion Mechanisms

In this section, we describe several properties of an optimal solution to (5). First, Proposition B.1 shows that for any feasible solution to (5), the probability of allocating a good (prior to observing w) is maximized when the sender exclusively targets the most accessible receiver n.

Proposition B.1. *Define $z_n \triangleq \min\{z \geq 0 : \mathbb{E}[w|w \geq z] \geq \alpha_n\}$, where α_n is receiver n's threshold value. For any feasible solution $\{q(i|w)\}$ of (5), we have $\sum_{i\in[n]} \int_0^1 q(i|w)g(w)dw \leq \mathbb{P}(w \geq z_n)$, with equality attained when the sender exclusively targets receiver n; that is, $q(n|w) = 1$ for any $w \geq z_n$, and $q(i|w) = 0$ for any $i \neq n$ or $w < z_n$.*

Proof. Since the threshold value α_i is strictly decreasing in the receiver index i by Assumption 4.3, the probability of receiving an offer, given by $\sum_{i\in[n]} \int_0^1 q(i|w)g(w)dw$, is maximized when the sender targets only receiver n, who has the lowest threshold α_n. That is, $q(i|w) = 0$ for all $i \neq n$ and $w \in [0, 1]$.

To see why, given any feasible solution $\{q(i|w)\}$ to (5), we can construct a new solution $\{\tilde{q}(i|w)\}$ by setting $\tilde{q}(n|w) = \sum_{i\in[n]} q(i|w)$ and $\tilde{q}(i|w) = 0$ for all $i < n$. Note that $\{\tilde{q}(i|w)\}$ is feasible to (5) and achieves the same acceptance probability. Moreover, if the original solution $\{q(i|w)\}$ assigns a positive probability to any receiver $i < n$, the participation constraint of receiver n will be loose under the new solution $\{\tilde{q}(i|w)\}$, which allows for further allocation of probability mass to receiver n without violating his participation constraint.

On the other hand, if the sender targets only receiver n, the acceptance probability is maximized with $q(n|w) = 1$ for all $w \geq z_n$ and $q(n|w) = 0$ otherwise, resulting in an acceptance probability of $\mathbb{P}(w \geq z_n)$. □

Second, Proposition B.2 shows that any optimal solution exhibits a cutoff structure. Specifically, there exists a threshold value $z \in [0, 1]$ such that a good is allocated if and only if its characteristics w exceeds z.

Proposition B.2. *Any optimal solution has a cutoff structure. That is, for any optimal solution $\{q^*(i|w)\}$ to (5), there exists a threshold value $z \in [0, 1]$ such that $\sum_{i\in[n]} \int_z^1 q^*(i|w)g(w)dw = \mathbb{P}(w \geq z)$ and $\sum_{i\in[n]} \int_0^z q^*(i|w)g(w)dw = 0$.*

Proof. Let $\{q(i|w)\}$ be a feasible solution of (5), and define $z \triangleq \sup\left\{z \in [0, 1] : \sum_{i\in[n]} \int_0^z q(i|w)g(w)dw = 0\right\}$ as the lower bound on the support of $\{q(i|w)\}$. If $\sum_{i\in[n]} \int_z^1 q(i|w)\, g(w)\, dw < \mathbb{P}(w \geq z)$, there exists a point $\tilde{z} \in (z, 1)$ satisfying:

$$\sum_{i\in[n]} \int_z^{\tilde{z}} q(i|w)\, g(w)\, dw = \int_{\tilde{z}}^1 \left(1 - \sum_{i\in[n]} q(i|w)\right) g(w)\, dw > 0.$$

We can create a new feasible solution $\{\tilde{q}(i|w)\}$ from $\{q(i|w)\}$ by transporting the mass of $\{q(i|w)\}$ from below $\tilde{z}$ to fill the "unoccupied" region above $\tilde{z}$; therefore, $\sum_{i\in[n]} \int_{\tilde{z}}^1 \tilde{q}(i|w)g(w)dw = \mathbb{P}(w \geq \tilde{z})$ and $\sum_{i\in[n]} \int_0^{\tilde{z}} \tilde{q}(i|w)g(w)dw = 0$. The two feasible solutions $\{\tilde{q}(i|w)\}$ and $\{q(i|w)\}$ have the same objective value because, by transporting, $\int_0^1 q(i|w)g(w)dw = \int_0^1 \tilde{q}(i|w)g(w)dw$ for any $i \in [n]$.

On the other hand, since $\{q(i|w)\}$ satisfies the participation constraints and we have shifted a positive mass of $\{q(i|w)\}$ from below $\tilde{z}$ to above $\tilde{z}$, the participation constraint for some receiver $i \in [n]$ must hold with strict inequality with $\{\tilde{q}(i|w)\}$. Given that $\tilde{z} > z \geq 0$, we can allocate some unallocated mass $w \in [0, \tilde{z})$ to this receiver without violating his participation constraint, thereby strictly increasing the sender's payoff. □

Finally, let $\{q^*(i|w)\}$ be an optimal solution to (5), and let $q_i^* \triangleq \int_0^1 q^*(i|w)g(w)dw$ denote the ex-ante probability that receiver i obtains the good. Without loss of generality, assume $q_i^* > 0$ for all $i \in [n]$, as receivers with $q_i^* = 0$ can be disregarded from consideration. Proposition B.3 shows that, for any optimal solution $\{q^*(i|w)\}$ to (5), the participation constraints for the first $n-1$ receivers always bind. Receiver n's participation constraint need not bind in general, but it must bind when $\sum_{i\in[n]} q_i^* < 1$.

Proposition B.3. *Let $\{q^*(i|w)\}$ be an optimal solution to (5). Define $q_i^* \triangleq \int_0^1 q^*(i|w)g(w)\, dw$ and assume $q_i^* > 0$ for all $i \in [n]$. The participation constraints for the first $n-1$ receivers always bind. Additionally, receiver n's participation constraint binds if $\sum_{i\in[n]} q_i^* < 1$.*

Proof. We first assume $\sum_{i\in[n]} q_i^* < 1$ and demonstrate that all receivers' participation constraints bind. Suppose instead that receiver j's participation constraint holds with strict inequality:

$$\int_0^1 w \cdot q^*(j|w)\, g(w)\, dw > \alpha_j \int_0^1 q^*(j|w)\, g(w)\, dw.$$

Then, since there is unallocated probability mass (as $\sum_{i\in[n]} q_i^* < 1$), we can allocate some of this mass to receiver j until his participation constraint binds, thereby strictly increasing the sender's expected payoff. This contradicts optimality.

We next assume $\sum_{i\in[n]} q_i^* = 1$ and show that the participation constraints of the first $n-1$ receivers must bind. If not, suppose the participation constraint for receiver $j \le n-1$ holds with strict inequality. Then, reallocating mass from receiver n to receiver j would strictly increase the sender's expected payoff, contradicting optimality. □

B.2 Proof of Lemma 4.2

Since the thresholds α_i are smaller than one by Assumption 4.3, it is straightforward to create a feasible solution to (5) where all participation constraints in (5) are satisfied with strict inequality. As a result, strong duality holds and an optimal dual variable $\boldsymbol{\mu}^*$ exists according to Theorem 1 in Section 8.6 of Luenberger (1997). Once strong duality is established, Bullet 2 follows directly from the optimality conditions (see Proposition 6.1.5 in Bertsekas 2016).

B.3 Proof of Theorem 4.3

In the proof, let $\boldsymbol{\mu}^*$ be an optimal dual variable for (5), and let $h(w;\boldsymbol{\mu}^*)$ denote the corresponding upper envelope function.

B.3.1 Preparation

As preparation for proving Theorem 4.3, we first show that if there exists an optimal solution $\{q^*(i|w)\}$ to (5) under which receiver j is considered by the sender—that is, $q_j^* \triangleq \int_0^1 q^*(j|w)\, g(w)\, dw > 0$—then the point (α_j, v_j) lies on the upper envelope function $h(w;\boldsymbol{\mu}^*)$ and is strictly within the linear segment containing receiver j. Furthermore, this segment coincides with receiver j's line $\ell_j(w;\mu_j^*)$. We formally state these results in Propositions B.4 and B.5, which respectively examine the cases of $\mu_j^* > 0$ and $\mu_j^* = 0$.

Proposition B.4. *Suppose $\mu_j^* > 0$. If there exists an optimal solution $\{q^*(i|w)\}$ to (5) such that $\int_0^1 q^*(j|w)\, g(w)\, dw > 0$, then the point (α_j, v_j) lies strictly within a linear segment of $h(w;\boldsymbol{\mu}^*)$, and this segment coincides with receiver j's line $\ell_j(w;\mu_j^*)$.*

Proof. Let $\{q^*(i|w)\}$ be an optimal solution to (5) with $\int_0^1 q^*(j|w)\, g(w)\, dw > 0$. Since $\{q^*(i|w)\}$ is an optimal solution to $V^{\mathrm{LR}}(\boldsymbol{\mu}^*)$ by Bullet 2 of Lemma 4.2, receiver j's line, $\ell_j(w;\mu_j^*)$, is a component of the envelope function $\bar{h}(w;\boldsymbol{\mu}^*)$ according to (7). Consequently, line $\ell_j(w;\mu_j^*)$ coincides with a linear segment i of $h(w;\boldsymbol{\mu}^*)$ on some interval $[z_i, z_{i-1}]$.

We now show that $\alpha_j \in (z_i, z_{i-1})$. Since the envelope function $h(w;\boldsymbol{\mu}^*)$ is convex, line $\ell_j(w;\mu_j^*)$ lies strictly below $h(w;\boldsymbol{\mu}^*)$ for all $w \in [0, z_i) \cup (z_{i-1}, 1]$. Therefore, by (7), we have $q^*(j|w) = 0$ for all $w \in [0, z_i) \cup (z_{i-1}, 1]$. On the other hand, since $\mu_j^* > 0$, receiver j's participation constraint binds under $\{q^*(i|w)\}$ by Bullet 2 of Lemma 4.2. Therefore, it must hold that $\alpha_j \in (z_i, z_{i-1})$. Otherwise, receiver j's participation constraint would not bind.

Finally, since line $\ell_j(w;\mu_j^*)$ passes through point (α_j, v_j) and coincides with segment i of $h(w;\boldsymbol{\mu}^*)$ on $w \in [z_i, z_{i-1}]$, point (α_j, v_j) lies strictly within segment i given that $\alpha_j \in (z_i, z_{i-1})$. □

Proposition B.5. *Suppose $\mu_j^* = 0$. If there exists an optimal solution $\{q^*(i|w)\}$ to (5) such that $\int_0^1 q^*(j|w)\, g(w)\, dw > 0$, then the point (α_j, v_j) lies strictly within a linear segment of $h(w;\boldsymbol{\mu}^*)$, which must be the first segment with range $[0, b]$ for some $b > 0$. Moreover, this segment coincides with receiver j's line $\ell_j(w;\mu_j^* = 0) = v_j$.*

Proof. Since $\mu_j^* = 0$, receiver j's line $\ell_j(w; \mu_j^* = 0) = v_j$ is constant (horizontal).

Let $\{q^*(i|w)\}$ be an optimal solution to (5) such that $\int_0^1 q^*(j|w)\, g(w)\, dw > 0$. By Bullet 2 of Lemma 4.2, $\{q^*(i|w)\}$ is also optimal to $V^{\text{LR}}(\boldsymbol{\mu}^*)$. Therefore, from (7), line $\ell_j(w; \mu_j^*)$ is part of the envelope function $\bar{h}(w; \boldsymbol{\mu}^*)$, and thus coincides with a linear segment of $h(w; \boldsymbol{\mu}^*)$. On the other hand, since $h(w; \boldsymbol{\mu}^*)$ is increasing and convex and line $\ell_j(w; \mu_j^*)$ is horizontal, line $\ell_j(w; \mu_j^*)$ must coincide with the first segment of $h(w; \boldsymbol{\mu}^*)$ on $[0, b]$ for some $b > 0$.

We now show that $\alpha_j < b$. Since the envelope function $h(w; \boldsymbol{\mu}^*)$ is convex, line $\ell_j(w; \mu_j^*)$ lies strictly below $h(w; \boldsymbol{\mu}^*)$ for any $w > b$. Therefore, by (7), we have $q^*(j|w) = 0$ for any $w > b$. On the other hand, receiver j's participation constraint must be satisfied under $\{q^*(i|w)\}$ by Bullet 2 of Lemma 4.2. Hence, we must have $\alpha_j < b$. Otherwise, receiver j's participation constraint would not hold.

Finally, point (α_j, v_j) lies strictly within the first segment of $h(w; \boldsymbol{\mu}^*)$, because $\alpha_j < b$ and line $\ell_j(w; \mu_j^*)$ passes through this point and coincides with the first segment of $h(w; \boldsymbol{\mu}^*)$ on $w \in [0, b]$. $\square$

B.3.2 Proof of Bullet Three

This result follows immediately from Propositions B.4 and B.5.

B.3.3 Proof of Bullets One and Two: Zero Slope Case

Suppose that $h(w; \boldsymbol{\mu}^*)$ is constant (i.e., has zero slope) on its first segment, with range $[0, b]$ for some $b > 0$. Let T_1 denote the set of receivers whose points (α_j, v_j) lie strictly within this segment. Since on this segment, $h(w; \boldsymbol{\mu}^*) = v_j$ for any $j \in T_1$, set T_1 contains only one receiver by Assumption 2.1.

Let j be the only receiver in set T_1. Since line $\ell_j(w; \mu_j^*)$ supports the envelope function $h(w; \boldsymbol{\mu}^*)$ on the interval $w \in [0, b]$, we must have $\mu_j^* = 0$. The lines corresponding to the other receivers have positive slopes (since they correspond to other segments) and lie strictly below $h(w; \boldsymbol{\mu}^*)$ on $[0, b]$ (since $h(w; \boldsymbol{\mu}^*)$ is convex). Therefore, by Bullet 2 of Lemma 4.2 and (7), any optimal solution $\{q^*(j|w)\}$ allocates goods with $w \in [0, b]$ exclusively to receiver j, and receiver j's participation constraint must hold.

B.3.4 Proof of Bullets One and Two: Positive Slope Case

Suppose segment i of $h(w; \boldsymbol{\mu}^*)$ has a positive slope. Let $[z_i, z_{i-1}] \subseteq [0, 1]$ denote its range, and let $T_i \subseteq [n]$ be the set of receivers whose points (α_j, v_j) lie on segment i.

Proposition B.4 implies that the lines $\ell_j(w; \mu_j^*)$ of all receivers $j \in T_i$ coincide with segment i. In contrast, the lines corresponding to the other receivers have different slopes (since they align with other segments) and lie strictly below $h(w; \boldsymbol{\mu}^*)$ on $[z_i, z_{i-1}]$ (since $h(w; \boldsymbol{\mu}^*)$ is convex). Therefore, by Bullet 2 of Lemma 4.2 and (7), any optimal solution $\{q^*(i|w)\}$ to (5) allocates goods with $w \in [z_i, z_{i-1}]$ exclusively to receivers in T_i, and the participation constraints bind for all receivers in T_i.

B.4 Proof of Proposition 5.1

Step One: Proving $\bar{V} \le V^{\text{CR}}$ We first prove that (9) is a relaxation of (5); therefore, $\bar{V} \le V^{\text{CR}}$. Specifically, let $\{q(i|w)\}$ be a feasible solution to (5). Define $q_i = \int_0^1 q(i|w) g(w) dw$ for any $i \in [n]$. We show that $\{q_i\}$ is feasible to (9). This, together with the fact that $\{q(i|w)\}$ and $\{q_i\}$ yield the same objective value, indicates that (9) is a relaxation of (5).

To show that $\{q_i\}$ is feasible to (9), first, note that $q_i \geq 0$ for any $i \in [n]$ because $q(i|w) \geq 0$ for any $i \in [n]$ and $w \in [0,1]$. Second,

$$\sum_{i\in[n]} q_i = \sum_{i\in[n]} \int_0^1 q(i|w)\, g(w)\, dw \leq \int_0^1 g(w)\, dw = 1,$$

where the inequality follows from the fact that $\sum_{i\in[n]} q(i|w) \leq 1$ for any $w \in [0,1]$.

Finally, we show $\{q_i\}$ is feasible to the first constraint in (9). To do so, let $q_{\leq k}(w) = \sum_{i\leq k} q(i|w)$ denote the probability that a good with characteristics w receives an offer from one of the top k receivers. Since $\{q(i|w)\}$ is a feasible solution to (5),

$$\alpha_i \int_0^1 q(i|w)\, g(w)\, dw \leq \int_0^1 w \cdot q(i|w)\, g(w)\, dw.$$

Summing over $i \leq k$ on both sides gives

$$\begin{aligned}
\sum_{i\leq k} \alpha_i q_i \leq & \int_0^1 w \cdot q_{\leq k}(w)\, g(w)\, dw \\
\leq & \int_{G^{-1}\left(1-\sum_{i\leq k} q_i\right)}^1 w \cdot g(w)\, dw \\
= & \mathbb{E}\left[w \cdot \mathbb{1}\left[G(w) \geq 1 - \sum_{i\leq k} q_i \right] \right] \\
= & \sum_{i\leq k} q_i \cdot \mathbb{E}\left[w \,\middle|\, G(w) \geq 1 - \sum_{i\leq k} q_i \right]
\end{aligned}$$

where the second inequality follows from the fact that $\int_0^1 q_{\leq k}(w)\, g(w)\, dw = \sum_{i\leq k} q_i$, and that the integration is maximized by taking $q_{\leq k}(w) = 1$ for all $w \geq G^{-1}\left(1 - \sum_{i\leq k} q_i\right)$ and $q_{\leq k}(w) = 0$ otherwise.

Step Two: Proving $V^{\mathrm{CR}} \leq \bar{V}$ We next prove that $V^{\mathrm{CR}} \leq \bar{V}$. Specifically, we show that for any feasible solution $\{q_i\}$ to (9), there exists a feasible solution $\{q(i|w)\}$ to (5) with the same objective value as $\{q_i\}$, thereby implying $V^{\mathrm{CR}} \leq \bar{V}$.

Let $\{q_i\}$ be feasible to (9). Since the participation condition (i.e., the first constraint) of (9) holds for $k = 1$, we can find a portion q_1 of goods whose mean quality just meets the threshold value α_1 of receiver 1. In other words, we can find a function $q(1|w) \geq 0$ satisfying:

$$\begin{aligned}
&\int_0^1 q(1|w)\, g(w)\, dw = q_1, \\
&\int_0^1 w \cdot q(1|w)\, g(w)\, dw = \alpha_1 \int_0^1 q(1|w)\, g(w)\, dw.
\end{aligned}$$

Now consider the remaining portion of goods. Since the participation condition of (9) holds for $k = 2$, within the remaining portion of goods, we can find a portion q_2 of goods whose mean quality just meets the threshold value α_2 of receiver 2. In other words, we can find a function $q(2|w) \geq 0$

satisfying:

$$\int_0^1 q(2|w)\,g(w)\,dw = q_2,$$
$$\int_0^1 w \cdot q(2|w)\,g(w)\,dw = \alpha_2 \int_0^1 q(2|w)\,g(w)\,dw,$$
$$q(2|w) \le 1 - q(1|w),\ \forall\, w \in [0,1].$$

Repeating the process, we can find qualified portions for all receivers, resulting in a set of $\{q(i|w)\}$ that is feasible to (5). Moreover, by construction, $\{q(i|w)\}$ and $\{q_i\}$ have the same objective value.

Step Three: Wrap-Up Combining the two steps, we have $\bar{V} = V^{\mathrm{CR}}$; that is, the optimal values of (5) and (9) are equal. Moreover, let $\{q^*(i|w)\}$ be an optimal solution to (5), and let $q_i^* = \int_0^1 q^*(i|w)g(w)dw$. Since $\{q_i^*\}$ is feasible to (9) and attains the same objective value as $\{q^*(i|w)\}$ by Step One, $\{q_i^*\}$ is optimal to (9). Conversely, if $\{q_i^*\}$ is an optimal solution to (9), then by Step Two, we can construct a feasible solution $\{q^*(i|w)\}$ to (5) satisfying $q_i^* = \int_0^1 q^*(i|w)g(w)dw$. This solution has an objective value $V^{\mathrm{CR}} = \bar{V}$, thus is optimal to (5).

B.5 Proof of Proposition 5.2

In this section, we prove that if $\boldsymbol{\lambda}^* = (\lambda_k^*)_{k\in[n]}$ is an optimal Lagrangian dual variable for (9), then $\{\mu_i\}$, with $\mu_i = \sum_{k\ge i} \lambda_k^*$, is an optimal Lagrangian dual variable for (5). Given these dual variables $\{\mu_i\}$, we further characterize the receivers' lines $\ell_j(w;\mu_j)$ and the upper envelope function $h(w;\boldsymbol{\mu})$. To achieve this, we first derive the optimality conditions for (9) in Section B.5.1.

B.5.1 Optimality Conditions for (9)

Let $\mathbf{q} = (q_i)_{i\in[n]} \in \mathbb{R}_+^n$ be a vector of allocation probabilities for the n receivers, and let $L(\mathbf{q},\boldsymbol{\lambda},\gamma)$ represent the Lagrangian function of (9) obtained by dualizing the participation constraints with dual variables $\boldsymbol{\lambda} = (\lambda_k)_{k\in[n]} \in \mathbb{R}_+^n$ and the constraint $\sum_{i\in[n]} q_i \le 1$ with dual variable $\gamma \ge 0$:

$$L(\mathbf{q},\boldsymbol{\lambda},\gamma) = \sum_{i=1}^n v_i q_i + \sum_{k\in[n]} \lambda_k \left(\int_{1-\sum_{i\le k} q_i}^1 G^{-1}(x)\,dx - \sum_{i\le k} \alpha_i q_i \right) + \gamma \left(1 - \sum_{i\in[n]} q_i \right).$$

Denote by $\mathbf{q}^* = (q_i^*)_{i\in[n]} \in \mathbb{R}_+^n$ an optimal solution to (9), and $\boldsymbol{\lambda}^* = (\lambda_k^*)_{k\in[n]} \in \mathbb{R}_+^n$ and $\gamma^* \ge 0$ optimal dual variables to (9). By the KKT conditions, the vector $\mathbf{q}^*$ solves the following Lagrangian problem:

$$\mathbf{q}^* \in \underset{\mathbf{q}\in\mathbb{R}_+^n,\ \sum_{i\in[n]} q_i \le 1}{\operatorname{argmax}} L(\mathbf{q},\boldsymbol{\lambda}^*,\gamma^*)\,.$$

Since $q_i^* > 0$ for all $i \in [n]$ (i.e., we consider only non-disregarded receivers), the first-order optimality conditions yield:

$$\frac{\partial L}{\partial q_i}(\mathbf{q}^*,\boldsymbol{\lambda}^*,\gamma^*) = v_i - \gamma^* + \sum_{k\ge i} \lambda_k^* \left(G^{-1}\left(1 - \sum_{j\le k} q_j^*\right) - \alpha_i \right) = 0, \quad \forall\, i \in [n]\,. \tag{15}$$

Finally, Proposition B.6 presents several preliminary properties of the optimal dual variables associated with (9).

Proposition B.6. *Any optimal solution* $\mathbf{q}^* \in \mathbb{R}_+^n$ *and optimal dual variables* $\boldsymbol{\lambda}^* \in \mathbb{R}_+^n$ *and* $\gamma^* \geq 0$ *of (9) satisfy the following properties:*

1. *The optimal dual variable* γ^* *satisfies* $\gamma^* \leq v_n$*;*
2. *If* $\sum_{i\in[n]} q_i^* < 1$*, then we have* $\lambda_n^* > 0$ *and* $\gamma^* = 0$*;*
3. *If* $\lambda_n^* = 0$*, then we have* $\sum_{i\in[n]} q_i^* = 1$ *and* $\gamma^* = v_n$*.*

Proof. Proof of Bullet One: If $\sum_{i\in[n]} q_i^* < 1$, then we have $\gamma^* = 0$ by complementary slackness. Otherwise, suppose $\sum_{i\in[n]} q_i^* = 1$. Then, (15) with $i = n$ implies that

$$\gamma^* = v_n - \lambda_n^* \alpha_n \leq v_n,$$

where the equality follows from $G^{-1}(0) = 0$ and the inequality from $\lambda_n^* \geq 0$.

Proof of Bullet Two: If $\sum_{i\in[n]} q_i^* < 1$, complementary slackness implies $\gamma^* = 0$. Additionally, setting $i = n$ in (15) yields:

$$\lambda_n^* \left(\alpha_n - G^{-1} \left(1 - \sum_{j\in[n]} q_j^* \right) \right) = v_n > 0\,.$$

We remark that $G^{-1}\left(1 - \sum_{j\in[n]} q_j^*\right) < \alpha_n$. To see this, note that any optimal persuasion mechanism exhibits a cutoff structure, such that a good is allocated if and only if its characteristics w exceeds a threshold value $z \in [0,1]$ (see Proposition B.2), and $G^{-1}\left(1 - \sum_{j\in[n]} q_j^*\right)$ corresponds to this threshold. Therefore, we must have $G^{-1}\left(1 - \sum_{j\in[n]} q_j^*\right) < \alpha_n$, because, otherwise, receiver n's participation constraint does not bind, allowing the sender to allocate some of the remaining unassigned mass to receiver n and thereby strictly increase the sender's expected payoff.[21] Consequently, the above equality implies that $\lambda_n^* > 0$.

Proof of Bullet Three: If $\lambda_n^* = 0$, we have $\sum_{i\in[n]} q_i^* = 1$ by Bullet 2. Additionally, setting $i = n$ in (15) gives $\gamma^* = v_n$. □

In what follows, we first prove Proposition 5.2 under the assumption that $\sum_{j\in[n]} q_j^* < 1$. The proofs for the remaining cases follow a similar argument.

B.5.2 Characterization of $h(w;\boldsymbol{\mu})$ when $\sum_{j\in[n]} q_j^* < 1$

Define dual variables $\boldsymbol{\mu} = (\mu_i)_{i\in[n]}$ by $\mu_i = \sum_{k\geq i} \lambda_k^*$ for each $i \in [n]$. In this section, we assume that $\sum_{j\in[n]} q_j^* < 1$ and verify the properties of the receivers' lines $\ell_j(w;\mu_j)$ and the upper envelope function $h(w;\boldsymbol{\mu})$ in Proposition 5.2.

When $\sum_{j\in[n]} q_j^* < 1$, Bullet 2 of Proposition B.6 implies that $\lambda_n^* > 0$ and $\gamma^* = 0$. Therefore, the first-order optimality condition (15) becomes

$$\frac{\partial L}{\partial q_i}\big(\mathbf{q}^*, \boldsymbol{\lambda}^*, \gamma^* = 0\big) = v_i + \sum_{k\geq i} \lambda_k^* \left(G^{-1} \left(1 - \sum_{j\leq k} q_j^* \right) - \alpha_i \right) = 0, \quad \forall\, i \in [n]\,. \tag{16}$$

[21] Such unallocated mass exists because $\sum_{i\in[n]} q_i^* < 1$ by assumption.

In addition, following the notation from Section 5.2, let

$$T \triangleq \left\{k \in [n] : \lambda_k^* > 0\right\}$$

denote the set of indices corresponding to positive entries in the optimal dual variable $\boldsymbol{\lambda}^*$. Since $\lambda_n^* > 0$, we have $n \in T$.

Suppose $T = \{t_1 < t_2 < \cdots < t_m = n\}$ consists of m receivers. These receivers partition the set of n receivers into m groups $\{T_i\}_{i\in[m]}$, where $T_1 = [t_1]$ and $T_i = [t_{i-1} + 1 : t_i]$ for $i \in [2 : m]$. Moreover, each group T_i contains exactly one element from T, which is its largest element.

If $k \in T$, complementary slackness implies that the participation constraint in (9) is binding for the top k receivers; that is,

$$\sum_{i\le k} \alpha_i q_i^* = \mathbb{E}\Big[w \cdot \mathbb{1}\Big[w \ge G^{-1}\Big(1 - \sum_{i\le k} q_i^*\Big)\Big]\Big]. \tag{17}$$

Finally, define $z_i \triangleq G^{-1}\left(1 - \sum_{j=1}^{t_i} q_j^*\right)$ for each $i \in [m]$ and set $z_0 = 1$, and define subinterval $I_i = [z_i, z_{i-1}]$ for each $i \in [m]$.

We now verify the properties of the receivers' lines $\ell_j(w; \mu_j)$ and the upper envelope function $h(w; \boldsymbol{\mu})$ characterized in Proposition 5.2.

Proof of Bullet One For any group T_i and element $k \in T_i$, subtracting both sides of the first constraint in (9) from both sides of (17) with $k = t_{i-1}$, and noting that the first constraint in (9) is binding with $k = t_i$, yields the following:

$$\begin{aligned} \sum_{j\in[t_{i-1}+1:k]} \alpha_j q_j^* &\le \mathbb{E}\Big[w \cdot \mathbb{1}\Big[G^{-1}\Big(1 - \sum_{j\le k} q_j^*\Big) \le w < z_{i-1}\Big]\Big], \forall\, k \in [t_{i-1} + 1 : t_i - 1], \\ \sum_{j\in T_i} \alpha_j q_j^* &= \mathbb{E}\Big[w \cdot \mathbb{1}\Big[z_i \le w < z_{i-1}\Big]\Big]. \end{aligned} \tag{18}$$

Additionally, we have $\mathbb{P}[z_i \le w < z_{i-1}] = \sum_{j\in T_i} q_j^*$.

To verify $\alpha_j \in (z_i, z_{i-1})$ for all $j \in T_i$, fix receiver $k = t_{i-1}+1$. The first inequality in (18) and the positivity of q_k imply $\alpha_k < z_{i-1}$. Otherwise, no goods from the interval $\left[G^{-1}\left(1 - \sum_{j\le k} q_j^*\right), z_{i-1}\right)$ would meet the hiring threshold α_k, contradicting the inequality.

Furthermore, subtracting the equality in (18) from the inequality in (18) evaluated at $k = t_i - 1$ yields:

$$\alpha_{t_i} q_{t_i}^* \ge \mathbb{E}\Big[w \cdot \mathbb{1}\Big[z_i \le w < G^{-1}\Big(1 - \sum_{j\le t_i - 1} q_j^*\Big)\Big]\Big].$$

Positivity of q_{t_i} implies $z_i < \alpha_{t_i}$. Otherwise, the interval $\left[z_i, G^{-1}\left(1 - \sum_{j\le t_i-1} q_j^*\right)\right)$ would contain only goods overqualified for receiver t_i and the inequality above cannot hold. Thus, we conclude that $\alpha_j \in (z_i, z_{i-1})$ for every $j \in T_i$.

We now show that within each group T_i, the lines $\ell_j(w; \mu_j)$ coincide for all $j \in T_i$ and pass through the points (α_j, v_j) for each $j \in T_i$. If the set T_i contains only one receiver, the result trivially holds. Now suppose that T_i contains multiple receivers (i.e., $t_{i-1} + 1 < t_i$). For any

$k \in [t_{i-1}+1 : t_i - 1]$, we have:

$$\mu_k = \sum_{j \geq k} \lambda_j^* = \mu_{t_i} = \frac{v_k - v_{t_i}}{\alpha_k - \alpha_{t_i}} \tag{19}$$

where the first equality follows from the definition of $\{\mu_i\}$ and the second equality follows from the fact that $\lambda_j^* = 0$ for any $j \in [t_{i-1}+1 : t_i - 1]$. The third equality is obtained by subtracting both sides of (15) with $i = t_i$ from both sides of the same equation with $i = k$. (19) implies that the points $\{(\alpha_j, v_j)\}_{j \in T_i}$ lie on a line, and the receivers' lines $\ell_j(w; \mu_j)$ for any $j \in T_i$ fully overlap and coincide with this line.

Proof of Bullet Two For ease of notation, we suppress the explicit dependence on the dual variables $\boldsymbol{\mu} = (\mu_j)_{j \in [n]}$ and denote

$$\ell_j(w) \triangleq \ell_j(w; \mu_j) = v_j + \mu_j(w - \alpha_j).$$

Based on Bullet 1, it suffices to show that: (i) the line $\ell_n(w)$ intersects the x-axis at $w = z_m > 0$, and (ii) for each $i \in [m-1]$, the lines $\ell_{t_i}(w)$ and $\ell_{t_{i+1}}(w)$ intersect at $w = z_i$.

First, from (16) with $i = n$, we obtain:

$$v_n + \lambda_n^*(z_m - \alpha_n) = v_n + \mu_n(z_m - \alpha_n) = 0,$$

where the first equality follows from $\mu_n = \lambda_n^*$ by definition. Thus, line $\ell_n(w)$ intersects the x-axis at $w = z_m$.

We now prove (ii) by induction. To start, note that

$$\mu_{t_i} = \sum_{j=i}^{m} \lambda_{t_j}^*, \ \forall i \in [m] \tag{20}$$

because $\lambda_k^* = 0$ for all $k \notin T$. We first show that (ii) holds for $i = m-1$. Since line $\ell_n(w)$ passes through the point (z_{m-1}, h_{m-1}) with

$$h_{m-1} = \mu_n \cdot (z_{m-1} - z_m), \tag{21}$$

it suffices to show that line $\ell_{t_{m-1}}(w)$ also passes through (z_{m-1}, h_{m-1}). We now verify this. Specifically, taking $i = t_{m-1}$ in (16) yields:

$$v_{t_{m-1}} + \lambda_n^* \cdot (z_m - \alpha_{t_{m-1}}) + \lambda_{t_{m-1}}^* \cdot (z_{m-1} - \alpha_{t_{m-1}}) = v_{t_{m-1}} + \mu_{t_{m-1}} \cdot \left(z_{m-1} - h_{m-1}/\mu_{t_{m-1}} - \alpha_{t_{m-1}}\right) = 0$$

where the first equality follows from (20) and (21). Therefore, it follows that:

$$v_{t_{m-1}} + \mu_{t_{m-1}} \cdot (z_{m-1} - \alpha_{t_{m-1}}) = h_{m-1},$$

implying that line $\ell_{t_{m-1}}(w)$ also passes through the point (z_{m-1}, h_{m-1}).

We now assume that (ii) holds for all $j \geq i+1$ and verify that it also holds for $j = i$. Given that (ii) holds for any $j \geq i+1$, line $\ell_{t_{i+1}}(w)$ passes through the point (z_i, h_i), where

$$h_i = \sum_{j=i}^{m-1} \mu_{j+1} \cdot (z_j - z_{j+1}). \tag{22}$$

To complete the induction step, it suffices to show that line $\ell_{t_i}(w)$ also passes through (z_i, h_i). To do so, take $i = t_i$ in (16); this gives:

$$v_{t_i} + \sum_{j=i}^{m} \lambda^*_{t_j} \cdot \left(z_j - \alpha_{t_i}\right) = v_{t_i} + \mu_{t_i} \cdot \Big(z_i - h_i/\mu_{t_i} - \alpha_{t_i}\Big) = 0,$$

where the first equality follows from (22) and the identity $\lambda^*_{t_j} = \mu_{t_j} - \mu_{t_{j+1}}$ for each $j \in [m]$ (letting $\mu_{t_{m+1}} = 0$) by (20). Consequently, we have:

$$v_{t_i} + \mu_{t_i} \cdot (z_i - \alpha_{t_i}) = h_i,$$

implying that line $\ell_{t_i}(w)$ also passes through the point (z_i, h_i). Therefore, (ii) holds for $j = i$, completing the induction.

Proof of Bullet Three Bullet 3 follows immediately from Bullets 1 and 2 and the fact that the dual variables $\{\mu_i\}$—which represent the slopes of the lines $\ell_i(w; \mu_i)$—decrease with index i due to the nonnegativity of $\{\lambda^*_k\}$. Note that by Bullet 3, the function $h(w; \boldsymbol{\mu})$ is nonnegative if and only if $w \geq z_m$. Therefore, since $\bar{h}(w; \boldsymbol{\mu}) = \max\{h(w; \boldsymbol{\mu}), 0\}$, it follows that $\bar{h}(w; \boldsymbol{\mu}) = h(w; \boldsymbol{\mu})$ for $w \geq z_m$, and $\bar{h}(w; \boldsymbol{\mu}) = 0$ otherwise.

B.5.3 Optimality of Dual Variable $\boldsymbol{\mu}$ when $\sum_{j\in[n]} q^*_j < 1$

In this section, we verify that $V^{\mathrm{LR}}(\boldsymbol{\mu}) = \bar{V}$ for the dual variable $\boldsymbol{\mu}$ defined in Proposition 5.2. Therefore, by the strong duality (Lemma 4.2), $\boldsymbol{\mu}$ is an optimal Lagrangian dual variable of (5).

Using the characterization of the upper envelope function $h(w; \boldsymbol{\mu})$ (see Bullets 2 and 3 in Proposition 5.2) and (7), a set of allocation probabilities $\{q(j|w)\}$ is optimal to $V^{\mathrm{LR}}(\boldsymbol{\mu})$ if and only if it satisfies:

$$\begin{aligned} &\sum_{j\in T_i} q(j|w) = 1, \; \forall\, w \in \big(z_i, z_{i-1}\big),\; i \in [m], \\ &\sum_{j\in[n]} q(j|w) = 0, \; \forall\, w < z_m. \end{aligned} \tag{23}$$

Moreover, by repeating the proof of Proposition 5.1 (specifically, Step Two in Section B.4), (18) implies that there exists an optimal solution $\{q^*(j|w)\}$ to $V^{\mathrm{LR}}(\boldsymbol{\mu})$ such that for any $i \in [m]$ and $j \in T_i$, we have:

$$\begin{aligned} &\int_{w\in I_i} q^*(j|w)\, g(w)\, dw = q^*_j, \\ &\int_{w\in I_i} w \cdot q^*(j|w)\, g(w)\, dw = \alpha_j \int_{w\in I_i} q^*(j|w)\, g(w)\, dw. \end{aligned} \tag{24}$$

We note that from (23), for any receiver $j \in T_i$, $q^*(j|w) = 0$ for any $w \notin I_i$. Together with (24), this implies that for any $j \in [n]$, we have:

$$\int_0^1 q^*(j|w)\, g(w)\, dw = q^*_j, \tag{25}$$

$$\int_0^1 w \cdot q^*(j|w)\, g(w)\, dw = \alpha_j \int_0^1 q^*(j|w)\, g(w)\, dw. \tag{26}$$

Therefore,

$$
\begin{aligned}
V^{\mathrm{LR}}(\boldsymbol{\mu}) &= \int_0^1 \sum_{j=1}^n \Big\{ v_j + \mu_j \big(w - \alpha_j\big) \Big\} \, q^*(j|w) \, g(w) \, dw \\
&= \sum_{j=1}^n v_j \int_0^1 q^*(j|w) \, g(w) \, dw \\
&= \sum_{j=1}^n v_j \cdot q_j^* = V^{\mathrm{CR}} = \bar{V},
\end{aligned}
\tag{27}
$$

where the first equality follows from the fact that $\{q^*(j|w)\}$ is optimal to $V^{\mathrm{LR}}(\boldsymbol{\mu})$, the second from (26), the third from (25), the fourth from the optimality of $\{q_j^*\}$ to (9), and the final one from Proposition 5.1.

B.5.4 Case Two: $\lambda_n^* > 0$ and $\sum_{j \in [n]} q_j^* = 1$

When $\lambda_n^* > 0$, we have $n \in T$. Suppose $T = \{t_1 < t_2 < \cdots < t_m = n\}$ consists of m receivers. Since $\sum_{j \in [n]} q_j^* = 1$, it follows that $z_m = G^{-1}\left(1 - \sum_{j=1}^n q_j^*\right) = 0$.

Note that the general first-order optimality condition (15) reduces to the simpler condition (16) if we define the modified offer values $v_i' = v_i - \gamma^*$ for all $i \in [n]$. Since the results in Section B.5.2 are derived based on (16), all properties of the receivers' lines $\ell_j(w; \mu_j)$ and the envelope function $h(w; \boldsymbol{\mu})$ remain valid with the modified values $\{v_i'\}$.

When transforming back from $\{v_i'\}$ to the original offer values $\{v_i\}$, the receivers' lines $\ell_j(w; \mu_j)$ and the envelope function $h(w; \boldsymbol{\mu})$ uniformly shift upward by γ^*. Consequently, the lines $\ell_j(w; \mu_j)$ for all $j \in T_m$ and the envelope function $h(w; \boldsymbol{\mu})$ intersect the y-axis at $\gamma^* \in [0, v_n]$.

Finally, the dual variable $\boldsymbol{\mu}$ constructed in Proposition 5.2 remains an optimal dual variable of (5), following the same proof in Section B.5.3.

B.5.5 Case Three: $\lambda_n^* = 0$

Suppose that Assumption 5.1 holds and $\lambda_n^* = 0$. Then Bullet 3 of Proposition B.6 implies that $\sum_{j \in [n]} q_j^* = 1$ and $\gamma^* = v_n$. In this case, Lemma B.7 shows that $\lambda_{n-1}^* > 0$, implying $n - 1 \in T$.

Lemma B.7. *Suppose Assumption 5.1 holds and $\lambda_n^* = 0$. Then, it follows that $\lambda_{n-1}^* > 0$.*

Proof. Taking $i = n - 1$ in (15) and noting that $\lambda_n^* = 0$ and $\gamma^* = v_n$, we have:

$$
\lambda_{n-1}^* \left(\alpha_{n-1} - G^{-1} \left(1 - \sum_{j \le n-1} q_j^* \right) \right) = v_{n-1} - \gamma^* = v_{n-1} - v_n > 0 \, .
$$

Given that $\lambda_{n-1}^* \ge 0$, it follows that $\lambda_{n-1}^* > 0$ and $\alpha_{n-1} > G^{-1}\left(1 - \sum_{j \le n-1} q_j^*\right)$. □

Suppose $T = \{t_1 < t_2 < \cdots < t_m = n - 1\}$ includes m receivers. These receivers partition the set of n receivers into $m + 1$ groups $\{T_i\}$, where $T_1 = [t_1]$, $T_i = [t_{i-1} + 1 : t_i]$ for all $i \in [2 : m]$, and $T_{m+1} = \{n\}$. Define the endpoints $z_i \triangleq G^{-1}\left(1 - \sum_{j=1}^{t_i} q_j^*\right)$ for each $i \in [m]$, and set $z_0 = 1$ and $z_{m+1} = 0$.

Applying the same approach as in Section B.5.4 – that is, reducing the general first-order optimality condition (15) to (16) by modifying the offer values to $v_i' = v_i - \gamma^*$ for all $i \in [n]$ – we can verify that all properties of the receivers' lines $\ell_j(w; \mu_j)$ and the envelope function $h(w; \boldsymbol{\mu})$ stated in Proposition 5.2 remain valid, with an additional feature that the envelope function $h(w; \boldsymbol{\mu}) = \gamma^* = v_n > 0$ is constant (horizontal) on the interval $[0, z_m]$.

Moreover, the dual variable $\boldsymbol{\mu}$ constructed in Proposition 5.2 continues to be optimal for (5). The proof follows the same argument as in Section B.5.3, with one modification: in (27), the second equality utilizes (26) and the fact that $\mu_n = \lambda_n^* = 0$.

Finally, note that in this case, we have $\mu_n = \lambda_n^* = 0$ and $\mu_i \geq \lambda_{n-1}^* > 0$ for any $i \leq n-1$. Therefore, the participation constraints for the first $n-1$ receivers are binding, whereas receiver n's participation constraint may not bind. This aligns with Proposition B.3.

C Optimal Mechanisms with Two Receivers

In this section, we consider two receivers $i \in \{1, 2\}$, with offer values $v_1 > v_2 > 0$ and hiring thresholds $\alpha_1 > \alpha_2 > 0$. We derive the optimal public persuasion mechanisms based on Bullet 2 of Lemma 4.2.

Define $w_0 \triangleq \mathbb{E}_{w \sim G(w)}[w]$ as the prior mean of the good's characteristic w. We assume that receiver 1 is selective; that is, $\alpha_1 > w_0$. Otherwise, the optimal mechanism is trivial since the sender can allocate the good to receiver 1 without revealing any information. Throughout this section, we also assume that receiver 2 is selective (i.e., $\alpha_2 > w_0$), as formally stated in Assumption C.1. The scenario in which receiver 2 is not selective yields similar optimal persuasion mechanisms but requires a separate discussion, which we provide in Section C.3.

Assumption C.1. Let $w_0 \triangleq \mathbb{E}_{w \sim G(w)}[w]$ denote the prior mean of the good's characteristic w. Both receivers 1 and 2 are selective; that is, their threshold values satisfy $0 < w_0 < \alpha_2 < \alpha_1 < 1$.

Finally, for a persuasion mechanism M, let $q_i(M)$ denote the probability that receiver i is allocated the good under persuasion mechanism M.

C.1 Preparation: Mechanisms Targeting a Single Receiver and Their Interpolation

We begin by analyzing two simple mechanisms—in which the sender prioritizes either receiver 1 or 2—and their interpolation, as preparation for characterizing an optimal mechanism in Section C.2.

Mechanism M_1: Prioritizing Receiver 1 First, we consider mechanism M_1, in which the sender prioritizes receiver 1 and recommends goods to receiver 2 only if suitable goods remain after targeting receiver 1. Specifically, define $\bar{z}_1 > 0$ such that $\mathbb{E}[w | w \geq \bar{z}_1] = \alpha_1$.[22] The sender sends the signal $s = 1$ if $w \geq \bar{z}_1$, resulting in allocation probability $q_1(M_1) = \mathbb{P}[w \geq \bar{z}_1]$. Then, two scenarios arise depending on the value of $\bar{z}_1$ relative to α_2:

- If $\bar{z}_1 > \alpha_2$: The sender can still persuade receiver 2 to extend an offer for some goods in the remaining pool after targeting receiver 1. Specifically, find a real value z_1 with $0 < z_1 < \alpha_2 < \bar{z}_1$ such that $\mathbb{E}[w | z_1 \leq w < \bar{z}_1] = \alpha_2$.[23] The sender sends the signal $s = 2$ if $z_1 \leq w < \bar{z}_1$, and the signal $s = \varnothing$ if $w < z_1$. Therefore, $q_2(M_1) = \mathbb{P}[z_1 \leq w < \bar{z}_1]$.

[22] $\bar{z}_1 > 0$ because $\alpha_1 > w_0$ by Assumption C.1.
[23] $z_1 > 0$ because $\alpha_2 > w_0$ by Assumption C.1.

- If $\bar{z}_1 \leq \alpha_2$: The sender cannot persuade receiver 2 to accept any remaining goods after targeting receiver 1. In this case, set $z_1 = \bar{z}_1$. The sender sends the signal $s = \varnothing$ when $w < z_1$, leading to $q_2(M_1) = 0$.

In both scenarios, the sender receives an offer if and only if $w \geq z_1$, which occurs with probability $\mathbb{P}[w \geq z_1]$.

Mechanism M_2: Prioritizing Receiver 2 Second, consider mechanism M_2, in which the sender exclusively targets receiver 2. Specifically, define $z_2 > 0$ such that $\mathbb{E}[w|w \geq z_2] = \alpha_2$. The sender sends signal $s = 2$ whenever $w \geq z_2$. Upon receiving this signal, only receiver 2 extends an offer. Since $z_2 < \bar{z}_1 < \alpha_1$, the sender can no longer persuade receiver 1 to extend an offer for goods in the remaining pool $[0, z_2)$ after targeting receiver 2. Therefore, the sender can only send the signal $s = \varnothing$ when $w < z_2$. As a result, $q_1(M_2) = 0$ and $q_2(M_2) = \mathbb{P}[w \geq z_2]$. The sender receives an offer if and only if $w \geq z_2$, which occurs with probability $\mathbb{P}[w \geq z_2]$.

Interpolation Between Mechanisms M_1 and M_2 In Proposition B.2 in the Appendix, we demonstrate that *every* optimal solution to (5) exhibits a cutoff structure: There exists a threshold value $z \in [0, 1]$ such that the good is allocated if and only if its quality w exceeds z. Clearly, any persuasion mechanism M with a cutoff value $z > z_1$ is suboptimal because mechanism M_1 yields a higher payoff for the sender. Conversely, the cutoff value z must satisfy $z \geq z_2$; otherwise, the participation constraint of at least one receiver would be violated. In Proposition C.1, we demonstrate that for any $z \in [z_2, z_1]$, there exists a persuasion mechanism with cutoff point z.

Proposition C.1. *For any cutoff $z \in [z_2, z_1]$, there exists a public persuasion mechanism M such that the good receives an offer if and only if $w \geq z$ and the participation constraints of both receivers bind. Moreover, under any such mechanism M, it holds that $q_1(M) = \mathbb{P}[w \geq z] \cdot \frac{\mathbb{E}[w|w \geq z] - \alpha_2}{\alpha_1 - \alpha_2} \geq 0$ and $q_2(M) = \mathbb{P}[w \geq z] \cdot \frac{\alpha_1 - \mathbb{E}[w|w \geq z]}{\alpha_1 - \alpha_2} \geq 0$.*

We prove Proposition C.1 in Section C.4. Intuitively, the probability of allocating the good to receiver 1 is highest when the sender primarily targets receiver 1 (using mechanism M_1). However, this also reduces the overall probability of receiving an offer because receiver 1 has higher acceptance standards. Conversely, the probability of receiving an offer is maximized when the sender exclusively targets the less selective receiver 2 (using mechanism M_2), which equals $\mathbb{P}[w \geq z_2]$. Proposition C.1 shows that any acceptance probability between these two extremes can be sustained by a mechanism that carefully balances the two receivers. In Section C.2, we show that any such mechanism can be optimal, depending on receiver 1's desirability (v_1) and hiring bar (α_1) relative to those of receiver 2. We conclude this section with a remark interpreting the probabilities $q_1(M)$ and $q_2(M)$ in Proposition C.1.

Remark C.1 (Interpreting Probabilities in Proposition C.1)**.** To interpret the probabilities $q_1(M)$ and $q_2(M)$, consider any public persuasion mechanism M characterized by a cutoff structure with threshold z. When the participation constraints of both receivers bind, the probabilities $q_1 \triangleq q_1(M)$ and $q_2 \triangleq q_2(M)$ must satisfy the following two linear equations:

$$\begin{aligned} q_1 + q_2 &= \mathbb{P}[w \geq z], \\ \alpha_1 q_1 + \alpha_2 q_2 &= (q_1 + q_2) \cdot \mathbb{E}[w|w \geq z]. \end{aligned} \tag{28}$$

The first equation follows from the condition that the good receives an offer (from either receiver 1 or 2) if and only if $w \geq z$. The second equation is derived from the binding participation constraints

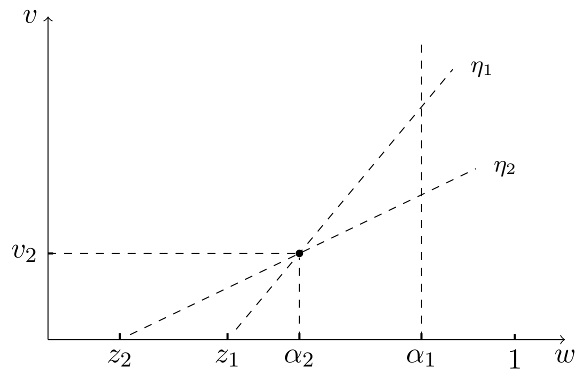


Figure 3: Visualization of the partition in Theorem C.2.

(i.e., $\mathbb{E}[w|s = i] = \alpha_i$) and the law of total expectation. These two equations uniquely determine the values of q_1 and q_2, as given in Proposition C.1. Conversely, consider a mechanism M that sends the signal $s = \varnothing$ if and only if $w < z$. In this case, if $\mathbb{P}[s = 1] = q_1$ and $\mathbb{E}[w \mid s = 1] = \alpha_1$, then $\mathbb{P}[s = 2] = q_2$ and $\mathbb{E}[w \mid s = 2] = \alpha_2$, and vice versa, provided that q_1 and q_2 satisfy the conditions stated in (28).

C.2 Optimal Mechanisms with Two Selective Receivers

In this section, we characterize optimal persuasion mechanisms with two selective receivers (i.e., Assumption C.1 holds). Intuitively, there is a trade-off: An offer from receiver 1 yields a higher payoff, but targeting receiver 1 more aggressively reduces the overall probability of securing an offer.

Notably, in the discussion preceding Proposition C.1, the cutoff value z—such that a good is allocated if and only if its quality $w \geq z$—satisfies $z \in [z_2, z_1]$ for any *reasonable* mechanism. This defines two lines, one of which (denoted by η_1) passes through the points $(z_1, 0)$ and (α_2, v_2), and the other (denoted by η_2) passes through the points $(z_2, 0)$ and (α_2, v_2), as illustrated in Figure 3. These two lines partition the value of $v_1 \in [v_2, \infty)$ into three regions, which determine the form of the optimal persuasion mechanisms.

Theorem C.2 characterizes the optimal persuasion mechanisms for the two-receiver case. Specifically, if the value of v_1 is sufficiently large (in particular, above line η_1), prioritizing receiver 1 is optimal. Alternatively, if v_1 is sufficiently small (i.e., below line η_2), it is optimal to allocate the good exclusively to receiver 2. Finally, if v_1 lies between the values given by the two lines, an optimal mechanism involves a nontrivial balance between the two receivers and exhibits the structure described in Proposition C.1.

Theorem C.2. *Under Assumptions 4.1 – 4.3 and C.1, the optimal public persuasion mechanism for two receivers is characterized as follows:*

1. *If* $v_1 \geq v_2 \cdot \left(\frac{\alpha_1 - z_1}{\alpha_2 - z_1}\right)$ *(i.e., the point* (α_1, v_1) *lies above line* η_1*), mechanism* M_1*, which prioritizes receiver* 1*, is the unique optimal mechanism.*

2. *If* $v_1 \leq v_2 \cdot \left(\frac{\alpha_1 - z_2}{\alpha_2 - z_2}\right)$ *(i.e., the point* (α_1, v_1) *lies below line* η_2*), mechanism* M_2*, which exclusively targets receiver* 2*, is the unique optimal mechanism.*

3. *Otherwise, any mechanism M satisfying Proposition C.1 with the cutoff value*

$$z^* \triangleq \alpha_2 - v_2 \cdot \left(\frac{\alpha_1 - \alpha_2}{v_1 - v_2} \right) \in [z_2, z_1],$$

representing the x-intercept of the line passing through points (α_2, v_2) and (α_1, v_1), is optimal. In other words, mechanism M satisfies the following:

(a) *It sends signal $s = \varnothing$ with probability one if $w < z^*$, and zero otherwise.*

(b) *Participation constraints bind; that is, $\mathbb{E}[w|s = i] = \alpha_i$ for each $i \in \{1, 2\}$.*

(c) *The allocation probabilities are as follows: $q_1(M) = \mathbb{P}[w \geq z^*] \cdot \frac{\mathbb{E}[w|w \geq z^*] - \alpha_2}{\alpha_1 - \alpha_2}$ and $q_2(M) = \mathbb{P}[w \geq z^*] \cdot \frac{\alpha_1 - \mathbb{E}[w|w \geq z^*]}{\alpha_1 - \alpha_2}$, as established in Proposition C.1.*

Moreover, this completely characterizes the set of all optimal public persuasion mechanisms.

We prove Theorem C.2 in Section C.5. In the proof, we identify a set of dual variables $\boldsymbol{\mu} \in \mathbb{R}^n_+$ that, together with the proposed mechanism, satisfy Bullet 2 of Lemma 4.2. This implies that the mechanism is optimal to (5), and that $\boldsymbol{\mu}$ is an optimal dual variable. Finally, in Section C.3, we characterize the optimal mechanisms when Assumption C.1 does not hold; the mechanisms have similar structures.

We note that the trade-off between the two receivers is nontrivial in Case 3 of Theorem C.2. In this scenario, the participation constraints of both receivers bind, and the optimal Lagrangian dual variables are $\mu_1^* = \mu_2^* = \frac{v_1 - v_2}{\alpha_1 - \alpha_2} > 0$. These values correspond to the slope of the line passing through the points (α_2, v_2) and (α_1, v_1). Consequently, the two receivers' lines, $\ell_1(w; \mu_1^*)$ and $\ell_2(w; \mu_2^*)$, completely coincide with this line (as visualized in Figure 6(b)). As a result, according to (7), any allocation $\{q(i|w)\}$ satisfying $q(1|w) + q(2|w) = 1$ for $w \geq z^*$ and $q(1|w) = q(2|w) = 0$ for $w < z^*$ is optimal to the Lagrangian $V^{\text{LR}}(\boldsymbol{\mu}^*)$. Provided that the probability mass of one is appropriately allocated between $q(1|w)$ and $q(2|w)$ for all $w \geq z^*$, ensuring that both receivers' participation constraints bind, it follows that each receiver i is allocated the good with probability q_i^* as specified in Proposition C.1, and that the mechanism $\{q(i|w)\}$ is optimal to (5) by Theorem C.2.

Although the aggregate allocation probabilities $\{q_i^*\}$ are unique by Proposition C.1, there are various ways to construct a set of probabilities $\{q(i|w)\}$ that satisfy Bullet 3 of Theorem C.2 and are thus optimal to (5). Below, we present two simple approaches to construct an optimal mechanism and illustrate them using Example C.1.

- (*Randomized Mechanism with Monotone Structure*) Let $q(1|w) = \frac{q_1^*}{\mathbb{P}[w \geq \bar{z}_1]} \leq 1$ for $w \geq \bar{z}_1$ and $q(1|w) = 0$ otherwise, where $\bar{z}_1 > 0$ satisfies $\mathbb{E}[w|w \geq \bar{z}_1] = \alpha_1$. Additionally, let $q(2|w) = 1 - q(1|w)$ for $w \geq z^*$ and $q(2|w) = 0$ otherwise. This defines a randomized persuasion mechanism that satisfies Bullet 3 of Theorem C.2 and is therefore optimal to (5). By construction, the sender's expected payoff, $v(w) \triangleq \sum_i v_i \, q(i|w)$, is increasing in w, which can be desirable in practice.[24]

[24]For example, in the student promotion context, an increasing expected payoff prevents students from strategically degrading their quality w for better positions.

- (*Deterministic Mechanism with a Double-Interval Structure*) Select an interval $\left[\underline{b}, \bar{b}\right] \subseteq \left[\bar{z}_1, 1\right]$ satisfying $\mathbb{P}\left[\underline{b} \le w \le \bar{b}\right] = q_1^*$ and $\mathbb{E}\left[w \mid \underline{b} \le w \le \bar{b}\right] = \alpha_1$.[25] Let $q(1|w) = 1$ for $w \in \left[\underline{b}, \bar{b}\right]$, $q(2|w) = 1$ for $w \in \left[z^*, \underline{b}\right) \cup \left(\bar{b}, 1\right]$, and $q(\varnothing|w) = 1$ for $w < z^*$. This defines a deterministic persuasion mechanism, as described in Candogan (2022). This mechanism satisfies Bullet 3 of Theorem C.2 and is therefore optimal to (5). Moreover, it exhibits a double-interval structure, with each signal associated with at most two intervals. Notably, the sender's expected payoff under this mechanism is not monotone in w.

Example C.1. Suppose $w \sim \text{Unif}[0, 1]$ follows a uniform distribution on $[0, 1]$, the sender's payoffs from the two receivers are $v_1 = 2$ and $v_2 = 1$, and the receivers' acceptance thresholds are $\alpha_1 = 0.9$ and $\alpha_2 = 0.7$. Given these parameters, $\bar{z}_1 = 0.8$, $z_1 = 0.6$, $z^* = 0.5$, $z_2 = 0.4$, and the optimal dual variables are $\mu_1^* = \mu_2^* = 5$. Figure 4(a) illustrates the two receivers' lines $\ell_1(w; \mu_1^*)$ and $\ell_2(w; \mu_2^*)$, which completely overlap. Additionally, we have $q_1^* = \frac{1}{8}$ and $q_2^* = \frac{3}{8}$. There are various ways to construct an optimal persuasion mechanism satisfying Bullet 3 of Theorem C.2. The previously described randomized persuasion mechanism is illustrated in Figure 4(b). The previously described deterministic persuasion mechanism is illustrated in Figure 4(c). In addition, an alternative deterministic persuasion mechanism with a double-interval structure can be constructed for this instance, where signal $s = 1$ is associated with two intervals, as illustrated in Figure 4(d).

C.3 Optimal Mechanisms with a Non-Selective Receiver 2

In this section, we derive the optimal public persuasion mechanisms for scenarios not covered in Section C.2.

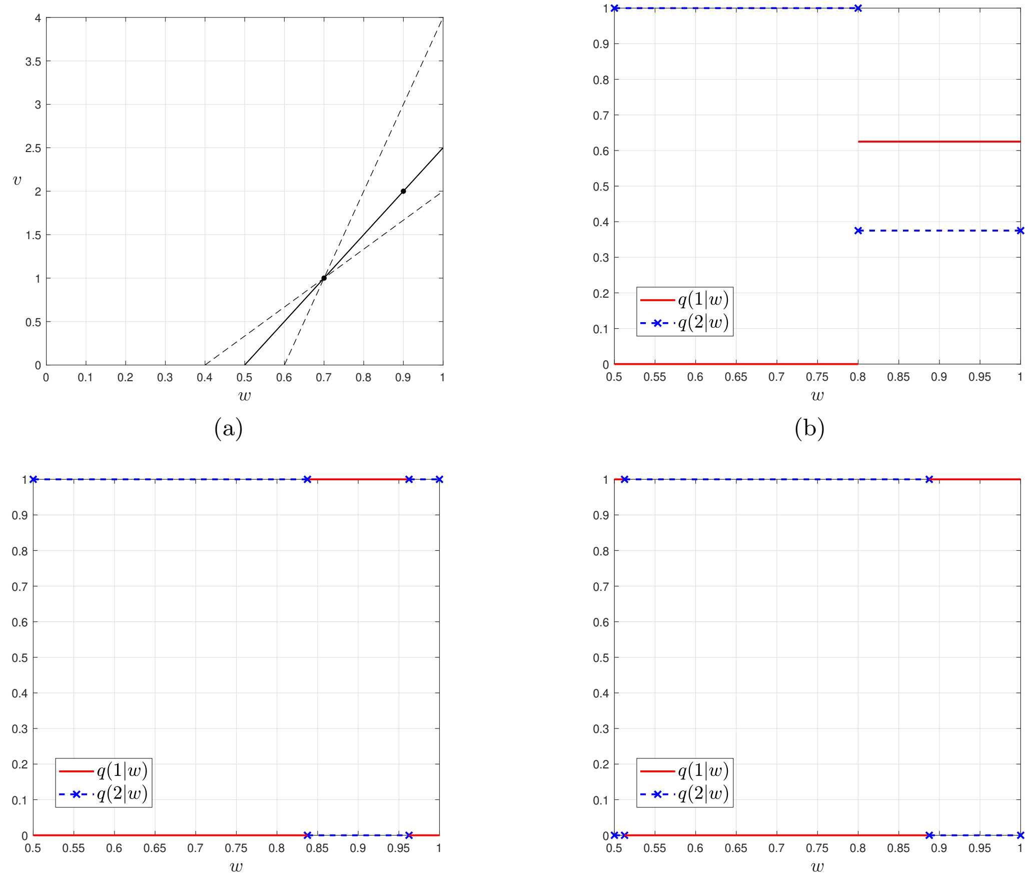
We begin by specifying two trivial scenarios and add assumptions to exclude them. Let $w_0 \triangleq \mathbb{E}_{w \sim G(w)}[w]$ denote the prior mean of the good's characteristic w. First, we assume receiver 1 is selective; that is, $\alpha_1 > w_0$. Otherwise, the optimal mechanism is trivial, as the sender can allocate the good to receiver 1 without revealing any information. Next, define $\bar{z}_1 > 0$ such that $\mathbb{E}[w|w \ge \bar{z}_1] = \alpha_1$.[26] We further assume that $\mathbb{E}[w|w < \bar{z}_1] < \alpha_2$. Otherwise, the optimal mechanism is straightforward: allocate goods with quality $w \in [\bar{z}_1, 1]$ to receiver 1 and those with quality $w \in [0, \bar{z}_1]$ to receiver 2. Finally, we assume receiver 2 is non-selective, meaning $\alpha_2 \le w_0$, which corresponds to the scenario not covered in Section C.2 (see Assumption C.1). We summarize these assumptions in Assumption C.2 and impose them throughout this section.

Assumption C.2. Let $w_0 \triangleq \mathbb{E}_{w \sim G(w)}[w]$ denote the prior mean of the good's characteristic w. Receiver 1's threshold value satisfies $\alpha_1 > w_0$. Moreover, define $\bar{z}_1 > 0$ such that $\mathbb{E}[w|w \ge \bar{z}_1] = \alpha_1$. Receiver 2's threshold value satisfies $\mathbb{E}[w|w < \bar{z}_1] < \alpha_2 \le w_0$.

We now characterize the set of optimal persuasion mechanisms under Assumption C.2. These mechanisms resemble those described in Section C.2, except that they either prioritize receiver 1 (Case 1 of Theorem C.2) or balance between the two receivers as described in Proposition C.1 (Case 3 of Theorem C.2), but never exclusively target receiver 2 (Case 2 of Theorem C.2).

Specifically, following the notation in Section C.2, define line η_1 as passing through the points $(z_1, 0)$ and (α_2, v_2), where z_1 is the threshold value of mechanism M_1 defined in Section C.1.[27] Line η_1 partitions the value of $v_1 \in [v_2, \infty)$ into two regions, as illustrated in Figure 5. If the value of

[25] Such an interval exists because $\mathbb{E}[w|w \ge \bar{z}_1] = \alpha_1$ and $\mathbb{P}[w \ge \bar{z}_1] \ge q_1^*$ (see Lemma C.4 in the Appendix).
[26] Note that $\bar{z}_1 > 0$ follows directly from the assumption $\alpha_1 > w_0$.
[27] We have $z_1 > 0$ by Assumption C.2.



(a) (b) (c) (d)

Figure 4: (a) Illustration of the two receivers' lines, $\ell_1(w; \mu_1^*)$ and $\ell_2(w; \mu_2^*)$ (solid), which completely overlap and pass through points $(0.7, 1)$ and $(0.9, 2)$, along with lines η_1 and η_2 from Theorem C.2 (dashed). (b) A randomized optimal persuasion mechanism, where $q(1|w) = 5/8$ for $w \in [0.8, 1]$, $q(2|w) = 3/8$ for $w \in [0.8, 1]$, and $q(2|w) = 1$ for $w \in [0.5, 0.8]$. (c) A deterministic optimal persuasion mechanism, where $q(1|w) = 1$ for $w \in [0.8375, 0.9625]$ (centered around 0.9 and with length 1/8) and $q(2|w) = 1$ for $w \in [0.5, 0.8375] \cup [0.9625, 1]$. (d) A deterministic optimal persuasion mechanism, where $q(2|w) = 1$ for $w \in [0.5125, 0.8875]$ (centered around 0.7 and with length 3/8) and $q(1|w) = 1$ for $w \in [0.5, 0.5125] \cup [0.8875, 1]$.

v_1 is sufficiently large (in particular, above line η_1), prioritizing receiver 1 is optimal. Otherwise, the optimal mechanism balances between the two receivers as described in Proposition C.1. We characterize these optimal persuasion mechanisms in Theorem C.3.

Theorem C.3. *Under Assumptions 4.1 – 4.3 and C.2, the optimal public persuasion mechanism for two receivers is characterized as follows.*

1. *If $v_1 \geq v_2 \cdot \frac{\alpha_1 - z_1}{\alpha_2 - z_1}$ (i.e., the point (α_1, v_1) lies above line η_1), then mechanism M_1, defined in Section C.1, which prioritizes receiver 1, is the unique optimal mechanism.*

2. *Otherwise, define line ℓ as the line passing through points (α_2, v_2) and (α_1, v_1). Let z^* be the x-intercept of line ℓ if it intersects the x-axis; otherwise, let $z^* = 0$ (in this case, line ℓ intersects the y-axis with an intercept in $[0, v_2]$). Any mechanism M satisfying Proposition C.1*

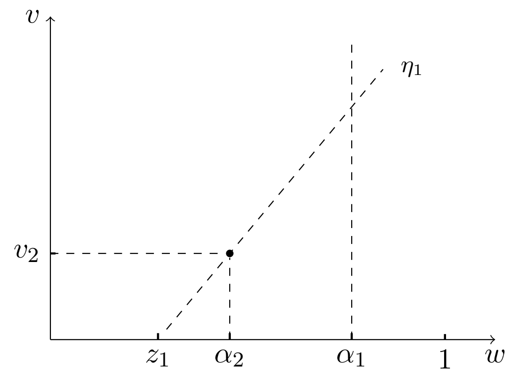


Figure 5: Visualization of the partition in Theorem C.3.

with cutoff value z^* is optimal. Furthermore, this fully characterizes the set of optimal public persuasion mechanisms.

We prove Theorem C.3 in Section C.6. The proof mimics the proof of Theorem C.2: we identify a set of dual variables $\boldsymbol{\mu} \in \mathbb{R}_+^n$, which, together with the proposed mechanism, satisfy Bullet 2 of Lemma 4.2. This indicates that the mechanism is optimal to (5), and $\boldsymbol{\mu}$ is an optimal dual variable.

C.4 Proof of Proposition C.1

For ease of notation, we drop the dependence on the mechanism M by letting $q_1 = q_1(M)$ and $q_2 = q_2(M)$. If the mechanism has a cutoff structure with a threshold z, and the participation constraints for both receivers bind, the following two linear equations must hold:

$$\begin{aligned} q_1 + q_2 &= \mathbb{P}[w \geq z], \\ \alpha_1 q_1 + \alpha_2 q_2 &= (q_1 + q_2) \cdot \mathbb{E}[w | w \geq z]. \end{aligned} \tag{29}$$

The first equation follows from the fact that the good is allocated (to either receiver 1 or 2) if and only if $w \geq z$, and the second equation follows from the cutoff structure, the law of total expectation:

$$\mathbb{E}[w|w \geq z] = \frac{q_1}{q_1 + q_2} \cdot \mathbb{E}[w|s = 1] + \frac{q_2}{q_1 + q_2} \cdot \mathbb{E}[w|s = 2],$$

and the fact that $\mathbb{E}[w|s = i] = \alpha_i$ by the binding participation constraints. The two equations in (29) determine the values of q_1 and q_2 as

$$\begin{aligned} q_1 &= \mathbb{P}[w \geq z] \cdot \frac{\mathbb{E}[w|w \geq z] - \alpha_2}{\alpha_1 - \alpha_2}, \\ q_2 &= \mathbb{P}[w \geq z] \cdot \frac{\alpha_1 - \mathbb{E}[w|w \geq z]}{\alpha_1 - \alpha_2}. \end{aligned} \tag{30}$$

Note that we have $\mathbb{E}[w|w \geq z] \in [\alpha_2, \alpha_1]$ when $z \in [z_2, z_1]$. Hence, $q_1, q_2 \in [0, 1]$ are well-defined probabilities.

We now construct public persuasion mechanisms M satisfying Proposition C.1. Such mechanisms must fulfill the following conditions:

1. $q(1|w) + q(2|w) = 1$ for all $w \geq z$, and $q(1|w) = q(2|w) = 0$ for all $w < z$;

2. $\mathbb{P}[s=1]=q_1$, and $\mathbb{P}[s=2]=q_2$;
3. $\mathbb{E}[w|s=1]=\alpha_1$, $\mathbb{E}[w|s=2]=\alpha_2$, and $\mathbb{E}[w|s=\varnothing]<\alpha_2$.

A feasible mechanism M can be constructed in multiple ways. For example, we can construct a deterministic persuasion mechanism by setting: $q(1|w)=1$ for $w\in T$, $q(2|w)=1$ for $w\in[z,1]\setminus T$, and $q(\varnothing|w)=1$ for $w<z$, for some subset $T\subseteq[z,1]$. To satisfy Proposition C.1, the subset T must meet these conditions:

1. $\mathbb{P}[w\in T]=q_1$ and $\mathbb{P}[w\in[z,1]\setminus T]=q_2$;
2. $\mathbb{E}[w|w\in T]=\alpha_1$, $\mathbb{E}[w|w\in[z,1]\setminus T]=\alpha_2$, and $\mathbb{E}[w|w<z]<\alpha_2$.

There are, again, various ways to construct such a subset T. For instance, T can be chosen as an interval $[\underline{b},\bar{b}]\subseteq[\bar{z}_1,1]$ that contains α_1 and satisfies

$$\mathbb{P}[\underline{b}\le w\le\bar{b}]=q_1 \quad\text{and}\quad \mathbb{E}[w\mid \underline{b}\le w\le\bar{b}]=\alpha_1.$$

The existence of such an interval $[\underline{b},\bar{b}]$ is guaranteed since $\mathbb{E}[w|w\ge\bar{z}_1]=\alpha_1$ and $\mathbb{P}[w\ge\bar{z}_1]\ge q_1$ (see Lemma C.4). Furthermore, conditions

$$\mathbb{P}[w\in[z,1]\setminus T]=q_2 \quad\text{and}\quad \mathbb{E}[w\mid w\in[z,1]\setminus T]=\alpha_2$$

hold by (29). Finally, we verify that $\mathbb{E}[w|w<z]\le\mathbb{E}[w|w<z_1]<\alpha_2$, where the second inequality follows from two scenarios: (i) if $\bar{z}_1\le\alpha_2$, then $z_1=\bar{z}_1\le\alpha_2$; (ii) if $\bar{z}_1>\alpha_2$, then $\mathbb{E}[w\mid w<z_1]<\mathbb{E}[w\mid z_1\le w<\bar{z}_1]=\alpha_2$.

Lemma C.4. *Let $q_1(z)$ denote the probability q_1 defined in* (30). *Then, $q_1(z)\le q_1(z_1)=\mathbb{P}[w\ge\bar{z}_1]$ for all $z\in[z_2,z_1]$.*

Proof. Since

$$\mathbb{E}[w\,|\,w\ge z_1]=\frac{\mathbb{P}[w\ge\bar{z}_1]}{\mathbb{P}[w\ge z_1]}\alpha_1+\frac{\mathbb{P}[z_1\le w\le\bar{z}_1]}{\mathbb{P}[w\ge z_1]}\alpha_2\,,$$

we have $q_1(z_1)=\mathbb{P}[w\ge\bar{z}_1]$ by (30). We next show that $q_1(z)\le q_1(z_1)$ for any $z\in[z_2,z_1]$.

From (30), we can express $q_1(z)$ as:

$$q_1(z)=\frac{1}{\alpha_1-\alpha_2}\int_z^1(w-\alpha_2)g(w)dw.$$

which yields the derivative:

$$\frac{dq_1(z)}{dz}=\frac{\alpha_2-z}{\alpha_1-\alpha_2}\cdot g(z).$$

Since $z_1\le\alpha_2$, $q_1(z)$ is increasing on $[z_2,z_1]$. Therefore, $q_1(z)\le q_1(z_1)=\mathbb{P}[w\ge\bar{z}_1]$ for all $z\in[z_2,z_1]$. □

C.5 Proof of Theorem C.2

In this proof, we identify a set of dual variables $\boldsymbol{\mu}\in\mathbb{R}^n_+$, which, together with the mechanism proposed in Theorem C.2, satisfies Bullet 2 of Lemma 4.2. This indicates that the mechanism is optimal to (5) and that $\boldsymbol{\mu}$ is an optimal dual variable.

Proof of Bullet Two Suppose

$$v_1 \le v_2 \cdot \frac{\alpha_1 - z_2}{\alpha_2 - z_2}\,,$$

that is, the point (α_1, v_1) lies below the line η_2. We construct the two receivers' lines ℓ_1 and ℓ_2 as follows.

Let ℓ_2 coincide with η_2 by choosing the dual variable

$$\mu_2 = \frac{v_2}{\alpha_2 - z_2}\,.$$

Next, let ℓ_1 lie strictly below ℓ_2 for all $w \in [z_2, 1]$. For example, this can be achieved by choosing

$$\mu_1 = \frac{v_1}{\alpha_1 - z_2}\,.$$

The resulting lines ℓ_1 and ℓ_2 are illustrated in Figure 6(a).

Since ℓ_2 dominates ℓ_1 pointwise, an optimal solution to the Lagrangian $V^{\mathrm{LR}}(\boldsymbol{\mu})$ with $\boldsymbol{\mu} = (\mu_1, \mu_2)$ will never allocate the good to receiver 1, irrespective of the good's quality w. It is easy to verify that the mechanism M_2 and the dual variable $\boldsymbol{\mu} = (\mu_1, \mu_2)$ satisfy Lemma 4.2 Bullet 2. Therefore, M_2 is optimal to (5), and $\boldsymbol{\mu} = (\mu_1, \mu_2)$ is an optimal dual solution.

Moreover, given $\boldsymbol{\mu} = (\mu_1, \mu_2)$, M_2 is the unique mechanism that satisfies Bullet 2 of Lemma 4.2.

Proof of Bullet Three Suppose

$$v_1 \in \left(v_2 \cdot \frac{\alpha_1 - z_2}{\alpha_2 - z_2},\ v_2 \cdot \frac{\alpha_1 - z_1}{\alpha_2 - z_1}\right),$$

which implies that the point (α_1, v_1) lies between the two lines η_1 and η_2. Define the dual variables

$$\mu_1 = \mu_2 = \frac{v_1 - v_2}{\alpha_1 - \alpha_2}\,,$$

so that the lines ℓ_1 and ℓ_2 completely overlap and pass through the points (α_2, v_2) and (α_1, v_1). These lines intersect the x-axis at $w = z^* \in [z_2, z_1]$, as illustrated in Figure 6(b).

It is easy to verify that any mechanism M feasible to Theorem C.2 Bullet 3, together with the dual variables $\boldsymbol{\mu} = (\mu_1, \mu_2)$, satisfies Bullet 2 of Lemma 4.2. Therefore, such a mechanism M is optimal to (5), and $\boldsymbol{\mu} = (\mu_1, \mu_2)$ is an optimal dual variable. Moreover, given the optimal dual variable $\boldsymbol{\mu} = (\mu_1, \mu_2)$, a mechanism M satisfies Bullet 2 of Lemma 4.2 if and only if it satisfies Bullet 3 of Theorem C.2.

Proof of Bullet One Suppose

$$v_1 \ge v_2 \cdot \frac{\alpha_1 - z_1}{\alpha_2 - z_1}\,,$$ [28]

which implies that the point (α_1, v_1) lies above line η_1. We construct the receivers' lines ℓ_1 and ℓ_2 by considering two cases: (i) $\bar{z}_1 \le \alpha_2$ and (ii) $\bar{z}_1 > \alpha_2$.

1. $z_1 = \bar{z}_1 \le \alpha_2$: Let line ℓ_1 pass through the points $(z_1, 0)$ and (α_1, v_1) by choosing the dual

[28] If $z_1 = \bar{z}_1 \le \alpha_2$, we set the right-hand side of the inequality to positive infinity.

variable

$$\mu_1 = \frac{v_1}{\alpha_1 - z_1} .$$

Next, let line ℓ_2 lie below ℓ_1 for all $w \in [z_1, 1]$. For example, this can be achieved by choosing

$$\mu_2 = \frac{v_2}{\alpha_2 - z_1} ,$$

which causes line ℓ_2 to coincide with line η_1, because the point (α_2, v_2) lies below line ℓ_1. The lines ℓ_1 and ℓ_2 are illustrated in Figure 6(c).

2. $z_1 < \alpha_2 < \bar{z}_1$: Let line ℓ_2 coincide with line η_1 by choosing the dual variable

$$\mu_2 = \frac{v_2}{\alpha_2 - z_1} .$$

Next, let line ℓ_1 pass through the points

$$\left(\bar{z}_1, \frac{v_2}{\alpha_2 - z_1}(\bar{z}_1 - \alpha_2) + v_2\right) \quad \text{and} \quad (\alpha_1, v_1)$$

by selecting the dual variable

$$\mu_1 = \frac{\frac{v_2}{\alpha_2 - z_1}(\bar{z}_1 - \alpha_2) + v_2 - v_1}{\bar{z}_1 - \alpha_1} .$$

It is easy to verify that line ℓ_1 intersects line ℓ_2 at $w = \bar{z}_1$ and that $\mu_1 > \mu_2$.[29] Moreover, line ℓ_1 lies above line ℓ_2 for all $w \in [\bar{z}_1, 1]$ and line ℓ_2 lies above line ℓ_1 for all $w \in [z_1, \bar{z}_1]$. The lines ℓ_1 and ℓ_2 are illustrated in Figure 6(d).

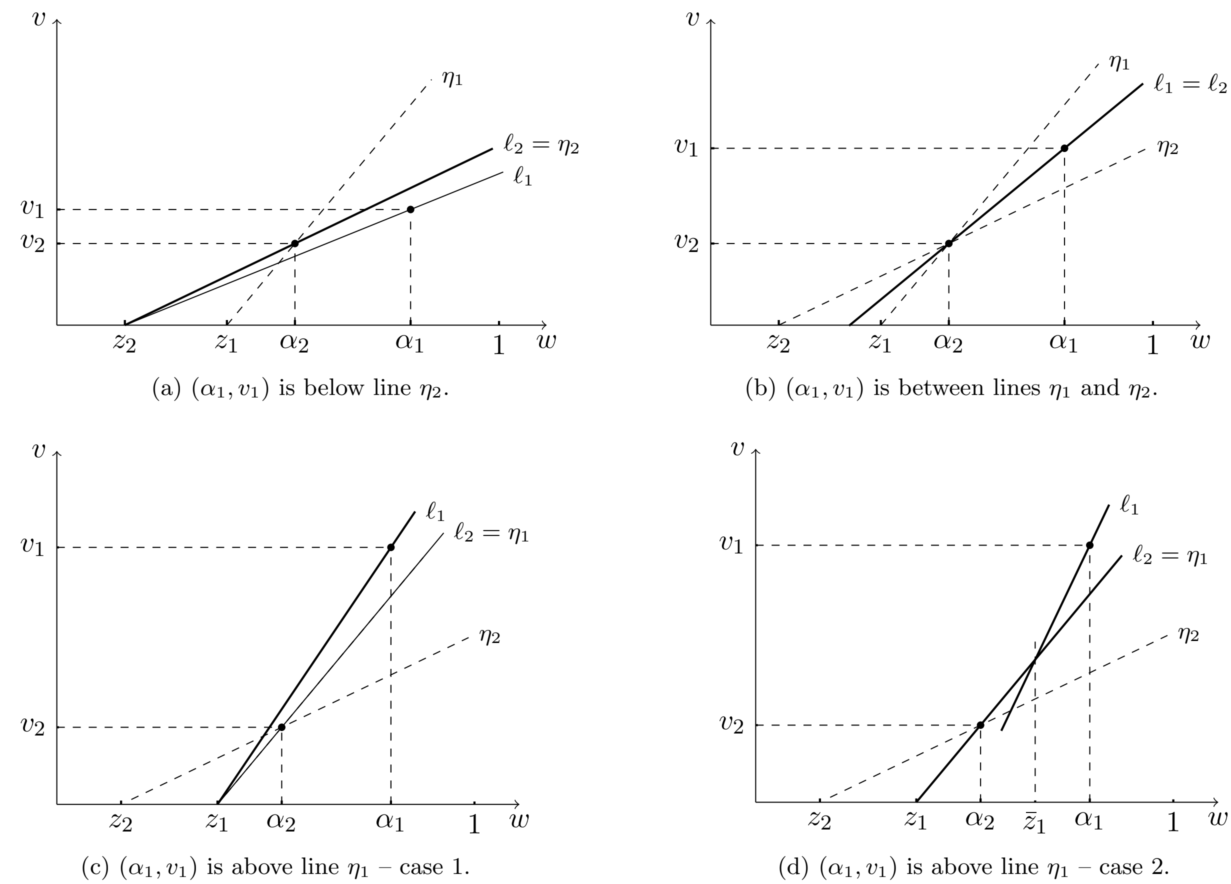
It can be easily verified that mechanism M_1 and the dual variables $\boldsymbol{\mu} = (\mu_1, \mu_2)$ satisfy Bullet 2 of Lemma 4.2 in both cases. Therefore, mechanism M_1 is optimal to problem (5), and $\boldsymbol{\mu} = (\mu_1, \mu_2)$ are optimal dual variables. Moreover, M_1 is the unique mechanism satisfying Bullet 2 of Lemma 4.2 given $\boldsymbol{\mu} = (\mu_1, \mu_2)$.

Complete Characterization of Optimal Persuasion Mechanisms Finally, we note that in all three cases above, given the optimal dual variables $\boldsymbol{\mu} = (\mu_1, \mu_2)$ identified in each case, a mechanism M satisfies Bullet 2 of Lemma 4.2 if and only if it meets the corresponding condition described above. Therefore, by Bullet 2 of Lemma 4.2, a persuasion mechanism M is optimal to (5) if and only if it satisfies Theorem C.2.

C.6 Proof of Theorem C.3

In this proof, we identify a set of dual variables $\boldsymbol{\mu} \in \mathbb{R}^n_+$, which, together with the mechanism proposed in Theorem C.3, satisfies Bullet 2 of Lemma 4.2. This indicates that the mechanism is optimal to (5) and that $\boldsymbol{\mu}$ is an optimal dual variable.

[29] Intuitively, $\mu_1 > \mu_2$ because the point (α_1, v_1) lies above line η_1.



(a) (α_1, v_1) is below line η_2.

(b) (α_1, v_1) is between lines η_1 and η_2.

(c) (α_1, v_1) is above line η_1 – case 1.

(d) (α_1, v_1) is above line η_1 – case 2.

Figure 6: Visualization of two receivers' associated lines.

Proof of Bullet Two Suppose $v_1 \leq v_2 \cdot \frac{\alpha_1 - z_1}{\alpha_2 - z_1}$,[30] which implies that the point (α_1, v_1) lies below line η_1. Define dual variables $\mu_1 = \mu_2 = \frac{v_1 - v_2}{\alpha_1 - \alpha_2} > 0$. Consequently, the two receivers' lines ℓ_1 and ℓ_2 coincide with line ℓ and pass through points (α_2, v_2) and (α_1, v_1), as illustrated in Figure 6(b) (note that line ℓ may either intersect the x-axis or intersect the y-axis below v_2 under Assumption C.2).

It is easy to verify that any mechanism M feasible to Bullet 2 of Theorem C.3, together with the dual variables $\boldsymbol{\mu} = (\mu_1, \mu_2)$ defined above, satisfies Bullet 2 of Lemma 4.2. Therefore, such a mechanism M is optimal to (5), and $\boldsymbol{\mu} = (\mu_1, \mu_2)$ is an optimal dual variable. Moreover, given the optimal dual variable $\boldsymbol{\mu} = (\mu_1, \mu_2)$, a mechanism M satisfies Bullet 2 of Lemma 4.2 if and only if it meets Bullet 2 of Theorem C.3.

Proof of Bullet One Suppose $v_1 \geq v_2 \cdot \frac{\alpha_1 - z_1}{\alpha_2 - z_1}$, which implies that the point (α_1, v_1) lies above line η_1. The proof is identical to the proof of Bullet 1 of Theorem C.2, presented in Section C.5.

Complete Characterization of Optimal Persuasion Mechanisms Finally, we note that given the optimal dual variables $\boldsymbol{\mu} = (\mu_1, \mu_2)$ identified in both cases, a mechanism M satisfies Bullet 2 of Lemma 4.2 if and only if it satisfies Theorem C.3. Therefore, by Bullet 2 of Lemma 4.2, a persuasion

[30] If $z_1 = \bar{z}_1 = \alpha_2$, we set the right-hand side of the inequality to positive infinity.

mechanism M is optimal to (5) if and only if it satisfies Theorem C.3.

D Constructing Optimal Mechanisms with Specific Structure

As discussed in Section 5.3, the general persuasion problem with n receivers decouples over subsets of receivers $\{T_i\}$, and we can build an optimal solution in an iterative way for each subset. Moreover, there exist multiple ways to construct an optimal mechanism, with the set of optimal persuasion mechanisms characterized by Theorem 4.3. In this section, we propose a specific allocation approach at each iteration step to obtain an optimal solution $\{q^*(j|w)\}$ to (5) that has a monotone structure (Section D.1). Additionally, we show that a deterministic persuasion mechanism exhibiting a double-interval structure, as described in Candogan (2022), can be easily derived using results derived from our dual approach (Section D.2).

D.1 Optimal Mechanism with Monotone Structure

In this section, we construct an optimal persuasion mechanism $\{q^*(j|w)\}$ iteratively that additionally satisfies a monotone property, as defined in Definition D.1. Specifically, for any $w \geq w'$, the distribution $q^*(\cdot|w)$ will first-order stochastically dominate the distribution $q^*(\cdot|w')$. Therefore, a good with a higher quality w is more likely to be in a better place, which is desirable in practice.[31,32]

Definition D.1 (Monotone Structure)**.** An optimal persuasion mechanism $\{q^*(j|w)\}$ satisfies a monotone property if, for any $w \geq w'$, the distribution $q^*(\cdot|w)$ first-order stochastically dominates the distribution $q^*(\cdot|w')$; in other words, we have $\sum_{k \leq i} q^*(k|w) \geq \sum_{k \leq i} q^*(k|w')$ for any $i \in [n]$.[33]

The monotone property automatically holds for two qualities w and w' from different intervals. Suppose $w \in I_i$ and $w' \in I_j$ with $i < j$. Since $\max T_i < \min T_j$, a good of quality w joins a better place for sure, which implies first-order stochastic dominance. Therefore, we only need to ensure the monotone structure for qualities within the same interval.

Algorithm 1 presents a way to construct an optimal mechanism $\{q(j|w)\}$ for $j \in T_i$ iteratively. The distribution $q(\cdot|w)$ from Algorithm 1 is first-order stochastically increasing in w; additionally, $q(j|w)$ is piecewise constant on $w \in I_i$ for any $j \in T_i$.

In Algorithm 1, $q_{\leq k}(w) = \sum_{j \leq k} q(j|w)$ represents the probability that a good of quality w receives an offer from one of the top k receivers. Note that for any $k \leq t_{i-1}$ and $w \in I_i$, we have $q_{\leq k}(w) = 0$. In each iteration, we allocate a fraction ρ_k of the remaining goods with quality at least b_k to receiver k. The values of ρ_k and b_k are selected to ensure that receiver k receives the good with probability q_k^*, and that the expected quality of the good, conditional on being allocated to receiver k, is exactly α_k (i.e., (24) holds for receiver k).

We note that the constructed sequence $\{b_k\}_{k \in T_i}$ decreases and partitions the interval I_i into subintervals $I_{ik} \triangleq [b_k, b_{k-1}]$ for $k \in T_i$. Additionally, the probabilities $q(j|w)$ equal a constant $q_j(k)$ on each subinterval $w \in I_{ik}$, where the values of $q_j(k)$ are specified as follows:

$$q(j|w) = q_j(k) = \begin{cases} 0 & k \geq j+1,\ w \in I_{ik} \\ \rho_j \cdot \Big(1 - q_{\leq j-1}(w)\Big) = \rho_j \cdot \prod_{\ell=k}^{j-1} \Big(1 - \rho_\ell\Big) & k \in [t_{i-1}+1 : j],\ w \in I_{ik} \end{cases}$$

[31] For instance, in the student promotion context, this monotone property prevents students from strategically degrading their "quality" w for better positions.

[32] We note that Arieli et al. (2023) also construct optimal persuasion mechanisms with a monotone structure when the distribution of posterior means exhibits a bi-pooling structure (Lemma 2 therein). Our construction is slightly more general by allowing for any distribution of posterior means sustained by an optimal persuasion mechanism.

[33] Note that an offer from a lower-indexed receiver provides a higher payoff to the sender by Assumption 2.1.

Algorithm 1: Optimal Persuasion Mechanism with Monotone Structure

Input: An optimal solution $\{q_i^*\}$ to (9).

Initialization: Set $b_{t_{i-1}} = z_{i-1}$ and initialize $q_{\le t_{i-1}}(w) = 0$ for all $w \in I_i$.

for $k \in T_i = [t_{i-1}+1 : t_i - 1]$ **do**

  Determine values $b_k \in [z_i, b_{k-1}]$ and $\rho_k \in [0,1]$ to satisfy condition (24) for receiver k by setting:

$$q(k|w) = \begin{cases} \rho_k \cdot (1 - q_{\le k-1}(w)), & \text{for all } w \in [b_k, z_{i-1}] \\ 0, & \text{for all } w < b_k \end{cases}$$

  Set $q_{\le k}(w) = q_{\le k-1}(w) + q(k|w)$ for all $w \in I_i$.

end

Set $q(t_i|w) = 1 - q_{\le t_i - 1}(w)$ for all $w \in I_i$ and define $b_{t_i} = z_i$.

Proposition D.1 demonstrates that the values of $\{\rho_k\}$ and $\{b_k\}$ in Algorithm 1 exist, and the allocation $\{q(j|w)\}$ returned by Algorithm 1 is optimal to (5) and satisfies the first-order stochastic increasing property.

Proposition D.1. *The allocation $\{q(j|w)\}$ returned by Algorithm 1 is optimal to (5) and satisfies the first-order stochastic increasing property.*

In Section D.3, we prove Proposition D.1 and demonstrate that the values of $\{\rho_k\}$ and $\{b_k\}$ can be easily identified. Note that when a group T_i contains two receivers, the allocation $\{q(j|w)\}$ returned by Algorithm 1 concurs with the randomized mechanism with a monotone structure described in Section C.2 for the two-receiver case.

D.2 Optimal Mechanism with Double-Interval Structure

In this section, we demonstrate that a deterministic persuasion mechanism with a double-interval structure, as described in Candogan (2022), can be easily derived using results from our dual analysis.

We first present two useful properties of the optimal solutions to (9) as established in Candogan (2022). Specifically, there exists an optimal solution $\{q_k^*\}$ in which each group T_i contains at most two receivers with a positive probability of q_k^*. Additionally, the optimal solution to (9) is unique if no three points of $\{(\alpha_i, v_i)\}_{i\in[n]}$ are collinear. We state these properties in Proposition D.2.

Proposition D.2. *The optimal solution of (9) satisfies the following two properties.*

1. *(Lemma 4 of Candogan 2022) Let $\{\lambda_k^*\}$ denote an optimal dual variable associated with the participation constraints in (9) and $\{T_i\}$ denote the corresponding partition of the n receivers as described in Section 5.2. There exists an optimal solution $\{q_k^*\}$ to (9) such that $|T_i \cap P| \le 2$ for any i, where $P \triangleq \{k \in [n] : q_k^* > 0\}$ denotes the set of positive entries of $\{q_k^*\}$. In other words, each set T_i contains at most two receivers with a positive probability of q_k^*.*
2. *(Appendix D of Candogan 2022) Problem (9) has a unique optimal solution $\{q_k^*\}$ if no three points of $\{(\alpha_i, v_i)\}_{i\in[n]}$ are collinear.*

We prove Proposition D.2 in Section D.4 based on results from our dual approach, which significantly simplifies the proof and makes both properties intuitive. For the first property, suppose a group T_i contains more than two receivers with positive probabilities. Since the points $\{(\alpha_j, v_j)\}_{j\in T_i}$ are collinear by Bullet 2 of Proposition 5.2, we can reallocate the probabilities of two non-adjacent receivers to an intermediate receiver until we drain the probability of one of the two original receivers, without changing the objective value. For the second property, since no three points of $\{(\alpha_i, v_i)\}_{i\in[n]}$ are collinear, Bullet 2 of Proposition 5.2 implies that any group T_i contains at most two receivers with positive probabilities. Furthermore, the values of these probabilities are uniquely determined by two linear equations analogous to (28). We provide more details in Section D.4.

Proposition D.2 implies that the general information design problem can be decomposed into separate design problems with two receivers, one for each group T_i. Applying the deterministic mechanism with a double-interval structure described in Section C.2 to each group T_i, we obtain the deterministic persuasion mechanism described in Candogan (2022).

D.3 Proof of Proposition D.1

In this section, we first prove that the values of $\{b_k\}$ and $\{\rho_k\}$ in Algorithm 1 exist and can be identified efficiently (Section D.3.1). We then show that the assignment probability $q(j|w)$ returned from Algorithm 1 is optimal to (5) and possesses the first-order stochastically increasing property (Section D.3.2).

D.3.1 Existence of $\{b_k\}$ and $\{\rho_k\}$

We prove by induction that the values of $\{\rho_k\}$ and $\{b_k\}$ in Algorithm 1 exist and can be computed efficiently.

Induction Step We first determine the values of $b_{t_{i-1}+1}$ and $\rho_{t_{i-1}+1}$. From (18), the following hold:

$$\mathbb{E}\left[w\middle|G^{-1}\Big(1-\sum_{j\le t_{i-1}+1} q_j^*\Big)\le w < z_{i-1}\right]\ge \alpha_{t_{i-1}+1}\,,$$

$$\mathbb{E}\Big[w\Big|z_i\le w<z_{i-1}\Big]=\sum_{j\in T_i}\alpha_j\cdot\frac{q_j^*}{\sum_{j\in T_i}q_j^*}\le \alpha_{t_{i-1}+1}\,,$$

where the inequality in the second line follows from the fact that $\alpha_{t_{i-1}+1}\ge\alpha_j$ for any $j\in T_i$. Therefore, there exists a value of $b_{t_{i-1}+1}$ satisfying that $z_i\le b_{t_{i-1}+1}\le G^{-1}\big(1-\sum_{j\le t_{i-1}+1}q_j^*\big)\le z_{i-1}$ such that $\mathbb{E}\big[w|b_{t_{i-1}+1}\le w\le z_{i-1}\big]=\alpha_{t_{i-1}+1}$. Additionally, let $\rho_{t_{i-1}+1}=q^*_{t_{i-1}+1}\,/\,\mathbb{P}[b_{t_{i-1}+1}\le w< z_{i-1}]\le 1$. The allocation probability $q(t_{i-1}+1|w)$ satisfies (24) by the setup of $b_{t_{i-1}+1}$ and $\rho_{t_{i-1}+1}$.

Iteration Step Let k be an integer with $k\in[t_{i-1}+2{:}t_i-1]$. Suppose that, for all $j\in[t_{i-1}+1{:}k-1]$, we have already determined the values of ρ_j and b_j such that the probability $q(j|w)$ satisfies (24). We now identify values of ρ_k and b_k so that the probability $q(k|w)$ also satisfies (24).

To achieve this, set $b_k=b\in[z_i,z_{i-1}]$ and $\rho_k=\rho\in\big[\underline{\rho},1\big]$, where $\underline{\rho}\triangleq\frac{q_k^*}{\sum_{\ell=k}^{t_i}q_\ell^*}$. Additionally, define the allocation probability as

$$q(k|w)=\begin{cases}\rho\cdot\big(1-q_{\le k-1}(w)\big), & w\in[b,z_{i-1}],\\ 0, & w\in[z_i,b).\end{cases}$$

Define the following two functions:

$$F\big(b,\rho\big) \triangleq \int_{w\in I_i} q(k|w)\, g(w)\, dw,$$
$$Q\big(b,\rho\big) \triangleq \int_{w\in I_i} w\cdot q(k|w)\, g(w)\, dw.$$

The allocation probability $q(k|w)$ satisfies (24) with $b_k = b$ and $\rho_k = \rho$ if and only if

$$F\big(b,\rho\big) = q_k^* \quad \text{and} \quad Q\big(b,\rho\big) = \alpha_k\, q_k^*\,.$$

Clearly, the function $F(b,\rho)$ is strictly increasing ρ and strictly decreasing in b. Therefore, for any $\rho \in [\underline{\rho}, 1]$, there exists a unique constant, denoted by $b(\rho)$, that satisfies $F(b(\rho),\rho) = q_k^*$. In particular, we have $b\big(\underline{\rho}\big) = z_i$, because

$$\begin{aligned} F\left(z_i,\underline{\rho}\right) &= \underline{\rho}\int_{w\in I_i} \big(1 - q_{\le k-1}(w)\big)\, g(w)\, dw \\ &= \underline{\rho}\left(\int_{w\in I_i} g(w)\, dw - \sum_{j=t_{i-1}+1}^{k-1} \int_{w\in I_i} q(j|w)\, g(w)\, dw\right) \\ &= q_k^*\,, \end{aligned}$$

where the first equality follows from the definition of $q(k|w)$, and the third equality follows from the fact that $\mathbb{P}[z_i \le w < z_{i-1}] = \sum_{j\in T_i} q_j^*$, that the probability $q(j|w)$ satisfies (24) for any $j \le k-1$, and from the definition of $\underline{\rho}$.

Furthermore, the function $b(\rho)$ is strictly increasing in ρ. Therefore, its inverse, denoted by $\rho(b)$, exists and is also strictly increasing. Now, define the function

$$Q(b) \triangleq Q(b,\rho(b))\,.$$

Since $F(b,\rho(b)) = q_k^*$ for any b, it suffices to find a value $b \in [z_i, b_{k-1}]$ satisfying that $Q(b) = \alpha_k q_k^*$, which we do now.

First, we note that the function $Q(b)$ is increasing. This is because, as b increases, we transport a fixed mass q_k^* to higher values, which increases the mean quality of the goods allocated.

Second, we examine the value of $Q(z_i)$. Specifically, the following holds:

$$\begin{aligned} Q(z_i) &= \underline{\rho}\int_{w\in I_i} w\,\big(1 - q_{\le k-1}(w)\big)\, g(w)\, dw \\ &= \underline{\rho}\left(\int_{w\in I_i} w\, g(w)\, dw - \sum_{j=t_{i-1}+1}^{k-1} \int_{w\in I_i} w\cdot q(j|w)\, g(w)\, dw\right) \\ &= \frac{q_k^*}{\sum_{\ell=k}^{t_i} q_\ell^*}\left(\sum_{\ell\in T_i} \alpha_\ell q_\ell^* - \sum_{\ell=t_{i-1}+1}^{k-1} \alpha_\ell q_\ell^*\right) \\ &= \frac{\sum_{\ell=k}^{t_i} \alpha_\ell q_\ell^*}{\sum_{\ell=k}^{t_i} q_\ell^*}\cdot q_k^* \\ &\le \alpha_k\, q_k^*\,, \end{aligned} \tag{31}$$

where the first equality uses $\rho(z_i) = \underline{\rho}$, the third equality follows from the second line of (18) and the fact that the probability $q(j|w)$ satisfies (24) for all $j \leq k-1$, and the inequality in the last line follows from the fact that α_ℓ decreases with index ℓ.

Finally, we derive two additional inequalities. Suppose that $b(1) \geq b_{k-1}$—that is, the "unoccupied" area to the right of b_{k-1} and above the function $q_{\leq k-1}(w)$ exceeds q_k^*. Then, we have

$$Q(b_{k-1}) = \alpha_{k-1}\, q_k^* > \alpha_k\, q_k^*\,, \tag{32}$$

because, in this case, $q(k|w) = c \cdot q(k-1|w)$ for some constant $c > 0$ and for all $w \in I_i$.

Alternatively, suppose that $b(1) \leq b_{k-1}$. Then, we have $q_{\leq k}(w) = 1$ for $w \in [b(1), z_{i-1}]$ and $q_{\leq k}(w) = 0$ for $w < b(1)$, which implies that $b(1) = G^{-1}\Big(1 - \sum_{j \leq k} q_j^*\Big)$. Consequently, the following holds:

$$\begin{aligned}
Q\big(b(1)\big) &= \int_{w \in I_i} w \cdot q_{\leq k}(w)\, g(w)\, dw - \int_{w \in I_i} w \cdot q_{\leq k-1}(w)\, g(w)\, dw \\
&= \int_{b(1)}^{z_{i-1}} w \cdot q_{\leq k}(w)\, g(w)\, dw - \sum_{j=t_{i-1}+1}^{k-1} \int_{w \in I_i} w \cdot q(j|w)\, g(w)\, dw \\
&= \mathbb{E}\left[w \cdot \mathbb{1}\left[G^{-1}\Big(1 - \sum_{j \leq k} q_j^*\Big) \leq w < z_{i-1}\right]\right] - \sum_{j=t_{i-1}+1}^{k-1} \int_{w \in I_i} w \cdot q(j|w)\, g(w)\, dw \\
&\geq \sum_{j=t_{i-1}+1}^{k} \alpha_j q_j^* - \sum_{j=t_{i-1}+1}^{k-1} \alpha_j q_j^* \\
&= \alpha_k\, q_k^*\,,
\end{aligned} \tag{33}$$

where the inequality follows from the first equation in (18) and the fact that the probability $q(j|w)$ satisfies (24) for all $j \leq k-1$.

Since function $Q(b)$ is continuous and strictly increasing in b, inequalities (31)–(33) guarantee the existence of a value $b_k \in \big[z_i, \min\{b_{k-1}, b(1)\}\big]$ such that $Q(b_k) = \alpha_k\, q_k^*$. Moreover, the value b_k can be efficiently computed via binary search. Let $\rho_k = \rho(b_k)$. With these choices of b_k and ρ_k, the resulting allocation probability $q(k|w)$ satisfies (24).

Final Step Let $q(t_i|w) = 1 - q_{\leq t_i - 1}(w)$ for any $w \in I_i$. Since $q(j|w)$ satisfies (24) for any $j \leq t_i - 1$, the second equation in (18) and the fact that $\mathbb{P}[z_i \leq w < z_{i-1}] = \sum_{j \in T_i} q_j^*$ imply that $q(t_i|w)$ also satisfies (24).

D.3.2 Optimality and FOSD Property

Let $\{q(j|w)\}$ denote the output of Algorithm 1. $\{q(j|w)\}$ is optimal to (5) according to Theorem 4.3.

We now prove that the distribution $q(\cdot|w)$ first-order stochastically increases with w on the interval I_i. By definition, this is equivalent to proving that the cumulative distribution function $q_{\leq k}(w)$ is increasing in w for any $k \in T_i$. We prove this by induction. First, $q_{\leq t_{i-1}}(w) = 0$ for any $w \in I_i$ by definition, which serves as the induction step. Next, suppose $q_{\leq k-1}(w)$ is increasing on $w \in I_i$ for some $k \in T_i$, we show that $q_{\leq k}(w)$ is also increasing. To do so, fix two points $w, w' \in I_i$ with $w' < w$. If $w' < b_k$, we have:

$$0 = q_{\leq k}(w') = q_{\leq k-1}(w') \leq q_{\leq k-1}(w) \leq q_{\leq k}(w),$$

where the first inequality is because $q_{\leq k-1}(w)$ increases with w. Alternatively, if $w' \geq b_k$, we have:

$$\begin{aligned}
q_{\leq k}(w') &= q_{\leq k-1}(w') + \rho_k \cdot \big(1 - q_{\leq k-1}(w')\big) \\
&= q_{\leq k-1}(w') + \rho_k \cdot \big(q_{\leq k-1}(w) - q_{\leq k-1}(w')\big) + \rho_k \cdot \big(1 - q_{\leq k-1}(w)\big) \\
&\leq q_{\leq k-1}(w') + q_{\leq k-1}(w) - q_{\leq k-1}(w') + \rho_k \cdot \big(1 - q_{\leq k-1}(w)\big) \\
&= q_{\leq k-1}(w) + \rho_k \cdot \big(1 - q_{\leq k-1}(w)\big) \\
&= q_{\leq k}(w),
\end{aligned}$$

where the inequality follows from the fact that $q_{\leq k-1}(w) \geq q_{\leq k-1}(w')$ and $\rho_k \leq 1$.

D.4 Proof of Proposition D.2

D.4.1 Proof of Bullet One

Let $\{\lambda_k^*\}$ denote an optimal dual variable for the participation constraints in (9) and $\{T_i\}$ denote the resulting partition of the n receivers as described in Section 5.2. For a feasible solution $\{q_k\}$ to (9), let

$$T_i\big(\{q_k\}\big) \triangleq \Big| T_i \cap \{k \in [n] : q_k > 0\} \Big|$$

denote the number of receivers in group T_i that have a positive probability q_k.

Let $\{q_k^*\}$ denote an optimal solution to (9). Lemma D.3 shows that if there exists a group T_i that satisfies $T_i\big(\{q_k^*\}\big) > 2$, we can find a new optimal solution $\{\tilde{q}_k\}$ to (9) that is closer to the desired one in Bullet 1.

Lemma D.3. *Let $\{q_k^*\}$ denote an optimal solution $\{q_k^*\}$ to (9). If there exists a subset T_i that satisfies $T_i\big(\{q_k^*\}\big) > 2$, we can find a new optimal solution $\{\tilde{q}_k\}$ to (9) such that (i) $\tilde{q}_k = q_k^*$ for any $k \notin T_i$, and (ii) $T_i\big(\{\tilde{q}_k\}\big) < T_i\big(\{q_k^*\}\big)$.*

Repeating the process in Lemma D.3 iteratively will eventually (in at most n steps) yield a desired optimal solution to (9) that satisfies Bullet 1.

Proof of Lemma D.3. From Proposition 5.1, there exists an optimal solution $\{q^*(j|w)\}$ to (5) such that the good is allocated to each receiver j with probability q_j^*. Suppose $T_i\big(\{q_k^*\}\big) > 2$. In the following, we modify $\{q^*(j|w)\}$ to create a new optimal solution $\{\tilde{q}(j|w)\}$ to (5) such that the good is allocated to each receiver j with probability $\tilde{q}_j$, where $\{\tilde{q}_j\}$ satisfies Lemma D.3. Then, $\{\tilde{q}_j\}$ is optimal to (9) again according to Proposition 5.1.

Assume $\{a, b, c\} \subseteq T_i\big(\{q_k^*\}\big)$, where a, b, and c denote indices of three distinct receivers. Without loss of generality, assume that $1 \leq a < b < c \leq n$. Therefore, $\alpha_a > \alpha_b > \alpha_c$. We consider the following two scenarios.

Case One Suppose

$$\frac{\alpha_a q_a^* + \alpha_c q_c^*}{q_a^* + q_c^*} = \alpha_b, \tag{34}$$

that is, the mean quality of the goods allocated to receivers a or c is precisely α_b, the acceptance bar of receiver b. Let

$$\tilde{q}(j|w) = \begin{cases} q^*(a|w) + q^*(b|w) + q^*(c|w) & \text{if } j = b, \\ 0 & \text{if } j \in \{a, c\}, \\ q^*(j|w) & \text{if } j \notin \{a, b, c\}. \end{cases}$$

(34) implies that the participation constraint for receiver b remains binding with $\tilde{q}(j|w)$. Therefore, $\tilde{q}(j|w)$ is optimal to (5) according to Theorem 4.3. Additionally, we have

$$\tilde{q}_j \triangleq \int_0^1 \tilde{q}(j|w)\, g(w)\, dw = \begin{cases} q_b^* + q_a^* + q_c^* & \text{if } j = b, \\ 0 & \text{if } j \in \{a, c\}, \\ q_j^* & \text{if } j \notin \{a, b, c\}. \end{cases}$$

As a result, $\{\tilde{q}_j\}$ satisfies Lemma D.3 because $\{\tilde{q}_j\}$ is optimal to (9) by Proposition 5.1 and $T_i\big(\{\tilde{q}_j\}\big) = T_i\big(\{q_j^*\}\big) - 2 < T_i\big(\{q_j^*\}\big)$ by construction.

Case Two Suppose (34) does not hold. Without loss of generality, assume that $\frac{\alpha_a q_a^* + \alpha_c q_c^*}{q_a^* + q_c^*} > \alpha_b$, which translates to $q_a^* > \underline{q}_a \triangleq q_c^* \cdot \frac{\alpha_b - \alpha_c}{\alpha_a - \alpha_b}$. Let $\rho_a \triangleq \underline{q}_a / q_a^* < 1$. Note that the following holds:

$$\frac{\alpha_a \underline{q}_a + \alpha_c q_c^*}{\underline{q}_a + q_c^*} = \alpha_b. \tag{35}$$

Let

$$\tilde{q}(j|w) = \begin{cases} \rho_a \cdot q^*(a|w) + q^*(b|w) + q^*(c|w) & \text{if } j = b, \\ (1 - \rho_a) \cdot q^*(a|w) & \text{if } j = a, \\ 0 & \text{if } j = c, \\ q^*(j|w) & \text{if } j \notin \{a, b, c\}. \end{cases}$$

(35) implies that the participation constraint for receiver b remains binding with $\tilde{q}(j|w)$. Therefore, $\tilde{q}(j|w)$ is optimal to (5) according to Theorem 4.3. Additionally, we have

$$\tilde{q}_j \triangleq \int_0^1 \tilde{q}(j|w)\, g(w)\, dw = \begin{cases} q_b^* + \rho_a \cdot q_a^* + q_c^* & \text{if } j = b, \\ (1 - \rho_a) \cdot q_a^* & \text{if } j = a, \\ 0 & \text{if } j = c, \\ q_j^* & \text{if } j \notin \{a, b, c\}. \end{cases}$$

As a result, $\{\tilde{q}_j\}$ satisfies Lemma D.3 because $\{\tilde{q}_j\}$ is optimal to (9) by Proposition 5.1 and $T_i\big(\{\tilde{q}_j\}\big) = T_i\big(\{q_j^*\}\big) - 1 < T_i\big(\{q_j^*\}\big)$ by construction. □

D.4.2 Proof of Bullet Two

We prove Bullet 2 based on our established results from the dual approach. Assume, without loss of generality, that there exists an optimal solution $\{q_k^*\}$ to (9) such that $q_k^* > 0$ for any $k \in [n]$. We then show that the values of $\{q_k^*\}$ are unique. To see that this assumption loses no generality, let

$$P_\varnothing = \Big\{k \in [n] : q_k^* = 0 \text{ for all optimal solutions } \{q_k^*\} \text{ to } (9)\Big\}$$

denote the set of receivers disregarded by all optimal solutions to (9). We can exclude the receivers in set $P_\varnothing$ without affecting anything. Meanwhile, define $P = [n] \setminus P_\varnothing$. Since (9) is a convex optimization problem, the set of optimal solutions is convex. This implies that there exists an optimal solution $\{q_k^*\}$ such that $q_k^* > 0$ for any $k \in P$.

Now, let $\{\lambda_k^*\}$ denote an optimal dual variable of (9). Let $\{T_i\}$ denote the partition of receivers described in Section 5.2. Since no three points of $\{(\alpha_i, v_i)\}_{i \in [n]}$ are collinear, any group T_i contains at most two receivers by Bullet 2 of Proposition 5.2. Fix a group T_i. First, suppose $T_i = \{k\}$ contains one receiver. Then, we have $q_k^* = \mathbb{P}[z_i \le w \le z_{i-1}]$, whose value is uniquely determined.

Second, suppose $T_i = \{k, j\}$ contains two receivers. Then, the values of q_k^* and q_j^* must satisfy

$$\begin{aligned} q_k^* + q_j^* &= \mathbb{P}[z_i \le w \le z_{i-1}], \\ \alpha_k q_k^* + \alpha_j q_j^* &= \mathbb{E}\Big[w \cdot \mathbb{1}\left[z_i \le w \le z_{i-1}\right]\Big], \end{aligned}$$

and therefore, are uniquely determined as well.

E Algorithm for Constructing the Upper Envelope Function $h(w; \boldsymbol{\mu}^*)$

In this section, we provide a step-by-step procedure for constructing the upper envelope function $h(w; \boldsymbol{\mu}^*)$ according to Proposition 5.2 and Remark 5.1, as summarized in Algorithm 2.

We note that, from a computational perspective, the optimal dual variables $\{\mu_i^*\}$, and hence the function $h(w; \boldsymbol{\mu}^*)$, can also be efficiently computed using subgradient-based methods, as discussed in Remark 4.1.

Algorithm 2: Characterizing the Upper Envelope Function $h(w;\boldsymbol{\mu}^*)$

`// Step 1: Solve the convex relaxation (9) and obtain primal-dual solutions`

Solve problem (9) and obtain an optimal solution $\{q_i^*\}$ and an optimal dual variable $\boldsymbol{\lambda}^* = (\lambda_k^*)_{k\in[n]} \in \mathbb{R}_+^n$ associated with the participation constraints.

`// Step 2: Remove receivers with zero probability (Assumption 5.1)`

Remove any receiver i with $q_i^* = 0$ and reindex so that $q_i^* > 0$ for all $i \in [n]$.

`// Step 3: Construct optimal dual variables` $\boldsymbol{\mu}^*$ `for problem (5)`

Define $\boldsymbol{\mu}^* = (\mu_i^*)_{i\in[n]} \in \mathbb{R}_+^n$ by setting $\mu_i^* = \sum_{k\geq i} \lambda_k^*$ for all $i \in [n]$.

`// Step 4: Identify binding participation constraints and form receiver groups`

Let $T = \{k \in [n] : \lambda_k^* > 0\}$ and order its elements as $T = \{t_1 < t_2 < \cdots < t_m\}$. Set $t_0 = 0$.

if $\lambda_n^* > 0$ **then**

  `// Case 1:` $\lambda_n^* > 0$ `(Proposition 5.2)`

  for $i \in [m]$ **do**

    $T_i = [t_{i-1} + 1 : t_i]$.

else

  `// Case 2:` $\lambda_n^* = 0$ `(Remark 5.1)`

  `// Then` $t_m = n - 1$ `and` $h(w;\boldsymbol{\mu}^*)$ `is horizontal on its leftmost interval`

  for $i \in [m]$ **do**

    $T_i = [t_{i-1} + 1 : t_i]$.

  Set $T_{m+1} = \{n\}$.

`// Step 5: Compute state breakpoints and associated intervals`

Set $z_0 = 1$. **for** $i \in [m]$ **do**

$$z_i = G^{-1}\left(1 - \sum_{j=1}^{t_i} q_j^*\right),$$

  and define state interval $I_i = [z_i, z_{i-1}]$.

if $\lambda_n^* = 0$ **then**

  Define the additional state interval $I_{m+1} = [0, z_m]$.

`// Step 6: Define receivers' lines within each group and assemble` $h(w;\boldsymbol{\mu}^*)$

for $i \in [m]$ **do**

  Pick any receiver $j \in T_i$ (all receivers in T_i share the same line). Define the line

$$\ell_i(w) = v_j + \mu_j^*\,(w - \alpha_j)$$

  Set $h(w;\boldsymbol{\mu}^*) = \ell_i(w)$ on interval I_i.

if $\lambda_n^* > 0$ **then**

  `// extend the last active line leftward`

  Set $h(w;\boldsymbol{\mu}^*) = \ell_m(w)$ on interval $[0, z_m]$; the line crosses the x-axis at $w = z_m$ if $z_m > 0$.

else

  Set $h(w;\boldsymbol{\mu}^*) = v_n$ on interval $[0, z_m]$. `// Leftmost segment is horizontal`